MARTI301

AI in Investment and Competitiveness

EU Business School · Masters Level 7 · Term 3 · 3 ECTS

Framework Sheet Library

Master Index · 24 Slibraries · 120 Reference Sheets

6	**Course Units** *covering strategy, ML, CX, finance, IP, investment*
24	**Slibraries (thematic clusters)** *4 per unit, colour-coded*
120	**Framework Reference Sheets** *5 per Slibrary, print-ready A4*
480+	**Harvard-referenced sources** *peer-reviewed and practitioner*

Design Philosophy

Each sheet follows a consistent nine-section layout: professional definition, who-developed-it, why-it-was-developed, a custom illustrative figure, how/when/by-whom-used (three columns), innovation versus the predecessor, related frameworks in the course, and Harvard-referenced sources. The structure enables students and practitioners to navigate the library as a reference work — treating the 120 sheets as a mental Slibrary (per Meier, 2025) that scales from single-framework lookup to full strategic reasoning across multiple connected tools.

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EU Business School · MARTI301

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Unit 1

AI Strategy & Governance

4 Slibraries · 20 framework sheets

SLIBRARY 1 · Strategic AI Framing — 5 sheets

1. EU AI Act Pyramid
2. NIST RMF
3. Perez S-Curve
4. AI Maturity Staircase
5. McKinsey High Performers

SLIBRARY 2 · Competitive Advantage — 5 sheets

1. Porter Five Forces
2. VRIO Framework
3. Value Chain
4. RBV · Dynamic Capabilities
5. Disruption S-Curves

SLIBRARY 3 · AI Programme Discipline — 5 sheets

1. CRISP-DM
2. DDED Funnel
3. Build-Buy-Borrow
4. AI Project Failure Funnel
5. Data Readiness

SLIBRARY 4 · Prediction & Judgment — 5 sheets

1. Prediction-Judgment
2. Centaur Model
3. Automation-Augmentation
4. BCG People Pillar
5. Human-in-the-Loop

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Unit 2

ML & Data-Driven Decisions

4 Slibraries · 20 framework sheets

SLIBRARY 5 · ML Foundations — 5 sheets

1. Three ML Paradigms
2. Bias-Variance
3. Train-Val-Test
4. Rule-Based vs ML
5. Accuracy-Interpretability

SLIBRARY 6 · MLOps & Lifecycle — 5 sheets

1. MLOps Loop
2. Cross-Validation
3. Model Selection
4. Ensemble Methods
5. Deployment Patterns

SLIBRARY 7 · Data as an Asset — 5 sheets

1. Data Value Chain
2. Data Moat
3. 1st / 3rd-Party Data
4. Feature Engineering
5. Data Network Effects

SLIBRARY 8 · Evaluation & Trust — 5 sheets

1. Confusion Matrix
2. ROC / AUC
3. SHAP Values
4. Drift Monitoring
5. Fairness Metrics

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Unit 3

Customer Experience & Marketing 6.0

4 Slibraries · 20 framework sheets

SLIBRARY 9 · Marketing Evolution — 5 sheets

1. Marketing 1.0 → 6.0
2. Metaverse
3. Phys-Digital Convergence
4. Generative AI in Marketing
5. Human-Centric Marketing

SLIBRARY 10 · Personalisation Engines — 5 sheets

1. Personalisation Maturity
2. Recommendation Systems
3. Realtime vs Batch
4. Privacy-Personalisation Trade-off
5. Uplift Modelling

SLIBRARY 11 · Customer Journey & AI — 5 sheets

1. Journey Map with AI
2. Next-Best-Action
3. Omnichannel
4. Sentiment Analysis
5. Customer Lifetime Value

SLIBRARY 12 · Trust & Brand — 5 sheets

1. Trust Framework
2. Brand Personality
3. Chatbot Maturity
4. Deepfake Risks
5. AI-Driven Loyalty

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Unit 4

Financial Side of AI · Risk Management

4 Slibraries · 20 framework sheets

SLIBRARY 13 · Cost-Benefit Analysis — 5 sheets

1. Cost-Benefit Waterfall
2. ROI Payback
3. Four Value Drivers
4. NPV Formula
5. Cost Breakdown

SLIBRARY 14 · TCO Framework — 5 sheets

1. TCO Iceberg
2. Cloud vs On-Prem
3. GPU Cost Curve
4. Hidden Cost Categories
5. FinOps Loop

SLIBRARY 15 · AI Risk Management — 5 sheets

1. AI Risk Taxonomy
2. SR 11-7 MRM
3. Risk Heatmap
4. Three Lines of Defence
5. AI Incident Response

SLIBRARY 16 · Responsible AI Scorecard — 5 sheets

1. Responsible AI Radar
2. Bias Audit Pipeline
3. RAI Governance Maturity
4. Red-Team Loop
5. Board Oversight Chain

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Unit 5

IP & Value Creation through AI

4 Slibraries · 20 framework sheets

SLIBRARY 17 · Copyright & Authorship — 5 sheets

1. Copyright Ownership Tree
2. Four Authorship Models
3. Getty v Stability
4. Training Data Provenance
5. Moral Rights

SLIBRARY 18 · AI Licensing Models — 5 sheets

1. Open-Closed Spectrum
2. AI Licence Types
3. API Tiering
4. Revenue Share Models
5. IP Risk Allocation

SLIBRARY 19 · Data IP Protection — 5 sheets

1. Data Rights Stack
2. Trade Secret vs Patent vs Copyright
3. EU Database Right
4. Data Licensing Taxonomy
5. Defensive Moats

SLIBRARY 20 · AI Monetisation Models — 5 sheets

1. Monetisation Canvas
2. AI Pricing Models
3. Platform vs Product
4. Ecosystem Capture Map
5. Value Capture Curve

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Unit 6

AI in Investment Management · Capstone

4 Slibraries · 20 framework sheets

SLIBRARY 21 · Algorithmic Trading & Quant — 5 sheets

1. Algo Trading Spectrum
2. Portfolio AI Stack
3. Factor Zoo
4. Risk Analytics · VaR
5. Investment AI Maturity

SLIBRARY 22 · Investment Process & Aladdin — 5 sheets

1. Investment Process Cycle
2. BlackRock Aladdin · Modules
3. Alternative Data
4. Backtesting Pipeline
5. ESG + AI Integration

SLIBRARY 23 · AI Adoption & Transformation — 5 sheets

1. Rogers Diffusion
2. Pilot-to-Scale
3. Transformation Flywheel
4. Talent Matrix
5. AI Readiness Radar

SLIBRARY 24 · Capstone Integration — 5 sheets

1. Six-Unit Integration
2. AI Strategy Canvas
3. Klarna Integrated Case
4. Practitioner Playbook
5. Future-Proof Leadership

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How to Use This Library

For students

Each Slibrary is designed as a mental reference — a 'Thumbnail Thinking' collection (Meier, 2025). Use individual sheets for framework lookup, entire Slibraries for thematic deep-dives, and the full library as a reference catalogue. The Related-Frameworks pills on every sheet are navigation links to the most relevant neighbouring ideas in the course.

For practising executives

The library functions as a decision-support reference. When facing an AI investment, strategy, or governance decision, the relevant framework(s) can be located by unit, keyword, or cross-reference. Sheet structure is optimised for quick comprehension: definition, visual, how/when/who used. Most sheets can be absorbed in under two minutes.

For MARTI301 assessments

The midterm and final briefs expect explicit application of named frameworks to the Klarna case (Slibrary 24 Sheet 3). Students should cite the specific sheet (e.g. 'Applying the TCO Iceberg from Slibrary 14 Sheet 1...') to demonstrate framework mastery. The resit brief similarly requires framework citation.

Reading order

Linear reading (Slibrary 1 through 24) follows the course's weekly sequence and is recommended for first-time readers. Topic-specific reading (e.g. IP concerns → Slibraries 17–20) is recommended for practitioners with focused questions. Cross-reference navigation via the Related-Frameworks pills produces exploration paths that reveal deeper connections.

Design conventions

Colour coding signals category: blue = strategy; violet/teal/pink = ML; orange = CX; gold/copper/red/purple = finance & risk; teal/indigo/emerald/rose = IP; amber/blue/purple/slate = investment. Each sheet's left-edge accent bar uses its Slibrary's signature colour. Harvard author-date referencing is used throughout, with full citations at the bottom of each sheet.

EU AI Act Risk Pyramid

Four risk tiers — four regulatory regimes for AI systems in the EU

PROFESSIONAL DEFINITION

The EU AI Act Risk Pyramid is the classification scheme that the EU Artificial Intelligence Act (Regulation (EU) 2024/1689) uses to assign regulatory obligations to AI systems on the basis of the risk they pose to health, safety, and fundamental rights. Systems are placed into one of four tiers — Unacceptable, High, Limited, or Minimal risk — and the obligations on providers and deployers scale with the tier (European Parliament and Council, 2024).

WHO DEVELOPED IT

The risk-tiered approach was proposed by the European Commission in its 2021 draft AI Act (European Commission, 2021) and adopted in Regulation (EU) 2024/1689 (European Parliament and Council, 2024). The underlying idea — proportionate, risk-based regulation — draws on product-safety legislation (Veale and Zuiderveen Borgesius, 2021).

WHY IT WAS DEVELOPED

The EU sought a regulatory instrument that would neither chill innovation by over-regulating low-risk systems nor leave dangerous applications unchecked. The pyramid resolves this tension by concentrating obligations where harm potential is greatest, banning a narrow set of unacceptable uses outright, and leaving the vast majority of AI systems lightly regulated.

FIGURE 1 · THE FOUR-TIER RISK PYRAMID

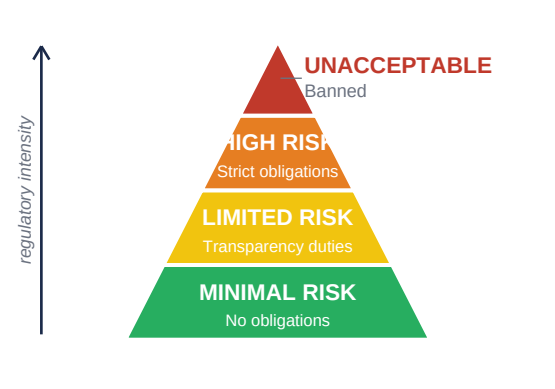

Figure 1. The four-tier EU AI Act risk pyramid. Unacceptable-risk systems are prohibited; high-risk systems face strict conformity assessment; limited-risk systems owe transparency duties; minimal-risk systems are unregulated (European Parliament and Council, 2024).

HOW IT IS USED

Each AI use-case is classified into one of the four tiers; the tier determines the conformity assessment, documentation, and post-market monitoring obligations that must be met before deployment in the EU.

WHEN IT IS USED

At AI project intake, before procurement decisions, during impact assessments, and whenever a system is materially modified or repurposed for a new context.

BY WHOM IT IS USED

Chief AI Officers, legal and compliance teams, risk managers, product owners of AI systems, and EU regulators supervising providers and deployers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with sector-specific regulation (GDPR for data, MDR for medical devices), the Risk Pyramid is a horizontal, cross-sector instrument: it applies to any AI system placed on the EU market irrespective of domain. Compared with principles-based guidance (OECD AI Principles, 2019), it is binding and enforceable, with fines up to €35m or 7% of global turnover for prohibited-use violations (European Parliament and Council, 2024).

RELATED FRAMEWORKS IN THIS COURSE

NIST AI RMF · *Slibrary 1* | **AI Maturity Staircase** · *Slibrary 1* | **AI Project Failure Funnel** · *Slibrary 3* | **Responsible AI Scorecard** · *Slibrary 16* | **Authorship Decision Tree** · *Slibrary 17*

SOURCES (HARVARD REFERENCING STYLE)

European Commission (2021) Proposal for a Regulation laying down harmonised rules on Artificial Intelligence (COM/2021/206 final). Brussels: European Commission.

European Parliament and Council (2024) Regulation (EU) 2024/1689 on Artificial Intelligence. Official Journal of the European Union, L series, 12 July 2024.

OECD (2019) Recommendation of the Council on Artificial Intelligence. OECD/LEGAL/0449. Paris: OECD.

Veale, M. and Zuiderveen Borgesius, F. (2021) 'Demystifying the Draft EU Artificial Intelligence Act', Computer Law Review International, 22(4), pp. 97–112.

NIST AI RMF Lifecycle Wheel

Govern · Map · Measure · Manage — the four core functions of trustworthy AI

PROFESSIONAL DEFINITION

The NIST AI Risk Management Framework (AI RMF 1.0) is a voluntary governance framework that organises trustworthy AI practice into four core functions: Govern (cross-cutting), Map (context and scope), Measure (assess and analyse risks), and Manage (prioritise and act). The wheel visualises these as a continuous lifecycle rather than a linear process (NIST, 2023).

WHO DEVELOPED IT

Published by the U.S. National Institute of Standards and Technology in January 2023, with a Generative AI Profile added in July 2024 (NIST, 2023; NIST, 2024). The framework was developed through open consultation with industry, civil society, and academic stakeholders.

WHY IT WAS DEVELOPED

Organisations needed a structured, voluntary framework to operationalise trustworthy AI principles. The RMF fills the gap between high-level ethics guidelines (OECD, 2019) and detailed technical standards, giving firms a repeatable process to identify, assess, and mitigate AI risks across the system lifecycle.

FIGURE 1 · THE NIST AI RMF WHEEL

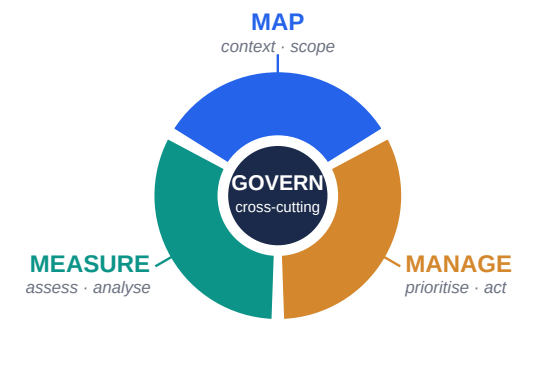

Figure 1. The NIST AI RMF wheel. Govern is cross-cutting and sits at the centre; Map, Measure, and Manage are the outer functions that cycle continuously through the AI system lifecycle (NIST, 2023).

HOW IT IS USED

Each function is operationalised through subcategories and example actions. Teams use the wheel to audit current practice, identify gaps, and allocate accountability for each function to named owners.

WHEN IT IS USED

At the start of AI programme design, during annual risk reviews, after incidents, and whenever a new AI capability (e.g. generative AI) is introduced into the organisation.

BY WHOM IT IS USED

AI risk officers, governance committees, internal audit, cybersecurity and privacy teams, and product leaders responsible for AI systems in regulated industries.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with ISO/IEC 42001 (AI Management System), which is certification-oriented, the NIST RMF is principles-and-practice oriented and freely adoptable. Compared with the EU AI Act, NIST is voluntary but internationally applicable — firms frequently adopt NIST to demonstrate reasonable care in jurisdictions without binding AI law. The RMF complements, rather than duplicates, regulatory compliance.

RELATED FRAMEWORKS IN THIS COURSE

EU AI Act Risk Pyramid · *Slibrary 1*
Responsible AI Scorecard · *Slibrary 16*
AI Project Lifecycle (DDED) · *Slibrary 3*
Bias-Variance Tradeoff · *Slibrary 5*

SOURCES (HARVARD REFERENCING STYLE)

NIST (2023) Artificial Intelligence Risk Management Framework (AI RMF 1.0), NIST AI 100-1. Gaithersburg, MD: National Institute of Standards and Technology.
NIST (2024) Artificial Intelligence Risk Management Framework: Generative AI Profile, NIST AI 600-1. Gaithersburg, MD: NIST.
ISO/IEC (2023) ISO/IEC 42001:2023 — Artificial Intelligence Management System. Geneva: International Organization for Standardization.
Tabassi, E. (2023) 'The AI RMF Playbook and its role in trustworthy AI', NIST Special Publication. Gaithersburg, MD: NIST.

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Perez Techno-Economic S-Curve

SLIBRARY 01 · SHEET 3 / 5

Irruption · Frenzy · Turning Point · Synergy · Maturity

PROFESSIONAL DEFINITION

Carlota Perez's Techno-Economic S-Curve models the diffusion of a technological revolution as two successive S-curves: an Installation Period (Irruption → Frenzy) driven by financial capital, a Turning Point crisis, and a Deployment Period (Synergy → Maturity) driven by productive capital. Each paradigm — from the Industrial Revolution to the current ICT/AI paradigm — follows the same structural pattern (Perez, 2002).

WHO DEVELOPED IT

The framework was developed by Venezuelan-British economist Carlota Perez in Technological Revolutions and Financial Capital (Perez, 2002), building on Schumpeter's theory of creative destruction (Schumpeter, 1942) and Freeman and Louçã's long-wave analysis (Freeman and Louçã, 2001).

WHY IT WAS DEVELOPED

Perez set out to explain why technological revolutions recur with similar financial and social dynamics across 250 years of capitalism. The S-curve showed that bubbles, crashes, and 'golden ages' are not accidents but structural features of how new general-purpose technologies are absorbed into the economy.

FIGURE 1 · TWO-PHASE DIFFUSION CURVE

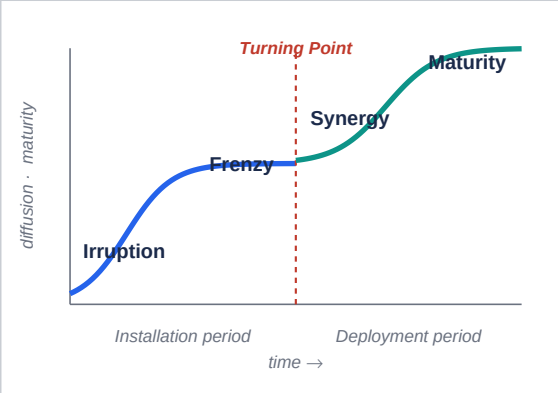

Figure 1. Perez's two-curve model of a technological revolution. The Installation period (blue) is dominated by financial capital and ends in a Turning Point crisis; the Deployment period (teal) is dominated by productive capital and diffuses the paradigm across the economy (Perez, 2002).

HOW IT IS USED

The current AI paradigm is mapped onto the curve to estimate where the economy sits (late-Frenzy, Turning Point, early-Synergy) and what investment posture — defensive, experimental, or offensive — is appropriate.

WHEN IT IS USED

For long-horizon corporate strategy, venture investment theses, technology adoption debates ('is AI a bubble?'), and policy analysis of industrial strategy.

BY WHOM IT IS USED

Corporate strategists, institutional investors, venture capitalists, think tanks, and policy-makers shaping industrial and innovation policy.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the Gartner Hype Cycle, which is descriptive and technology-specific, Perez's curve is explanatory and paradigm-specific — it situates a single revolution within a century-scale cycle of financial and productive capital. Compared with Rogers' Diffusion of Innovations, Perez adds the financial-capital dimension that explains why Frenzy periods end in crashes.

RELATED FRAMEWORKS IN THIS COURSE

AI Maturity Staircase · *Slibrary 1* | **Disruption S-Curves** · *Slibrary 2* | **Porter's Five Forces** · *Slibrary 2* | **Change Management Flywheel** · *Slibrary 24*

SOURCES (HARVARD REFERENCING STYLE)

Perez, C. (2002) Technological Revolutions and Financial Capital: The Dynamics of Bubbles and Golden Ages. Cheltenham: Edward Elgar.
Freeman, C. and Louçã, F. (2001) As Time Goes By: From the Industrial Revolutions to the Information Revolution. Oxford: Oxford University Press.
Rogers, E.M. (2003) Diffusion of Innovations. 5th edn. New York, NY: Free Press.
Schumpeter, J.A. (1942) Capitalism, Socialism and Democracy. New York, NY: Harper & Brothers.

AI Maturity Staircase

Stagnating · Emerging · Scaling · Future-built

PROFESSIONAL DEFINITION

The AI Maturity Staircase is a four-level diagnostic that classifies an organisation's AI capability on a single ordered scale from Stagnating (no value) through Emerging and Scaling to Future-built (value at scale). It converts the multidimensional reality of AI adoption into a single positioning statement and a clear direction of travel (Boston Consulting Group, 2025).

WHO DEVELOPED IT

The four-level classification is the diagnostic backbone of BCG's Build for the Future study (Boston Consulting Group, 2024; 2025), based on a 41-dimension maturity score validated across ~1,250 firms globally. It synthesises earlier staged-adoption models (CMMI Institute, 2018; Westerman, Bonnet and McAfee, 2014).

WHY IT WAS DEVELOPED

Executives needed a single, defensible answer to 'where are we on AI?' that was benchmarkable across peers. BCG's research found that only 5% of firms achieve value at scale and 60% achieve no material value — a gap that widens each year. The staircase makes the gap visible and the path upward explicit.

FIGURE 1 · THE FOUR-STEP MATURITY LADDER

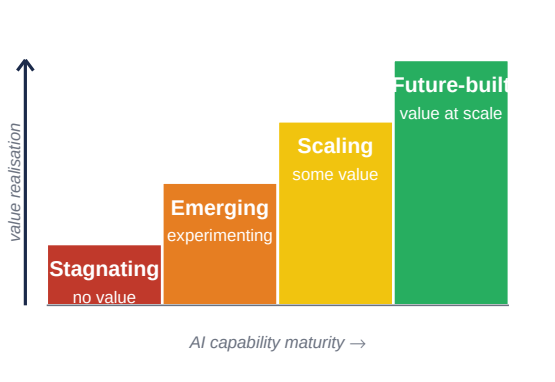

Figure 1. The four-step AI Maturity Staircase. Each step represents a qualitatively different stage of value realisation; Future-built firms generate 1.7× revenue growth and 3.6× three-year TSR versus laggards (Boston Consulting Group, 2025).

HOW IT IS USED

Leadership self-assesses the organisation against the 41 BCG dimensions or a tailored subset, positions itself on one step, and identifies the two or three lifts required to advance to the next step within a defined horizon.

WHEN IT IS USED

At strategic planning cycles, before board discussions of AI spend, for investor communication, and when benchmarking against peers in the same sector.

BY WHOM IT IS USED

CEOs, boards, Chief AI Officers, transformation offices, strategy consultancies, and institutional investors assessing AI readiness of portfolio companies.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with generic digital maturity models (e.g. MIT-Capgemini Digital Maturity, Westerman et al., 2014), the BCG staircase is AI-specific, empirically grounded, and tied to financial outcomes (TSR, EBIT margin, patents). Compared with the Gartner AI Maturity Model (five levels, 2023), BCG's four-level scheme is simpler and linked to a named top tier — 'future-built' — that names the destination rather than describing it abstractly.

RELATED FRAMEWORKS IN THIS COURSE

EU AI Act Risk Pyramid · *Slibrary 1*
Perez Techno-Economic S-Curve · *Slibrary 1*
AI Readiness Assessment Radar · *Slibrary 24*
Investment AI Maturity Model · *Slibrary 24*

SOURCES (HARVARD REFERENCING STYLE)

Boston Consulting Group (2024) Where's the Value in AI? Build for the Future 2024. Boston, MA: BCG.
Boston Consulting Group (2025) The Widening AI Value Gap. Boston, MA: BCG.
CMMI Institute (2018) CMMI V2.0: Model for Development. Pittsburgh, PA: ISACA/CMMI Institute.
Westerman, G., Bonnet, D. and McAfee, A. (2014) Leading Digital: Turning Technology into Business Transformation. Boston, MA: Harvard Business Review Press.

McKinsey AI High Performers

The practice gap between value-capturing firms and the rest

PROFESSIONAL DEFINITION

The McKinsey AI High Performers chart is a paired-bar visualisation that compares the share of 'AI high performers' — firms attributing at least 20% of EBIT to AI — against the share of other respondents across a set of management practices. The gap between the two bars on any practice identifies where imitators should invest to close the distance (McKinsey, 2024; 2025).

WHO DEVELOPED IT

Developed by McKinsey's QuantumBlack AI practice and published annually in the State of AI survey since 2017 (McKinsey, 2017–2025). The 2025 edition surveyed 1,993 respondents across industries and geographies (McKinsey, 2025).

WHY IT WAS DEVELOPED

The chart was designed to answer a recurring executive question: 'what do AI leaders actually do differently?' By ranking practices by the size of the high-performer/other gap, it converts a large survey dataset into a short, imitable list of leadership behaviours.

FIGURE 1 · HIGH PERFORMERS VS. OTHERS

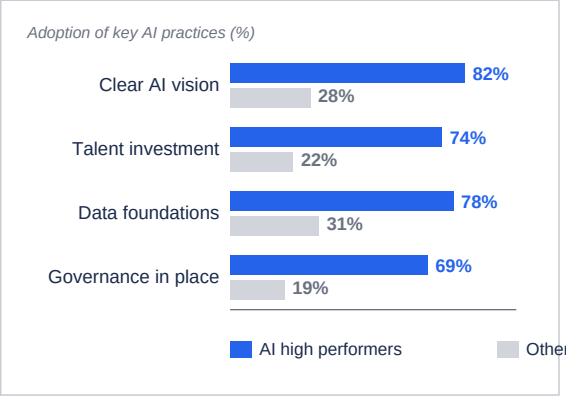

Figure 1. Adoption of four AI practices among AI high performers (blue) vs other respondents (grey). The largest gaps appear in governance and talent — the practices hardest to copy (McKinsey, 2025).

HOW IT IS USED

Executives read the chart as a gap analysis: for each practice with a large high-performer / other gap, assess current state, set a 12-month target, and assign a named owner.

WHEN IT IS USED

At strategic reviews, during annual AI programme planning, and when benchmarking investment allocation against industry peers.

BY WHOM IT IS USED

CEOs, CIOs, Chief AI Officers, transformation leads, and strategy consultants using the McKinsey data as an external benchmark.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with descriptive adoption reports (e.g. Stanford AI Index, 2024), the High Performers chart is prescriptive — it identifies which practices correlate with value capture rather than simply measuring diffusion. Compared with the BCG Widening Value Gap framing, which is outcome-led, McKinsey's lens is practice-led: it tells executives what to do, not only where they stand.

RELATED FRAMEWORKS IN THIS COURSE

AI Maturity Staircase · *Slibrary 1* | **NIST AI RMF** · *Slibrary 1* | **Porter's Five Forces** · *Slibrary 2* | **BCG People Investment Thesis** · *Slibrary 4*

SOURCES (HARVARD REFERENCING STYLE)

McKinsey (2024) The State of AI in Early 2024: Gen AI Adoption Spikes and Starts to Generate Value. New York, NY: McKinsey & Company.
McKinsey (2025) The State of AI in 2025: Agents, Innovation, and Transformation. New York, NY: McKinsey & Company.
Stanford HAI (2024) Artificial Intelligence Index Report 2024. Stanford, CA: Stanford University Human-Centered AI Institute.
Chui, M., Hall, B., Mayhew, H., Singla, A. and Sukharevsky, A. (2023) 'The State of AI in 2023: Generative AI's Breakout Year', McKinsey Quarterly.

Porter's Five Forces

The structural attractiveness of an industry in five forces

PROFESSIONAL DEFINITION

Porter's Five Forces is a framework for analysing the structural attractiveness of an industry by decomposing competitive pressure into five interacting forces: the threat of new entrants, the bargaining power of buyers, the threat of substitutes, the bargaining power of suppliers, and the intensity of rivalry among existing competitors (Porter, 1980). The stronger these forces, the lower the average profitability attainable in the industry.

WHO DEVELOPED IT

Developed by Harvard Business School professor Michael E. Porter in Competitive Strategy (Porter, 1980), drawing on industrial organisation economics and the structure-conduct-performance paradigm (Bain, 1956; Mason, 1939). Porter later extended the analysis in On Competition (Porter, 2008).

WHY IT WAS DEVELOPED

Strategists in the 1970s had case-based intuitions but no systematic lens for comparing industries. Porter's framework reduced industry analysis to five quantifiable forces, turning strategy formulation from art into a structured diagnostic that could be taught, taught again, and applied consistently across sectors (Porter, 1980).

FIGURE 1 · THE FIVE-FORCE DIAMOND

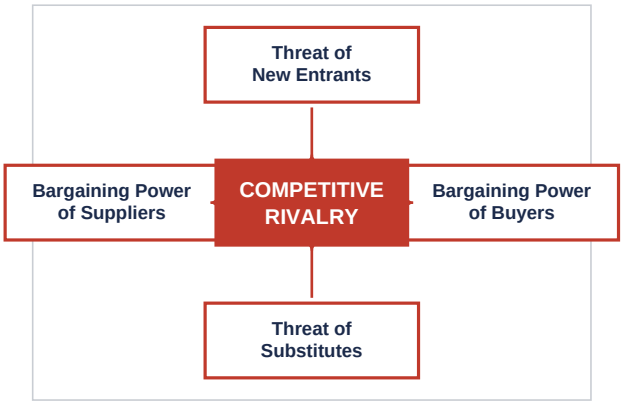

Figure 1. The five structural forces that shape industry profitability. All four peripheral forces flow inward toward the central rivalry, which is itself amplified or dampened by the four (Porter, 1980).

HOW IT IS USED

Analysts score each force as low / medium / high based on industry evidence, then draw strategic implications: if barriers to entry are low, incumbents invest in scale; if buyer power is high, they pursue differentiation.

WHEN IT IS USED

Market entry decisions, M&A target screening, annual strategy reviews, and whenever a technology shift (such as AI-driven disintermediation) alters one or more forces.

BY WHOM IT IS USED

Strategy consultants, corporate strategists, investment analysts, venture capitalists, and MBA students applying industry analysis in any commercial context.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with SWOT analysis, which collects internal and external factors as a loose list, Five Forces is explanatory: it tells you why profitability differs across industries. Compared with PESTEL, which scans macro environment, Five Forces focuses on the industry boundary where margin is actually made. Porter and Heppelmann's (2014) extension adapts the framework for data and network-effect industries, where substitution and entrant dynamics are reshaped by platform economics.

RELATED FRAMEWORKS IN THIS COURSE

VRIO Framework · *Slibrary 2* | **Porter's Value Chain** · *Slibrary 2* | **Resource-Based View** · *Slibrary 2* | **Disruption S-Curves** · *Slibrary 2* | **Perez Techno-Economic S-Curve** · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

Porter, M.E. (1980) Competitive Strategy: Techniques for Analyzing Industries and Competitors. New York, NY: Free Press.
Porter, M.E. (1985) Competitive Advantage: Creating and Sustaining Superior Performance. New York, NY: Free Press.
Porter, M.E. (2008) On Competition. Updated and expanded edn. Boston, MA: Harvard Business Review Press.
Porter, M.E. and Heppelmann, J.E. (2014) 'How Smart, Connected Products Are Transforming Competition', Harvard Business Review, November 2014, pp. 64–88.

VRIO Framework

Valuable · Rare · Inimitable · Organised — the test for sustained advantage

PROFESSIONAL DEFINITION

VRIO is a four-question filter that tests whether a firm resource or capability can be a source of sustained competitive advantage. A resource passes only if it is Valuable (exploits opportunities or neutralises threats), Rare (held by few competitors), costly to Imitate (hard to copy), and the firm is Organised to capture its value. A 'no' at any gate reduces the resource to a lesser outcome (Barney and Hesterly, 2020).

WHO DEVELOPED IT

Developed by Jay B. Barney as VRIN in 'Firm Resources and Sustained Competitive Advantage' (Barney, 1991), refined to VRIO in Gaining and Sustaining Competitive Advantage (Barney, 1995; Barney and Hesterly, 2020). The framework operationalises the broader Resource-Based View of the firm (Wernerfelt, 1984; Penrose, 1959).

WHY IT WAS DEVELOPED

Porter's Five Forces explained why some industries are more profitable than others, but not why some firms outperform rivals within the same industry. Barney's VRIO filled the gap by turning the RBV into a decision tool: which of our resources actually give us durable advantage? (Barney, 1991).

FIGURE 1 · THE FOUR-GATE FILTER

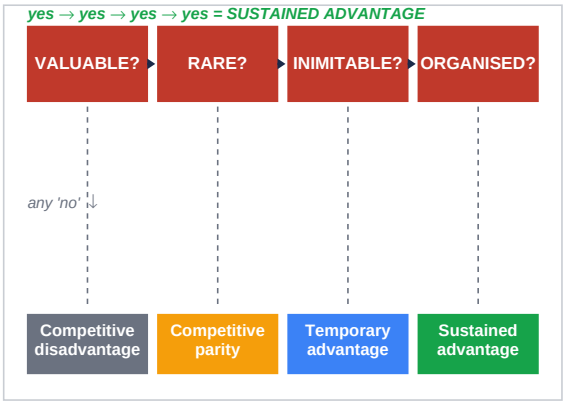

Figure 1. The VRIO filter. Each gate passed raises the strategic value of the resource; only resources that pass all four gates produce sustained competitive advantage (Barney and Hesterly, 2020).

HOW IT IS USED

Each candidate resource or capability is walked through the four gates. The framework surfaces resources that deserve protection and investment, and those that offer only temporary or parity-level outcomes.

WHEN IT IS USED

Strategic reviews, M&A due diligence, AI capability audits ('is our data asset truly rare?'), and whenever a firm must justify investment in a proprietary resource.

BY WHOM IT IS USED

Corporate strategists, HR business partners auditing talent, technology leaders assessing proprietary systems, and investors screening for durable moats.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the original RBV (Wernerfelt, 1984), which was conceptual, VRIO is procedural — it gives practitioners a sequence of yes/no questions. Compared with Porter's Five Forces, which is industry-level, VRIO is firm-level: the two are complementary, not substitutes. For AI, VRIO is used to test whether a proprietary dataset, a talent pool, or a model architecture is a genuine source of advantage or merely a necessary cost of doing business (Davenport and Ronanki, 2018).

RELATED FRAMEWORKS IN THIS COURSE

Porter's Five Forces · *Slibrary 2* | **Porter's Value Chain** · *Slibrary 2* | **Resource-Based View** · *Slibrary 2*

Data Readiness Ladder · *Slibrary 3* | **AI Readiness Assessment Radar** · *Slibrary 24*

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Porter's Value Chain

Primary and support activities that create the margin

PROFESSIONAL DEFINITION

The Value Chain disaggregates the firm into nine generic activities — five primary (inbound logistics, operations, outbound logistics, marketing & sales, service) and four support (firm infrastructure, HR, technology development, procurement) — each of which contributes to the margin the firm earns. Competitive advantage arises from performing one or more of these activities better or more cheaply than rivals (Porter, 1985).

WHO DEVELOPED IT

Introduced by Michael E. Porter in Competitive Advantage (Porter, 1985) as the firm-level complement to the industry-level Five Forces. Later refined by practitioners including Rappa (2010) and extended to digital value networks by Stabell and Fjeldstad (1998).

WHY IT WAS DEVELOPED

Strategy textbooks of the early 1980s treated the firm as a black box. Porter's value chain opened the box: it gave strategists a repeatable way to locate where cost advantages are created and where differentiation can be built, and to identify the activity linkages through which those advantages compound.

FIGURE 1 · PRIMARY AND SUPPORT ACTIVITIES

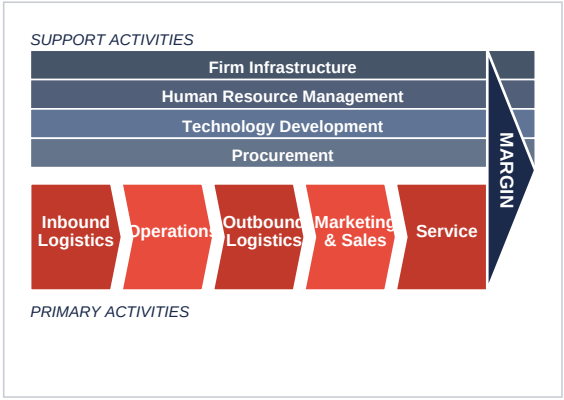

Figure 1. The value chain. Five primary activities flow left-to-right; four support activities span the top. The triangular 'Margin' captures the gap between total value created and total cost of activities (Porter, 1985).

HOW IT IS USED

The firm's activities are mapped onto the chain, then benchmarked against competitors to identify where advantage is built or destroyed. AI initiatives are now typically positioned on specific activities rather than deployed horizontally.

WHEN IT IS USED

Strategic cost analysis, AI use-case prioritisation, operating-model design, and post-merger integration when two activity sets must be rationalised.

BY WHOM IT IS USED

Operations leaders, strategy teams, management consultants, transformation offices, and Chief AI Officers mapping where AI can most profitably be embedded.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the earlier 'functional view' of the firm (marketing, finance, production as silos), the value chain emphasises horizontal flow: activities are linked, and linkage management is itself a source of advantage. Compared with process maps, the chain is strategic rather than operational — it asks not 'how do we do this?' but 'where do we win?'. Applied to AI (Agrawal, Gans and Goldfarb, 2018), the chain identifies activities where prediction is a bottleneck.

RELATED FRAMEWORKS IN THIS COURSE

Porter's Five Forces · *Slibrary 2* | **VRIO Framework** · *Slibrary 2* | **Resource-Based View** · *Slibrary 2* | **CRISP-DM Cycle** · *Slibrary 3* | **Data Value Chain** · *Slibrary 7*

SOURCES (HARVARD REFERENCING STYLE)

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Resource-Based View

Resources → Capabilities → Core Competencies → Sustained Advantage

PROFESSIONAL DEFINITION

The Resource-Based View (RBV) holds that sustained competitive advantage originates from the firm's bundle of heterogeneous, immobile resources rather than from industry position. Resources are combined into capabilities; a subset of capabilities become core competencies; and those that satisfy VRIN conditions (Valuable, Rare, Inimitable, Non-substitutable) deliver advantage that persists over time (Barney, 1991).

WHO DEVELOPED IT

First articulated by Edith Penrose in The Theory of the Growth of the Firm (Penrose, 1959). Formalised for strategy by Wernerfelt (1984), extended by Barney (1991), and linked to core competencies by Prahalad and Hamel (1990). The dynamic extension — how capabilities are created and reconfigured — was developed by Teece, Pisano and Shuen (1997) and refined in Teece (2007).

WHY IT WAS DEVELOPED

Industrial-organisation theories of strategy attributed performance differences to industry structure, but empirical studies (Rumelt, 1991) showed firm-level effects were far larger. The RBV filled the gap by locating advantage inside the firm, in its idiosyncratic resource stocks accumulated over time.

FIGURE 1 · THE NESTED RESOURCE LAYERS

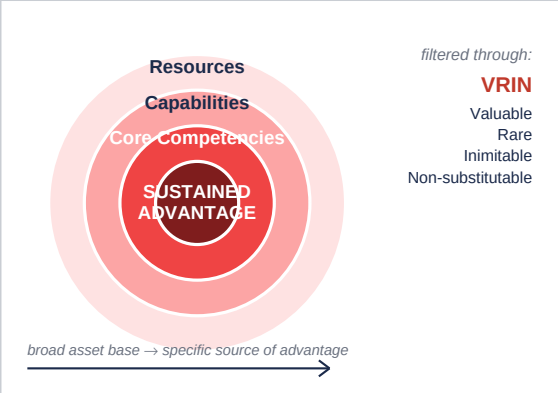

Figure 1. The RBV as nested layers. A broad resource base narrows through capabilities into a small set of core competencies; only those passing the VRIN filter yield sustained advantage (Barney, 1991; Teece, 2007).

HOW IT IS USED

A firm inventories its resources, groups them into capabilities, identifies the two or three core competencies that differentiate it, and uses VRIN to filter which of these genuinely sustain advantage.

WHEN IT IS USED

Strategic planning, portfolio reviews, capability-building programmes, and whenever 'build vs buy vs partner' decisions must be justified against an existing base.

BY WHOM IT IS USED

Executive teams, strategy offices, organisational development functions, and consultants guiding capability-led transformation programmes.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with Porter's industry-structure view, the RBV treats the firm — not the industry — as the unit of analysis. Compared with the earlier 'distinctive competence' literature (Selznick, 1957), the RBV adds economic rigour by specifying the VRIN conditions. The Dynamic Capabilities extension (Teece, 2007) further addresses how resources are sensed, seized, and reconfigured in environments of rapid change — making the RBV directly applicable to firms navigating AI-driven disruption.

RELATED FRAMEWORKS IN THIS COURSE

Porter's Five Forces · *Slibrary 2* | **VRIO Framework** · *Slibrary 2* | **Porter's Value Chain** · *Slibrary 2* | **Disruption S-Curves** · *Slibrary 2* | **AI Maturity Staircase** · *Slibrary 1*

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Disruption S-Curves

SLIBRARY 02 · SHEET 5 / 5

Incumbents plateau while entrants accelerate — the innovator's crossover

PROFESSIONAL DEFINITION

The Disruption S-Curves model visualises technological succession as two overlapping S-shaped performance curves: an incumbent technology that rises, matures, and plateaus; and an entrant technology that starts below the performance threshold but improves faster, eventually crossing over. Christensen (1997) showed that well-managed incumbents routinely lose to disruptors not through poor execution, but by listening too closely to their existing customers.

WHO DEVELOPED IT

The technology S-curve was popularised by Richard Foster in Innovation: The Attacker's Advantage (Foster, 1986), drawing on earlier work by Fisher and Pry (1971). Clayton Christensen extended it into the Theory of Disruptive Innovation in The Innovator's Dilemma (Christensen, 1997), distinguishing sustaining from disruptive innovation. Christensen, Raynor and McDonald (2015) later refined and clarified the theory.

WHY IT WAS DEVELOPED

Business historians had observed that industry leaders frequently fall when new technologies arise — despite superior resources and management. Christensen's insight was that incumbents' rational focus on their most profitable customers causes them to underinvest in initially inferior but rapidly improving entrant technologies.

FIGURE 1 · INCUMBENT AND ENTRANT TRAJECTORIES

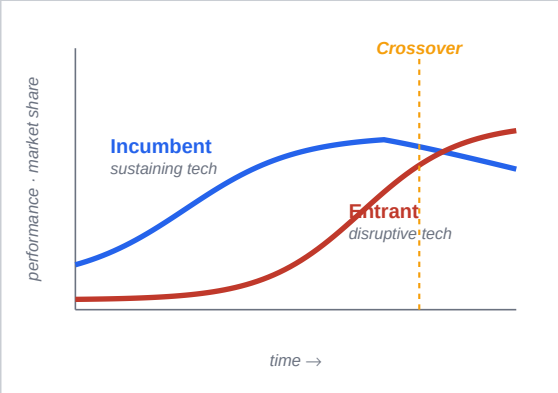

Figure 1. The incumbent (blue) delivers sustaining performance improvements until its curve plateaus. The entrant (red) starts below the adequacy threshold but accelerates and eventually crosses over (Christensen, 1997).

HOW IT IS USED

Analysts plot the incumbent and candidate entrant technologies on a common performance axis over time, identify the likely crossover, and use that timing to trigger incumbent defensive moves or entrant investment decisions.

WHEN IT IS USED

Technology strategy reviews, venture investment theses, corporate R&D allocation, and scenario planning during technology transitions such as generative AI or agentic AI succession.

BY WHOM IT IS USED

CTOs, innovation directors, venture capitalists, corporate venture teams, and strategy consultants advising on disruption risk.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the single technology S-curve (Foster, 1986), the disruption model adds the second curve and the crossover concept, making the framework predictive rather than descriptive. Compared with Perez's techno-economic paradigm, which operates at century-scale, Christensen's curves work at industry-scale — typically 5 to 20 years. In AI, the model explains why incumbents with strong predictive-ML operations may still be disrupted by entrants building on foundation-model and agentic capabilities.

RELATED FRAMEWORKS IN THIS COURSE

Porter's Five Forces · *Slibrary 2* | **VRIO Framework** · *Slibrary 2* | **Resource-Based View** · *Slibrary 2*
Perez Techno-Economic S-Curve · *Slibrary 1* | **AI Maturity Staircase** · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

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CRISP-DM Cycle

Six iterative phases from business question to deployed model

PROFESSIONAL DEFINITION

CRISP-DM (Cross-Industry Standard Process for Data Mining) is a six-phase, iterative process model that structures data mining and, by extension, machine learning projects. The phases — Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment — are executed cyclically, with feedback loops that push teams back to earlier phases whenever findings warrant (Chapman et al., 2000). It is the most widely adopted process model in industrial data-science practice.

WHO DEVELOPED IT

Developed in 1996 by a consortium of five companies — Teradata, SPSS (now IBM), Daimler AG, NCR, and OHRA — and published as an open standard in Chapman et al. (2000). Updates have been proposed (CRISP-DM 2.0 draft; ASUM-DM by IBM, 2015) but the original six-phase model remains dominant in practitioner surveys (KDnuggets, 2014–2023).

WHY IT WAS DEVELOPED

Early data-mining projects routinely failed because business stakeholders, data engineers, and modellers each worked from their own mental models of the workflow. CRISP-DM provided a shared vocabulary and a disciplined handoff sequence, making projects auditable, repeatable, and transferable across teams (Chapman et al., 2000).

FIGURE 1 · THE SIX PHASES

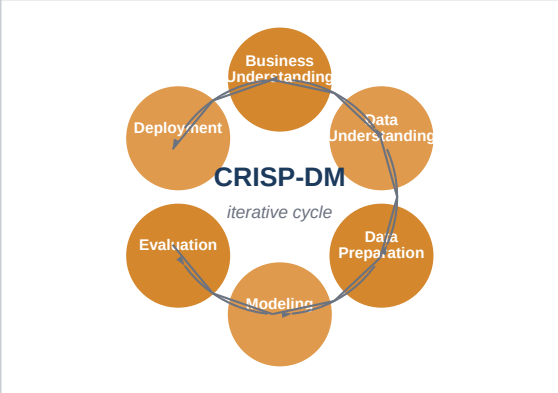

Figure 1. The CRISP-DM cycle. Arrows between phases indicate the expected forward flow; feedback loops (not shown) allow the team to return to any earlier phase when findings require it (Chapman et al., 2000).

HOW IT IS USED

Each project defines entry and exit criteria for all six phases, holds a review at the end of each, and explicitly documents whether the outcome justifies moving forward, iterating, or terminating the project.

WHEN IT IS USED

Applicable to virtually any supervised or unsupervised ML project; especially valuable when the business problem is underspecified, when data quality is unknown, or when multiple stakeholders must agree on scope.

BY WHOM IT IS USED

Data scientists, ML engineers, analytics leads, and the business sponsors they serve; also used by auditors reviewing AI systems for regulatory conformance.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the earlier KDD process (Fayyad, Piatetsky-Shapiro and Smyth, 1996), CRISP-DM added the explicit Business Understanding and Deployment phases, anchoring the workflow to business value rather than model accuracy. Compared with modern MLOps pipelines, CRISP-DM is method-agnostic and does not prescribe tooling — a strength for cross-industry adoption, but a limitation for highly automated production systems, where frameworks such as Microsoft TDSP or Google ML Ops complement it.

RELATED FRAMEWORKS IN THIS COURSE

DDED Stage-Gate Funnel · *Slibrary 3* | **Build vs. Buy** · *Slibrary 3* | **AI Project Failure Funnel** · *Slibrary 3* | **Data Readiness Ladder** · *Slibrary 3* | **NIST AI RMF** · *Slibrary 1*

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DDED Stage-Gate Funnel

SLIBRARY 03 · SHEET 2 / 5

Design · Develop · Evaluate · Deploy — with go/hold/kill gates between stages

PROFESSIONAL DEFINITION

The DDED Stage-Gate Funnel is a four-stage lifecycle model for AI projects — Design, Develop, Evaluate, Deploy — with formal gate reviews between each stage at which a project is continued, held, or killed. The funnel narrows at each gate, reflecting the principle that uncertainty is resolved and commitment increases as the project advances (Cooper, 2008).

WHO DEVELOPED IT

The Stage-Gate model was developed by Robert G. Cooper for new-product development (Cooper, 1990; 2008). Its DDED adaptation for AI projects is taught in the MARTI301 course and aligns with Microsoft's Team Data Science Process (TDSP) and BCG's AI delivery methodology (BCG, 2024).

WHY IT WAS DEVELOPED

Without formal gates, AI projects routinely continue past the point at which evidence suggests they should be stopped. The gated funnel imposes disciplined decision points that surface risk early, preserve budget, and force explicit sponsorship at each incremental investment (Cooper, 2008).

FIGURE 1 · FOUR STAGES, THREE GATES

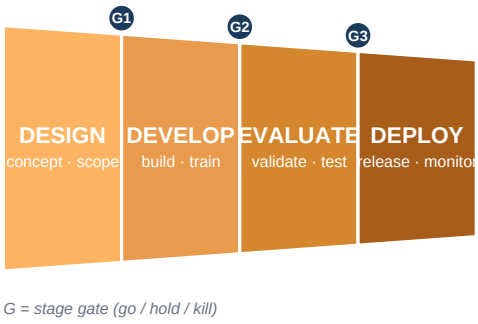

Figure 1. The DDED funnel. Each stage is separated by a gate (G1, G2, G3) at which a go/hold/kill decision is taken; the funnel narrows as commitment rises and uncertainty falls (Cooper, 2008).

HOW IT IS USED

Each gate defines mandatory deliverables — a business case at G1, a working prototype at G2, validated metrics at G3 — and a named decision-maker who can authorise continuation, pause, or termination.

WHEN IT IS USED

For AI initiatives requiring multi-year, multi-million-euro commitments; less suited to rapid experimentation, where Lean Startup-style build–measure–learn loops are more efficient.

BY WHOM IT IS USED

Transformation offices, AI programme managers, portfolio owners, and steering committees approving AI investments.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the CRISP-DM cycle, which is iterative and learning-oriented, DDED is decision-oriented — it answers 'should we continue?' rather than 'what did we learn?'. Compared with Lean Startup, DDED imposes firmer commitment thresholds, making it more suitable for capital-intensive AI programmes where pivot cost is high. Many firms run CRISP-DM inside each DDED stage, treating them as complementary rather than competing models (Cooper, 2008; BCG, 2024).

RELATED FRAMEWORKS IN THIS COURSE

CRISP-DM Cycle · *Slibrary 3* | **Build vs. Buy** · *Slibrary 3* | **AI Project Failure Funnel** · *Slibrary 3* | **NIST AI RMF** · *Slibrary 1* | **Pilot-to-Scale Staircase** · *Slibrary 24*

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Build vs. Buy Matrix

Strategic importance × internal capability — four AI sourcing strategies

PROFESSIONAL DEFINITION

The Build vs. Buy Matrix is a 2×2 framework that maps each AI initiative on two dimensions — strategic importance to the firm, and the firm's current internal capability — and prescribes one of four sourcing strategies: Build internally, Partner with a specialist, Buy as SaaS, or consume via API. The right choice depends on both axes, not just cost (Davenport and Ronanki, 2018; McKinsey, 2024).

WHO DEVELOPED IT

The generic build-vs-buy 2×2 has roots in transaction-cost economics (Williamson, 1981) and IT outsourcing strategy (Lacity and Willcocks, 1998). Its AI-specific adaptation was codified by Davenport and Ronanki in 'Artificial Intelligence for the Real World' (Davenport and Ronanki, 2018), with further refinement in McKinsey's State of AI series (McKinsey, 2024; 2025).

WHY IT WAS DEVELOPED

As AI capabilities proliferated — foundation models, SaaS copilots, bespoke models — firms needed a decision rule to avoid two common errors: building commodity capabilities internally (wasting talent) and buying strategic capabilities externally (ceding moats). The 2×2 forces explicit reasoning on both axes before committing capital.

FIGURE 1 · THE SOURCING MATRIX

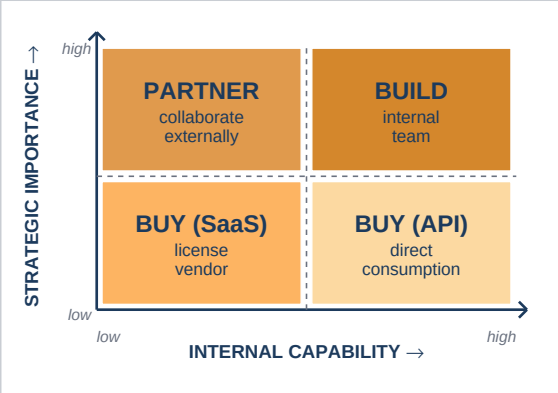

Figure 1. The Build vs. Buy matrix. High strategic importance favours Build or Partner; low strategic importance favours Buy (SaaS or API). Internal capability determines which of the two applies on each axis (Davenport and Ronanki, 2018).

HOW IT IS USED

Each candidate AI use-case is scored on both axes and placed in a quadrant. The quadrant dictates the sourcing strategy and identifies the supporting investments (talent, tools, partnerships) required.

WHEN IT IS USED

AI portfolio reviews, vendor selection decisions, technology roadmapping, and when evaluating whether to in-source a previously outsourced AI capability.

BY WHOM IT IS USED

CIOs, Chief AI Officers, enterprise architects, procurement teams, and strategy consultants supporting AI sourcing decisions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with a pure cost-benefit analysis, the matrix captures the strategic dimension — some capabilities are worth building even at higher cost because they constitute the firm's moat. Compared with the RBV (Slibrary 2), which asks whether a resource is VRIN, Build vs. Buy translates that question into an actionable sourcing choice. In the foundation-model era, the matrix must also factor in vendor lock-in, model drift, and data leakage risks (McKinsey, 2025).

RELATED FRAMEWORKS IN THIS COURSE

CRISP-DM Cycle · *Slibrary 3* | **DDED Stage-Gate Funnel** · *Slibrary 3* | **VRIO Framework** · *Slibrary 2* | **Resource-Based View** · *Slibrary 2* | **AI Readiness Assessment Radar** · *Slibrary 24*

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AI Project Failure Funnel

From pilot to value — where AI initiatives die

PROFESSIONAL DEFINITION

The AI Project Failure Funnel visualises the steep attrition from initial AI pilots to durable value. At each stage — leaving pilot, reaching production, delivering measurable ROI, and sustaining that value at scale — a further cohort of projects falls out. Industry studies consistently find that only a small minority of AI initiatives clear all four gates (MIT Sloan Management Review, 2024; Boston Consulting Group, 2025).

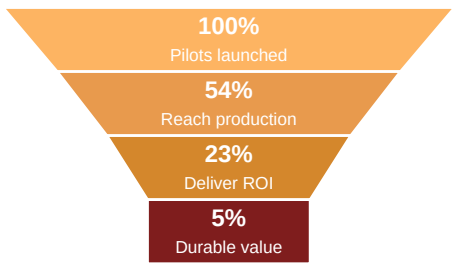
WHO DEVELOPED IT

The funnel is a composite picture assembled from MIT Sloan Management Review and BCG's Build for the Future series (BCG, 2024; 2025), McKinsey's State of AI survey (McKinsey, 2024), and Gartner's AI project benchmarks (Gartner, 2023). No single author claims the framework; it is a pedagogical synthesis of industry data.

WHY IT WAS DEVELOPED

Executives hearing AI success stories often underestimate the base-rate failure. The funnel reframes the conversation from 'how do we launch AI?' to 'how do we avoid being the 95% that fails to capture value?' — directing investment toward the weakest stage of the funnel rather than adding more pilots at the top.

FIGURE 1 · ATTRITION FROM PILOT TO VALUE



indicative attrition rate (MIT Sloan, 2024; BCG, 2025)

Figure 1. Indicative attrition at each stage of AI project delivery. Only ~5% of pilots create durable value at scale (BCG, 2025). The numbers vary by industry but the shape is remarkably consistent across surveys.

HOW IT IS USED

Teams benchmark their own portfolio against the industry funnel, identify the stage where their attrition is steepest, and invest disproportionately in closing that gap — usually data foundations, change management, or evaluation rigour.

WHEN IT IS USED

Annual AI programme reviews, board-level conversations about AI spend, and diagnostic sessions when leadership is frustrated with slow AI returns.

BY WHOM IT IS USED

CAIOs, transformation offices, boards, institutional investors assessing AI readiness, and consultants diagnosing AI execution gaps.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-number 'X% of AI projects fail' headlines, the funnel locates failure at a specific stage, which points to a specific remedy. Compared with BCG's Widening Value Gap framing (Slibrary 1), which classifies firms by maturity, the funnel classifies projects by stage — the two views are complementary: firm maturity predicts where in the funnel a firm loses projects (BCG, 2025).

RELATED FRAMEWORKS IN THIS COURSE

CRISP-DM Cycle · *Slibrary 3*
DDED Stage-Gate Funnel · *Slibrary 3*
AI Maturity Staircase · *Slibrary 1*
Pilot-to-Scale Staircase · *Slibrary 24*
Planning vs. Implementation Gap · *Slibrary 24*

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Data Readiness Ladder

Five levels from raw data to AI-ready

PROFESSIONAL DEFINITION

The Data Readiness Ladder is a five-level classification that describes an organisation's data estate from Raw (L1) through Consolidated (L2), Cleansed (L3), and Enriched (L4) to fully AI-Ready (L5). Each rung specifies the technical and governance conditions a dataset must satisfy; AI initiatives are only feasible at the rung that matches their data quality requirement (DAMA International, 2017; Microsoft, 2022).

WHO DEVELOPED IT

The ladder is a synthesis of several adjacent models: DAMA-DMBOK's data quality dimensions (DAMA International, 2017), Google Cloud's Data Maturity Model (Google Cloud, 2022), and Microsoft's Data Estate Maturity assessment (Microsoft, 2022). The five-rung version is the form most commonly taught in executive AI education.

WHY IT WAS DEVELOPED

AI projects frequently fail at the data stage — not because the model is wrong, but because the data cannot support the use-case (MIT Sloan, 2024). The ladder gives executives a vocabulary to talk about data quality without needing engineering detail, and directs investment to the specific rung blocking the programme.

FIGURE 1 · FIVE LEVELS OF DATA READINESS

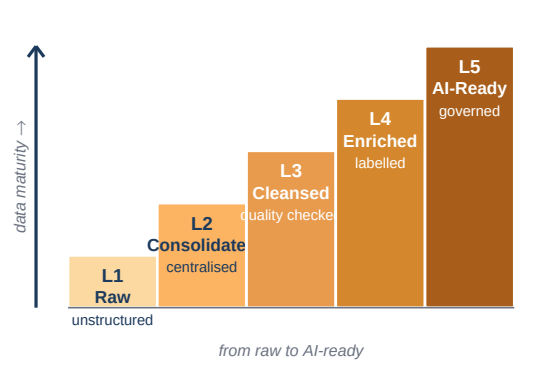

Figure 1. The Data Readiness Ladder. Each rung adds quality controls, governance, and integration capabilities; production AI requires L4 at minimum, and real-time AI systems typically require L5 (Microsoft, 2022).

HOW IT IS USED

A data officer audits each critical dataset and places it on a rung. The gap between current state and the rung required by the target AI use-case becomes the data investment plan, with cost and timeline estimates per rung.

WHEN IT IS USED

Before approving any production AI investment; during annual data strategy reviews; and when a failed AI project is being retrospectively diagnosed.

BY WHOM IT IS USED

Chief Data Officers, enterprise architects, data governance leads, and Chief AI Officers negotiating data readiness requirements with business owners.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with generic data quality frameworks (DAMA DMBOK), the ladder is AI-specific: it explicitly ties each rung to the class of AI use-case it enables. Compared with the DIKW pyramid (Data–Information–Knowledge–Wisdom), which is epistemological, the ladder is operational — it prescribes technical and governance practices per rung. The framework is directly testable: a team can measure their current rung and plan their ascent with concrete engineering tasks (Microsoft, 2022).

RELATED FRAMEWORKS IN THIS COURSE

CRISP-DM Cycle · *Slibrary 3* | **Build vs. Buy** · *Slibrary 3* | **AI Project Failure Funnel** · *Slibrary 3* | **Data Value Chain** · *Slibrary 7* | **AI Readiness Assessment Radar** · *Slibrary 24*

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Prediction-Judgment Split

AI predicts · humans judge · together they decide

PROFESSIONAL DEFINITION

The Prediction-Judgment Split decomposes every decision into two economic activities: prediction (inferring the most likely state of the world) and judgment (weighing how much we value each possible outcome). AI makes prediction cheap, abundant, and accurate; judgment remains scarce and expensive. As prediction becomes a commodity, the economic value of complementary human judgment rises (Agrawal, Gans and Goldfarb, 2018).

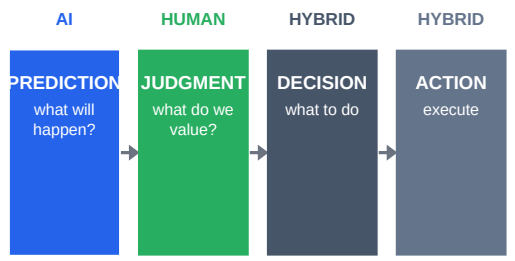
WHO DEVELOPED IT

Formalised by Ajay Agrawal, Joshua Gans, and Avi Goldfarb at the Rotman School of Management in Prediction Machines (Agrawal, Gans and Goldfarb, 2018) and extended in Power and Prediction (Agrawal, Gans and Goldfarb, 2022). The economic framing draws on decision theory (Savage, 1954) and the Schumpeterian literature on general-purpose technologies (Bresnahan and Trajtenberg, 1995).

WHY IT WAS DEVELOPED

Executives routinely debate 'will AI replace humans?' in blunt, all-or-nothing terms. The split reframes the question economically: AI reduces the cost of prediction, which raises the economic value of the judgment tasks that remain human. This shifts the strategic conversation from replacement to complementarity (Agrawal, Gans and Goldfarb, 2018).

FIGURE 1 · THE FOUR-STEP DECISION FLOW



Prediction is cheap; judgment is scarce (Agrawal, Gans & Goldfarb, 2018)

Figure 1. Every decision is a chain: AI-driven Prediction feeds human Judgment, which shapes a Decision, which is converted into Action. AI absorbs the prediction step; humans retain judgment (Agrawal, Gans and Goldfarb, 2018).

HOW IT IS USED

Leaders decompose each decision workflow into prediction and judgment components, identify where prediction dominates the cost, and invest AI in those components while protecting the judgment steps that remain human.

WHEN IT IS USED

Operating-model design, automation ROI assessment, AI literacy programmes, and reskilling decisions — anywhere the balance of AI and human effort in a workflow must be re-examined.

BY WHOM IT IS USED

COOs, Chief AI Officers, HR strategists, management consultants, and policy makers shaping workforce transition programmes.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the 'AI will automate jobs' narrative, the split is an economist's correction: it separates tasks within jobs and predicts which ones get cheaper rather than which jobs disappear. Compared with the Centaur Model (next sheet), which is organisational, the split is economic — it uses the language of marginal cost and value to guide investment. For leaders, it is the most actionable way to think about where AI belongs in a workflow (Agrawal, Gans and Goldfarb, 2022).

RELATED FRAMEWORKS IN THIS COURSE

Centaur Model · *Slibrary 4* | **Automation-Augmentation Spectrum** · *Slibrary 4* | **BCG People Investment Thesis** · *Slibrary 4* | **HITL Feedback Loop** · *Slibrary 4* | **Porter's Value Chain** · *Slibrary 2*

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Centaur Model

Human + AI together outperforms either alone

PROFESSIONAL DEFINITION

The Centaur Model describes a human-AI collaboration pattern in which a human operator works with an AI system as a single combined decision-maker, each contributing their distinctive strengths: AI handles pattern recognition at scale; the human provides strategic judgment, context, and adjustment. The combined 'centaur' — named after the mythical half-human half-horse — consistently outperforms either the human alone or the AI alone in tasks requiring both raw processing and contextual reasoning (Kasparov, 2017; Daugherty and Wilson, 2018).

WHO DEVELOPED IT

The term was popularised by former World Chess Champion Garry Kasparov after his 1997 loss to IBM's Deep Blue, when he observed that human-AI pairs were out-playing both solo humans and solo computers in freestyle chess tournaments (Kasparov, 2017). The business-strategy version was codified by Paul Daugherty and H. James Wilson in Human + Machine (Daugherty and Wilson, 2018).

WHY IT WAS DEVELOPED

Kasparov and others observed that the dichotomy 'human vs machine' misread the pattern: the strongest chess entities were human-computer teams. Extending this to business, Daugherty and Wilson argued that the real productivity frontier comes from deliberate complementarity — designing roles so that AI extends human reach rather than replacing humans in tasks they still do better.

FIGURE 1 · THE HUMAN-AI COMPLEMENTARITY

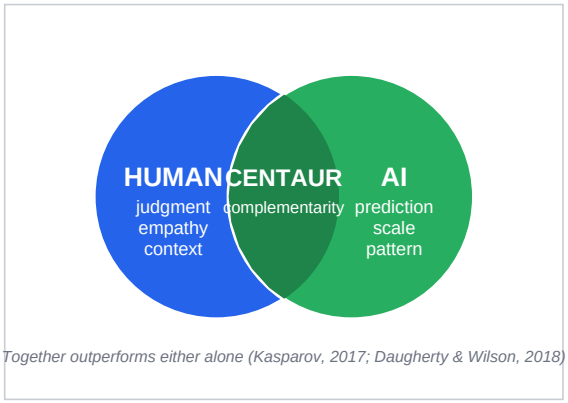

Figure 1. The Centaur Model as a Venn diagram. Human and AI circles overlap in a complementarity zone; the combined entity outperforms either alone on tasks that span both domains (Kasparov, 2017).

HOW IT IS USED

Role designers identify which elements of a job require prediction-at-scale (AI side), which require judgment or empathy (human side), and which require both (centaur zone) — then build tooling that supports centaur work rather than automating or de-skilling it.

WHEN IT IS USED

When designing AI-enabled roles, reskilling programmes, customer-service operations, medical decision support, and any knowledge-work process where neither full automation nor pure human effort is optimal.

BY WHOM IT IS USED

Organisational designers, HR leaders, learning-and-development teams, product managers of AI-enabled tools, and clinical informatics leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with pure automation narratives, the centaur reframes the goal from replacing humans to compounding their output. Compared with 'AI as tool' metaphors, it emphasises a tighter, iterative partnership in which both parties learn from each other. For firms, the model implies workflow redesign and centaur-friendly tooling rather than just dropping AI into legacy processes (Daugherty and Wilson, 2018).

RELATED FRAMEWORKS IN THIS COURSE

Prediction-Judgment Split · *Slibrary 4*
Automation-Augmentation Spectrum · *Slibrary 4*
HITL Feedback Loop · *Slibrary 4*
BCG People Investment Thesis · *Slibrary 4*
AI Maturity Staircase · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

Daugherty, P.R. and Wilson, H.J. (2018) Human + Machine: Reimagining Work in the Age of AI. Boston, MA: Harvard Business Review Press.
Kasparov, G. (2017) Deep Thinking: Where Machine Intelligence Ends and Human Creativity Begins. New York, NY: PublicAffairs.
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Automation-Augmentation Spectrum

SLIBRARY 04 · SHEET 3 / 5

From AI replaces human to AI enhances human — a five-point design choice

PROFESSIONAL DEFINITION

The Automation-Augmentation Spectrum places every AI-enabled workflow on a continuous scale between two poles: full automation, where AI acts without human involvement, and full human control, where AI advises but does not act. Between those poles lie supervised, collaborative, and advisory configurations that distribute authority differently between human and machine (Brynjolfsson and McAfee, 2017; Daugherty and Wilson, 2018).

WHO DEVELOPED IT

The two-pole framing is associated with Erik Brynjolfsson and Andrew McAfee's work at MIT (Brynjolfsson and McAfee, 2017; Brynjolfsson, 2022). The five-point taxonomy used here synthesises their spectrum with Parasuraman, Sheridan and Wickens' earlier human-automation interaction model (Parasuraman, Sheridan and Wickens, 2000) and Daugherty and Wilson's application to business workflows (Daugherty and Wilson, 2018).

WHY IT WAS DEVELOPED

Treating every AI deployment as either 'automation' or 'augmentation' obscures the most important design choice — where on the spectrum the workflow should sit. Brynjolfsson (2022) argued that defaulting to automation leads to a 'Turing trap' — optimising for imitating human labour rather than augmenting it — and foregoes most of the economic upside of AI.

FIGURE 1 · FIVE POINTS ON THE SPECTRUM

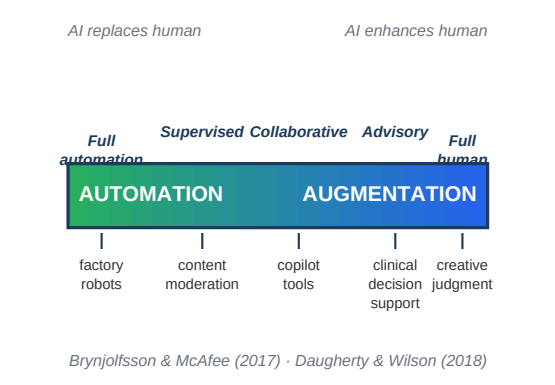

Figure 1. The automation-augmentation spectrum with five representative deployment patterns. Each point trades off speed and scale (left) against judgment and contextual nuance (right) (Brynjolfsson and McAfee, 2017).

HOW IT IS USED

Teams classify each AI use-case against the five-point scale, check whether its position reflects a considered design choice or an accident of implementation, and reposition it if the current point is sub-optimal for the task.

WHEN IT IS USED

AI use-case design reviews, operating-model redesign, risk-tolerance discussions, and when regulators or ethics committees require explicit rationale for the level of human oversight.

BY WHOM IT IS USED

AI product designers, ethics officers, risk managers, operations leaders, and regulators reviewing AI deployments in safety-critical industries.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with binary 'automate or not' debates, the spectrum exposes intermediate designs that are often the best answer — most production AI systems operate in the supervised or collaborative zones. Compared with human-factors engineering's 10-level scale (Parasuraman et al., 2000), the five-point version is more usable in executive settings while preserving the core insight: human oversight comes in gradations, not on/off states.

RELATED FRAMEWORKS IN THIS COURSE

Prediction-Judgment Split · *Slibrary 4* | **Centaur Model** · *Slibrary 4* | **HITL Feedback Loop** · *Slibrary 4*

EU AI Act Risk Pyramid · *Slibrary 1* | **NIST AI RMF** · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

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BCG People Investment Thesis

Future-built firms invest more in people, not less

PROFESSIONAL DEFINITION

The BCG People Investment Thesis is the empirical finding, across ~1,250 firms in the Build for the Future study, that AI-leading firms systematically out-spend their peers on human capital — upskilling, talent acquisition, and AI literacy — rather than substituting capital for labour. The finding inverts the common assumption that AI adoption reduces headcount investment: instead, it raises it (Boston Consulting Group, 2024; 2025).

WHO DEVELOPED IT

Developed by Boston Consulting Group's Henderson Institute and Build for the Future research team (BCG, 2024; 2025), based on a 41-dimension maturity score and detailed spend data from ~1,250 firms. Earlier supporting evidence comes from the World Economic Forum's Future of Jobs reports (WEF, 2023; 2025) and the McKinsey State of AI survey (McKinsey, 2025).

WHY IT WAS DEVELOPED

Executives and boards routinely ask 'should we cut training and hiring because AI will replace these workers?' BCG's data provides a counter-evidence response: the firms generating the most AI value are also the firms spending the most on their people. The thesis turns people investment into a measurable correlate of AI success, not a separate cost centre.

FIGURE 1 · FUTURE-BUILT VS. LAGGARD PEOPLE SPEND

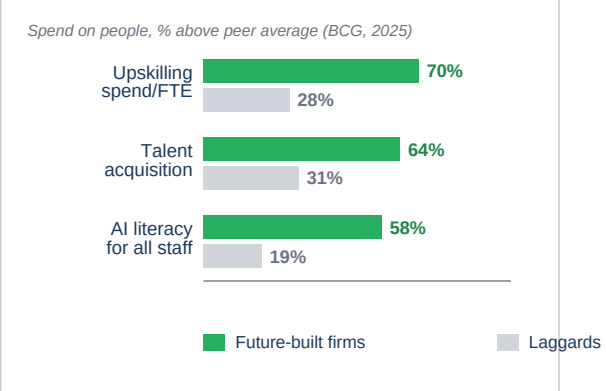

Figure 1. Share of firms above peer-average spend on three people-investment dimensions. Future-built firms (green) lead laggards (grey) by 2–3× on each dimension (Boston Consulting Group, 2025).

HOW IT IS USED

Leadership benchmarks its own people-spend ratios against the Future-built cohort, identifies the investment dimension with the largest relative gap, and builds a multi-year plan to close it as part of the broader AI programme.

WHEN IT IS USED

Annual HR strategy reviews, board-level conversations on AI workforce transition, investor communication on AI readiness, and union or works-council dialogue on AI-era employment commitments.

BY WHOM IT IS USED

CHROs, Chief AI Officers, transformation leaders, CFOs allocating transformation budgets, and boards assessing AI readiness.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-statistic headlines ('X% of firms are reskilling'), the thesis links people investment directly to measured AI value creation — making it strategically actionable. Compared with WEF forecasts on future job changes, BCG's data is backward-looking and empirical, showing which firms are already doing it. The thesis also complements Unit 1's AI Maturity Staircase: people spend differs sharply between maturity steps (BCG, 2025).

RELATED FRAMEWORKS IN THIS COURSE

Prediction-Judgment Split · *Slibrary 4* | **Centaur Model** · *Slibrary 4* | **Automation-Augmentation Spectrum** · *Slibrary 4*

AI Maturity Staircase · *Slibrary 1* | **McKinsey AI High Performers** · *Slibrary 1*

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World Economic Forum (2025) The Future of Jobs Report 2025. Geneva: World Economic Forum.

Human-in-the-Loop Feedback

AI predicts · human corrects · model retrains · accuracy rises

PROFESSIONAL DEFINITION

The Human-in-the-Loop (HITL) Feedback Loop is a continuous-learning architecture in which AI model outputs are reviewed by human operators whose corrections become labelled training data that is fed back into the model. Over successive cycles, model accuracy rises and the human correction rate falls — embedding continuous improvement directly into the operating workflow (Monarch, 2021; Holstein et al., 2019).

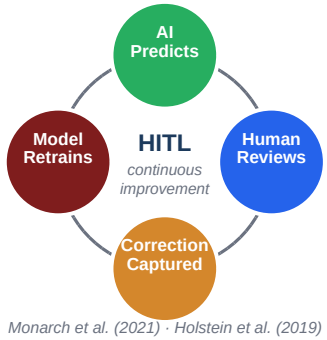
WHO DEVELOPED IT

The HITL concept has roots in active learning research (Settles, 2009) and participatory machine learning. Robert (Munro) Monarch codified the production pattern in Human-in-the-Loop Machine Learning (Monarch, 2021). Related research on human-AI interaction design was developed by Holstein et al. (2019) and Amershi et al. (2019) at Microsoft Research.

WHY IT WAS DEVELOPED

Static AI models degrade over time as the world drifts away from their training data. HITL turns every human correction in production into a new training signal, creating a self-improving system without requiring separate labelling projects. It also builds in an ongoing audit trail of human oversight, which is increasingly required for regulatory compliance under the EU AI Act.

FIGURE 1 · THE FOUR-STEP CONTINUOUS LOOP



Monarch et al. (2021) · Holstein et al. (2019)

Figure 1. The HITL cycle: AI predicts, a human reviews, corrections are captured as training data, and the model retrains. Accuracy rises and human effort falls over successive cycles (Monarch, 2021).

HOW IT IS USED

Teams instrument every AI output with a review gate, route uncertain or high-impact predictions to a human for correction, store each correction as training data, and establish a scheduled retraining cadence with gated deployment.

WHEN IT IS USED

Whenever an AI system operates in production with real consequences — content moderation, claims processing, clinical diagnostics, fraud detection — and particularly in regulated industries requiring demonstrable human oversight.

BY WHOM IT IS USED

ML engineers, MLOps teams, operations leaders, compliance officers, and domain experts providing corrections; also auditors verifying that the feedback loop is operating as designed.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with static deployment ('train once, deploy forever'), HITL turns the production system itself into a training system, compounding value per correction. Compared with purely autonomous AI, HITL provides an ongoing compliance-ready oversight mechanism — one that the EU AI Act (Slibrary 1) and NIST AI RMF both explicitly recommend for high-risk systems (EU Parliament and Council, 2024; NIST, 2023).

RELATED FRAMEWORKS IN THIS COURSE

Prediction-Judgment Split · *Slibrary 4* | **Centaur Model** · *Slibrary 4* | **Automation-Augmentation Spectrum** · *Slibrary 4*

NIST AI RMF · *Slibrary 1* | **CRISP-DM Cycle** · *Slibrary 3*

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Three ML Paradigms

Supervised · Unsupervised · Reinforcement — three data regimes, three business uses

PROFESSIONAL DEFINITION

The Three ML Paradigms is the canonical taxonomy of machine learning algorithms by the type of data signal they consume: Supervised learning uses labelled input-output pairs to learn a prediction function; Unsupervised learning discovers latent structure in unlabelled data; Reinforcement Learning learns action policies by interacting with an environment and receiving rewards. The three paradigms fit different business problems and demand different data investments (Goodfellow, Bengio and Courville, 2016).

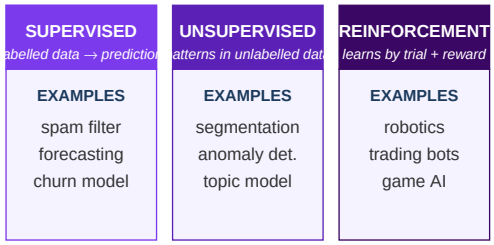
WHO DEVELOPED IT

The tripartite classification emerged from the convergence of statistical learning theory (Vapnik, 1995), unsupervised methods in cluster analysis (MacQueen, 1967), and reinforcement learning from operations research and psychology (Sutton and Barto, 1998; 2018). The modern business framing is most clearly articulated in Agrawal, Gans and Goldfarb (2018) and Russell and Norvig's Artificial Intelligence: A Modern Approach (Russell and Norvig, 2021).

WHY IT WAS DEVELOPED

Without a clean taxonomy, executives would conflate quite different AI techniques — treating recommendation systems, clustering, and trading bots as a single category. The three-paradigm classification gives business leaders a vocabulary to match AI techniques to problem types and to understand why each requires a different data investment and operating model.

FIGURE 1 · THE THREE PARADIGMS



Three ML paradigms · different data · different business uses

Figure 1. The three ML paradigms with representative business use cases. Each paradigm requires a different type of data asset and maps to different operating models (Goodfellow, Bengio and Courville, 2016; Sutton and Barto, 2018).

HOW IT IS USED

Teams classify each candidate AI use-case into one of the three paradigms, which in turn determines the data pipeline, labelling cost, feedback-loop design, and typical evaluation metrics appropriate for the project.

WHEN IT IS USED

At AI portfolio intake, when defining data labelling budgets, when scoping reinforcement learning experiments, and whenever a business problem is being translated into a technical AI specification.

BY WHOM IT IS USED

AI product managers, data scientists, Chief AI Officers, and the business sponsors they advise on technology selection.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the earlier 'statistics vs AI' division, the three paradigms reflect how modern ML practitioners actually think about problems — by data regime, not by mathematical lineage. Compared with rule-based expert systems, all three paradigms share the property that they learn from data rather than being programmed explicitly, making them adaptable to changes in the data-generating process (Russell and Norvig, 2021).

RELATED FRAMEWORKS IN THIS COURSE

Bias-Variance Tradeoff · *Slibrary 5* | **Train-Validate-Test Split** · *Slibrary 5* | **Rule-based vs ML Decision** · *Slibrary 5* | **CRISP-DM Cycle** · *Slibrary 3* | **Centaur Model** · *Slibrary 4*

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Bias-Variance Tradeoff

The U-curve at the heart of model selection

PROFESSIONAL DEFINITION

The Bias-Variance Tradeoff decomposes a model's expected prediction error into three parts: squared bias (error from wrong model assumptions), variance (error from sensitivity to training data), and irreducible noise. As model complexity rises, bias falls but variance rises — producing a U-shaped total error curve with a 'sweet spot' of optimal complexity (Hastie, Tibshirani and Friedman, 2009).

WHO DEVELOPED IT

The decomposition was formalised by Geman, Bienenstock and Doursat (Geman et al., 1992) and became a foundational concept in statistical learning theory through the work of Hastie, Tibshirani and Friedman (2009) and Vapnik (1995). The practical business framing appears in virtually every modern ML textbook and in James et al. (2013).

WHY IT WAS DEVELOPED

Before the tradeoff was formalised, practitioners tended to maximise training accuracy, producing models that fit training data perfectly but failed on new cases — overfitting. The tradeoff gave a principled reason to prefer simpler models when data is limited and a vocabulary for diagnosing failure modes ('high bias' vs 'high variance').

FIGURE 1 · THE U-CURVE OF TOTAL ERROR

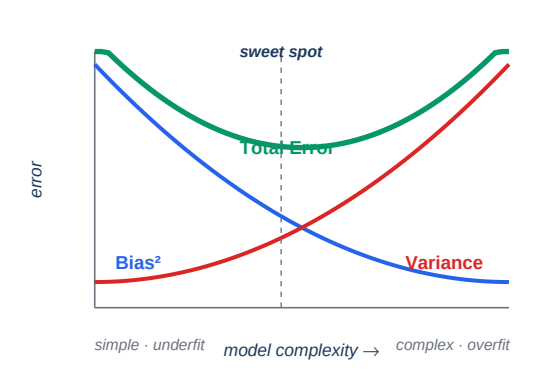

Figure 1. Bias² falls and variance rises with complexity; total error is their sum and is minimised at the sweet spot — the optimal model complexity for the given data (Hastie, Tibshirani and Friedman, 2009).

HOW IT IS USED

Data scientists diagnose model failures using the tradeoff: low training accuracy signals high bias (simplify the model or add features); high training but low test accuracy signals high variance (regularise or gather more data).

WHEN IT IS USED

At every model selection decision, during cross-validation, when debugging a deployed model that is under-performing, and when setting regularisation strength.

BY WHOM IT IS USED

Data scientists, ML engineers, and any business stakeholder reading model performance reports who must understand why 'more complex' is not always 'better'.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the earlier 'best model on training data wins' view, the tradeoff reveals why models that memorise data generalise poorly. Compared with simple cross-validation, it gives a conceptual account of why CV works — the held-out data estimates the variance component. The tradeoff is the single most cited concept in applied machine learning education (Hastie, Tibshirani and Friedman, 2009).

RELATED FRAMEWORKS IN THIS COURSE

Three ML Paradigms · *Slibrary 5* | **Train-Validate-Test Split** · *Slibrary 5* | **Accuracy-Interpretability** · *Slibrary 5* | **Cross-Validation** · *Slibrary 6* | **Model Drift Monitoring** · *Slibrary 8*

SOURCES (HARVARD REFERENCING STYLE)

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Train-Validate-Test Split

Three sets, three jobs — the discipline of honest ML evaluation

PROFESSIONAL DEFINITION

The Train-Validate-Test Split is the discipline of partitioning a dataset into three disjoint subsets: Train (to learn model parameters), Validation (to tune hyperparameters and select models), and Test (to produce an honest final performance estimate). A typical split is 70 / 15 / 15 but ratios vary with dataset size. The golden rule: never train — or tune — on test data (Hastie, Tibshirani and Friedman, 2009).

WHO DEVELOPED IT

The three-way split was formalised in the statistical learning literature (Stone, 1974; Hastie, Tibshirani and Friedman, 2009) and popularised by the machine learning community during the 1990s deep-learning resurgence (Goodfellow, Bengio and Courville, 2016). Kaggle's competition format made the three-set discipline standard practice across industry.

WHY IT WAS DEVELOPED

Without a separate test set, reported accuracy is inflated: the model has implicitly 'seen' the data through hyperparameter tuning, producing over-optimistic metrics. The three-way split creates an untouched holdout that gives executives a credible, vendor-independent performance number.

FIGURE 1 · THE THREE-WAY SPLIT

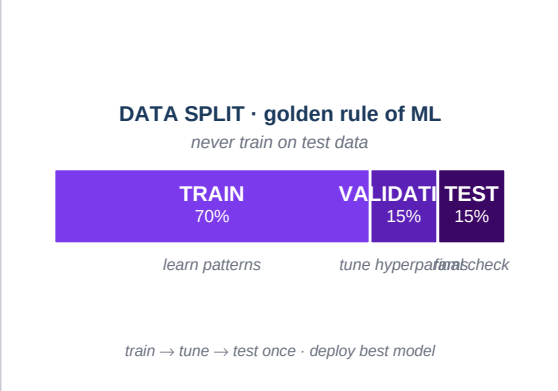

Figure 1. The Train-Validate-Test split. Train learns parameters; Validation tunes hyperparameters and selects among models; Test is touched once, at the end (Hastie, Tibshirani and Friedman, 2009).

HOW IT IS USED

Data is randomly (or stratified) partitioned at the start of the project; model selection and tuning happen on Train+Validation; Test is locked away and used only for the final evaluation report.

WHEN IT IS USED

On every supervised learning project; small-data projects may substitute k-fold cross-validation for the Validation set while retaining a held-out Test set.

BY WHOM IT IS USED

Data scientists, ML engineers, and reviewers auditing reported model metrics for honesty — including investors and regulators evaluating AI vendors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with reporting only training accuracy, the three-way split converts performance from a vendor claim into an auditable number. Compared with two-way (train/test) splits, the three-way split separates the two distinct uses of held-out data — model selection and final evaluation — that were previously conflated.

RELATED FRAMEWORKS IN THIS COURSE

Three ML Paradigms · *Slibrary 5* | **Bias-Variance Tradeoff** · *Slibrary 5* | **Cross-Validation** · *Slibrary 6* | **Confusion Matrix** · *Slibrary 8* | **CRISP-DM Cycle** · *Slibrary 3*

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Stone, M. (1974) 'Cross-Validatory Choice and Assessment of Statistical Predictions', Journal of the Royal Statistical Society B, 36(2), pp. 111–147.

Rule-Based vs. ML Decision

When NOT to use machine learning — the first question every CAIO should ask

PROFESSIONAL DEFINITION

The Rule-Based vs. ML Decision is a simple diagnostic that asks, for any automation candidate, whether the underlying pattern is clear and stable enough to be captured in explicit if-then rules or whether it requires a machine-learned model. Rules are cheaper, more auditable, and more predictable; ML is more flexible but more expensive and less transparent. The diagnostic protects against the common anti-pattern of applying ML where rules would suffice (Agrawal, Gans and Goldfarb, 2018; Breck et al., 2017).

WHO DEVELOPED IT

Articulated informally in the software engineering community (Hulten, 2018) and codified in Google's 'Rules of Machine Learning' (Zinkevich, 2017) and the 'ML Test Score' (Breck et al., 2017). The business framing — when ML is over-engineering — is developed in Agrawal, Gans and Goldfarb (2018) and Davenport (2018).

WHY IT WAS DEVELOPED

Early corporate AI programmes frequently deployed ML models for problems that could have been solved with a handful of business rules — wasting budget, obscuring auditability, and introducing unnecessary drift risk. The diagnostic forces teams to justify why ML is needed before incurring its costs.

FIGURE 1 · THE DECISION TREE

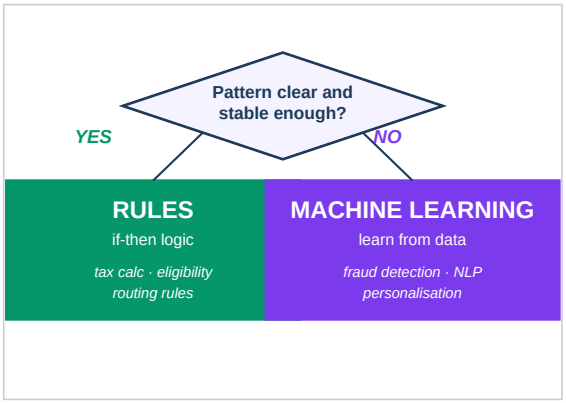

Figure 1. A simple diagnostic: if the pattern is clear and stable, rules suffice; otherwise, ML's adaptive capacity earns its cost. Start with the rules branch as the default (Zinkevich, 2017).

HOW IT IS USED

Each automation candidate is scored on pattern clarity and stability. Clear + stable patterns go to rules; ambiguous or shifting patterns go to ML. Hybrid cases use rules for the common path and ML only for the residual.

WHEN IT IS USED

At the business-case stage of every AI initiative, before data scientists are engaged and before model infrastructure is procured.

BY WHOM IT IS USED

AI product managers, solution architects, CAIOs, and the business owners commissioning automation work; also procurement teams evaluating vendor pitches.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the AI-first default that dominated 2017–2022, the diagnostic restores a sense of proportion: ML is a tool with a cost profile, not an identity. Compared with pure cost-benefit analysis, it focuses the conversation on the specific condition — pattern stability — that separates the two approaches. The diagnostic has become standard in ML Ops literature (Zinkevich, 2017; Breck et al., 2017).

RELATED FRAMEWORKS IN THIS COURSE

Three ML Paradigms · *Slibrary 5* | **Build vs. Buy** · *Slibrary 3* | **CRISP-DM Cycle** · *Slibrary 3* | **Cost-Benefit Framework** · *Slibrary 13* | **Accuracy-Interpretability** · *Slibrary 5*

SOURCES (HARVARD REFERENCING STYLE)

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Breck, E., Cai, S., Nielsen, E., Salib, M. and Sculley, D. (2017) 'The ML Test Score: A Rubric for ML Production Readiness and Technical Debt Reduction', IEEE International Conference on Big Data.

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Accuracy-Interpretability Tradeoff

The efficient frontier of model selection

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PROFESSIONAL DEFINITION

The Accuracy-Interpretability Tradeoff is the empirical observation that, in most supervised learning contexts, the most accurate models (deep neural networks, gradient-boosted trees) are also the least interpretable, while the most interpretable models (linear regression, decision trees, rule lists) are less accurate. The relationship traces an efficient frontier along which practitioners must choose (Rudin, 2019; Molnar, 2022).

WHO DEVELOPED IT

The empirical observation was formalised in the interpretable-ML literature, notably by Cynthia Rudin (Rudin, 2019) and Christoph Molnar (Molnar, 2022). Related concepts appear in Doshi-Velez and Kim (2017). The debate has sharpened as regulators — notably the EU AI Act — increasingly require explanations for high-risk AI decisions.

WHY IT WAS DEVELOPED

Firms deploying AI in regulated industries (finance, healthcare, insurance) frequently discovered that their most accurate models could not be defended to regulators or customers. The tradeoff framework gives executives a principled way to pick the right point on the frontier for each use-case — depending on stakes, regulation, and acceptable error.

FIGURE 1 · THE EFFICIENT FRONTIER

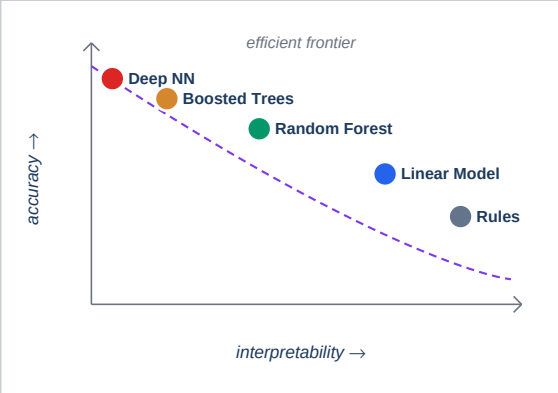

Figure 1. Model families positioned on the accuracy-interpretability frontier. High-stakes, regulated contexts pull selection toward the lower-right; discovery and personalisation tasks pull toward the upper-left (Rudin, 2019).

HOW IT IS USED

For each candidate use-case, the team asks two questions: how costly is a wrong decision, and how costly is an unexplainable decision? The answers position the use-case on the frontier and constrain which model families are admissible.

WHEN IT IS USED

Model selection reviews, regulatory submissions, customer-facing AI deployments, and any decision where audit trails matter as much as accuracy.

BY WHOM IT IS USED

Data scientists, model risk managers, compliance officers, product managers, and boards setting risk tolerance for AI deployments.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the 'maximise accuracy' default, the tradeoff explicitly prices the cost of a black-box decision. Compared with post-hoc explainability tools (SHAP, LIME), the tradeoff asks a harder question: should we start with an interpretable model? Rudin (2019) argues that in high-stakes contexts, interpretable models should be the default — an argument increasingly reflected in EU AI Act guidance.

RELATED FRAMEWORKS IN THIS COURSE

Three ML Paradigms · *Slibrary 5* | **Bias-Variance Tradeoff** · *Slibrary 5* | **Rule-based vs ML** · *Slibrary 5* | **Model Explainability** · *Slibrary 8* | **EU AI Act Risk Pyramid** · *Slibrary 1*

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MLOps Loop

Data → Model → Deploy → Monitor — the continuous machine-learning lifecycle

PROFESSIONAL DEFINITION

The MLOps Loop is the continuous-delivery lifecycle for machine-learning systems in production: data ingestion and preparation feed model training; trained models are packaged and deployed; deployed models are monitored for drift and performance; monitoring feedback triggers retraining on refreshed data. Unlike traditional software, ML systems have two inputs — code and data — and both must be versioned, tested, and monitored (Sculley et al., 2015; Google Cloud, 2021).

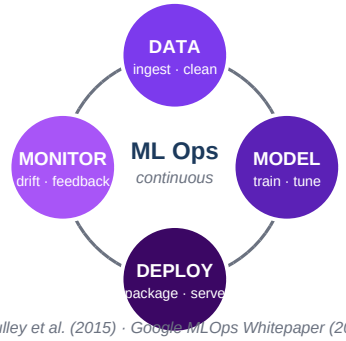
WHO DEVELOPED IT

The MLOps discipline emerged from Google's 2015 'Hidden Technical Debt in Machine Learning Systems' paper (Sculley et al., 2015) and was formalised in Google Cloud's MLOps Whitepaper (Google Cloud, 2021), Microsoft's MLOps Maturity Model (Microsoft, 2022), and AWS's ML Well-Architected Framework (AWS, 2023). It builds on DevOps (Humble and Farley, 2010) with ML-specific extensions.

WHY IT WAS DEVELOPED

ML projects succeed in notebooks but fail in production because data drifts, dependencies rot, and performance silently degrades. MLOps brings the discipline of DevOps — automated testing, continuous delivery, observability — to ML systems, converting one-off experiments into reliable production software.

FIGURE 1 · THE FOUR-PHASE CONTINUOUS LOOP



Sculley et al. (2015) · Google MLOps Whitepaper (2021)

Figure 1. The MLOps loop. Data feeds Model, Model is Deployed, Deployment is Monitored, and Monitoring signals return to Data for the next cycle (Sculley et al., 2015; Google Cloud, 2021).

HOW IT IS USED

Teams instrument each of the four stages with automated pipelines, CI/CD for both code and data, monitoring dashboards, and retraining triggers — reducing mean time to detection and mean time to recovery for model issues.

WHEN IT IS USED

For any ML system that runs in production for more than six months, any model whose inputs or targets drift over time, and any regulated ML deployment requiring demonstrable lifecycle controls.

BY WHOM IT IS USED

ML engineers, data engineers, DevOps and SRE teams, platform engineering teams, and compliance officers verifying the lifecycle controls of regulated AI systems.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional software DevOps, MLOps explicitly treats data as a versioned, testable artifact alongside code. Compared with notebook-centric data science, MLOps forces the discipline of reproducibility and observability from day one. The loop is increasingly required by regulators for high-risk AI systems under the EU AI Act.

RELATED FRAMEWORKS IN THIS COURSE

CRISP-DM Cycle · *Slibrary 3* | **Model Selection Tree** · *Slibrary 6* | **Ensemble Methods** · *Slibrary 6* | **HITL Feedback Loop** · *Slibrary 4* | **Model Drift Monitoring** · *Slibrary 8*

SOURCES (HARVARD REFERENCING STYLE)

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Cross-Validation

k-fold CV — the honest-evaluation discipline for small datasets

PROFESSIONAL DEFINITION

Cross-Validation (CV) is a resampling technique for estimating model performance when the dataset is too small to afford a full train-validate-test split. In k-fold CV the data is partitioned into k equal folds; the model is trained k times, each time using a different fold as the validation set and the rest for training. The k validation scores are averaged, producing a more stable estimate than any single split (Stone, 1974; Kohavi, 1995).

WHO DEVELOPED IT

Formalised in the statistical literature by Stone (1974) and further developed by Kohavi (1995), Efron and Tibshirani (1993), and Hastie, Tibshirani and Friedman (2009). It is the default validation strategy in most ML tooling including scikit-learn, Caret, and MLR.

WHY IT WAS DEVELOPED

A single random train-test split can produce highly variable performance estimates for small datasets — good luck on one split, bad luck on another. CV averages across k splits, producing a lower-variance estimate and squeezing more statistical power out of limited data.

FIGURE 1 · 5-FOLD CROSS-VALIDATION

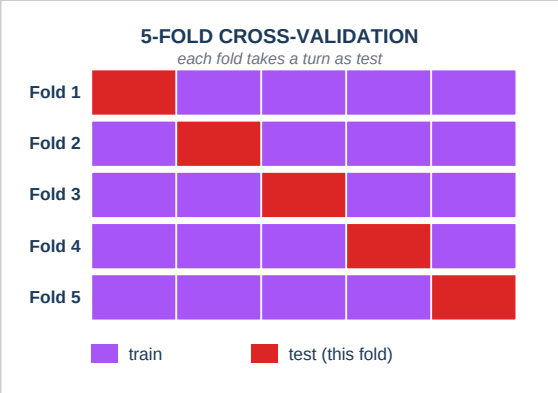

Figure 1. 5-fold cross-validation. Each row is one iteration; the red cell is the test fold for that iteration, and the purple cells are training folds. The mean of the five test scores is the CV estimate (Kohavi, 1995).

HOW IT IS USED

For each model/hyperparameter setting, run k-fold CV on the training+validation data, average the k scores, and select the setting with the best mean performance (optionally adjusted for standard error). Final test set is still held out.

WHEN IT IS USED

Whenever data is limited (< 10,000 rows) or the signal is weak; also for time-series data (where time-aware CV variants like TimeSeriesSplit apply); less needed for very large datasets where a simple train-validate-test split suffices.

BY WHOM IT IS USED

Data scientists, statisticians, and researchers publishing ML results — CV is the minimum evidence bar for peer-reviewed ML claims.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with a single train-test split, CV extracts more information from limited data and produces lower-variance estimates. Compared with repeated random splits, k-fold guarantees every data point appears in the test set exactly once. Stratified and time-aware variants extend the method to class-imbalanced and temporal data.

RELATED FRAMEWORKS IN THIS COURSE

Bias-Variance Tradeoff · *Slibrary 5* | **Train-Validate-Test Split** · *Slibrary 5* | **MLOps Loop** · *Slibrary 6* | **Ensemble Methods** · *Slibrary 6* | **Confusion Matrix** · *Slibrary 8*

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Model Selection Decision Tree

A quick tree from problem type to first-choice algorithm

PROFESSIONAL DEFINITION

The Model Selection Decision Tree is a pragmatic flowchart that takes a supervised learning problem (classification vs regression, tabular vs unstructured, interpretability required or not) and maps it to a first-choice algorithm family. It formalises the experienced practitioner's heuristic of starting with a simple baseline appropriate to the data shape, then adding complexity only when baseline performance is insufficient (Kuhn and Johnson, 2013).

WHO DEVELOPED IT

No single author — the tree synthesises practitioner heuristics documented in Kuhn and Johnson (2013), scikit-learn's algorithm cheat sheet (Pedregosa et al., 2011), and Andrew Ng's Machine Learning Yearning (Ng, 2018). It is the shortest path from problem definition to productive experimentation.

WHY IT WAS DEVELOPED

Newcomers to ML are tempted to reach for deep neural networks for every problem — wasting compute, introducing unneeded complexity, and often losing to simpler baselines. The tree enforces the principle 'start simple, earn complexity', which repeatedly yields better outcomes in tabular business settings.

FIGURE 1 · STARTING-POINT SELECTION

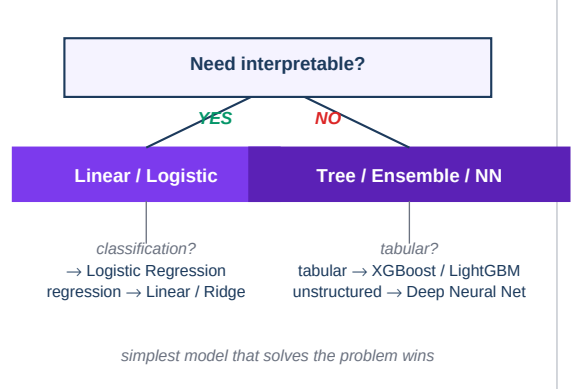

Figure 1. A compact decision tree: if interpretability is required, start with linear or logistic models; otherwise, tree ensembles for tabular data, deep networks for unstructured data (Kuhn and Johnson, 2013; Ng, 2018).

HOW IT IS USED

The team walks the tree for their problem, implements the first-choice algorithm as a baseline, and measures it against business requirements. Only if the baseline is inadequate does the team escalate to a more complex algorithm family.

WHEN IT IS USED

At the start of every supervised learning project, during solution architecture reviews, and when onboarding new team members to a standardised starting protocol.

BY WHOM IT IS USED

Data scientists, ML engineers, AI solution architects, and product managers reviewing proposed technical approaches.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the 'always deep learning' default, the tree forces explicit reasoning about data shape and interpretability. Compared with no starting protocol, it accelerates teams through the baseline-establishment phase of every project. For MBA-level leaders, the tree demystifies algorithm selection without requiring a statistics PhD.

RELATED FRAMEWORKS IN THIS COURSE

Three ML Paradigms · *Slibrary 5* | **Bias-Variance Tradeoff** · *Slibrary 5* | **MLOps Loop** · *Slibrary 6* | **Accuracy-Interpretability** · *Slibrary 5* | **Cross-Validation** · *Slibrary 6*

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Ensemble Methods

Bagging · Boosting · Stacking — three ways to turn many weak models into one strong model

PROFESSIONAL DEFINITION

Ensemble Methods combine multiple models — 'weak learners' — into a single stronger predictor by aggregating their outputs. The three major families differ in how they train the learners: Bagging trains them in parallel on different data samples and averages (e.g. Random Forest); Boosting trains them sequentially, each correcting the previous one's errors (e.g. XGBoost); Stacking trains a meta-model on the outputs of heterogeneous base models (Dietterich, 2000; Breiman, 2001).

WHO DEVELOPED IT

Ensemble theory was formalised by Dietterich (2000) and Schapire (1990). Bagging was introduced by Breiman (1996); Random Forests by Breiman (2001); AdaBoost by Freund and Schapire (1997); gradient boosting by Friedman (2001); XGBoost by Chen and Guestrin (2016); and stacking by Wolpert (1992).

WHY IT WAS DEVELOPED

No single model family dominates all problems (the 'no free lunch' theorem), and individual models have different error patterns. Ensemble methods exploit these differences — errors that are uncorrelated across models average out, while correct predictions reinforce each other. The result is consistently higher accuracy with controllable variance.

FIGURE 1 · THREE ENSEMBLE FAMILIES

Figure 1. Bagging trains in parallel (Random Forest); Boosting trains sequentially (XGBoost, LightGBM); Stacking combines models via a meta-model (common on Kaggle winning entries) (Dietterich, 2000).

HOW IT IS USED

Teams typically start with a strong ensemble baseline — Random Forest or gradient-boosted trees — for tabular data, tune hyperparameters via cross-validation, and move to deeper architectures only if tabular ensembles are insufficient.

WHEN IT IS USED

On virtually every tabular supervised learning problem; Random Forest and gradient boosting are the de-facto defaults in industrial data-science practice (Kaggle State of Data Science, 2023).

BY WHOM IT IS USED

Data scientists, ML engineers, quantitative analysts, and any team facing a tabular prediction problem where accuracy matters.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-model approaches, ensembles consistently win in practice — Kaggle competitions are almost always won by ensembles of ensembles. Compared with deep learning on tabular data, gradient-boosted trees remain competitive or superior at a fraction of the compute cost (Grinsztajn, Oyallon and Varoquaux, 2022). For tabular business problems, ensembles are the correct first move.

RELATED FRAMEWORKS IN THIS COURSE

Model Selection Tree · *Slibrary 6* | **Bias-Variance Tradeoff** · *Slibrary 5* | **Cross-Validation** · *Slibrary 6* | **MLOps Loop** · *Slibrary 6* | **Deployment Patterns** · *Slibrary 6*

SOURCES (HARVARD REFERENCING STYLE)

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ML Deployment Patterns

Batch · Real-time API · Streaming · Edge — four ways to serve a model

PROFESSIONAL DEFINITION

ML Deployment Patterns is the classification of how trained models are made available to downstream consumers, organised by latency profile. Batch scoring runs predictions on schedule (hours to days); Real-time API returns predictions on request in under a second; Streaming consumes continuous event streams; Edge deploys models onto devices without server round-trips. Each pattern has distinct cost, complexity, and capability profiles (Huyen, 2022; Kreuzberger et al., 2023).

WHO DEVELOPED IT

The four-pattern taxonomy is synthesised from practitioner literature — Chip Huyen's Designing Machine Learning Systems (Huyen, 2022), Kreuzberger, Kühl and Hirschl's MLOps architectural review (Kreuzberger et al., 2023), and the major cloud vendors' ML reference architectures (AWS, 2023; Google Cloud, 2022).

WHY IT WAS DEVELOPED

ML teams frequently default to the deployment pattern they are familiar with — typically a REST API — even when batch or streaming would be far cheaper or more accurate. Making the pattern choice explicit forces teams to reason about actual latency requirements before building infrastructure.

FIGURE 1 · FOUR DEPLOYMENT PATTERNS

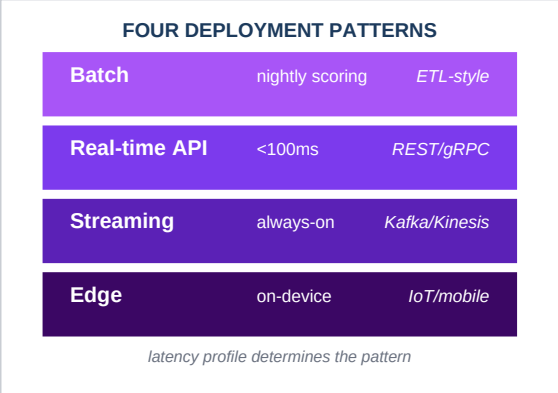

Figure 1. The four common ML deployment patterns, ordered from highest (batch, hours+) to lowest (edge, microseconds) latency tolerance. Latency requirements drive pattern selection (Huyen, 2022).

HOW IT IS USED

For each model, the team identifies the actual latency requirement of the business use-case, maps it to one of the four patterns, and sizes the infrastructure accordingly — avoiding over-engineering batch jobs as APIs or under-engineering streaming requirements as batch.

WHEN IT IS USED

At solution architecture reviews, during procurement of ML platform services, and when refactoring an existing deployment to reduce cost or meet new SLAs.

BY WHOM IT IS USED

ML architects, platform engineers, SREs supporting ML systems, FinOps teams tracking ML infrastructure cost, and solution architects from cloud vendors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with pattern-free deployment, the four-pattern taxonomy prevents misspending on over-provisioned infrastructure. Compared with legacy batch-only or API-only defaults, it acknowledges the full spectrum of modern deployment contexts — including on-device ML for mobile and IoT. The taxonomy is increasingly reflected in cloud pricing models, which charge very differently per pattern.

RELATED FRAMEWORKS IN THIS COURSE

MLOps Loop · *Slibrary 6* | **Model Drift Monitoring** · *Slibrary 8* | **TCO Framework** · *Slibrary 14* | **Build vs. Buy** · *Slibrary 3* | **HITL Feedback Loop** · *Slibrary 4*

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Data Value Chain

Collect · Store · Process · Analyse · Act — Porter for the data economy

PROFESSIONAL DEFINITION

The Data Value Chain is a five-stage model — Collect, Store, Process, Analyse, Act — that traces how raw data is transformed into business action and, ultimately, into value. It is Porter's value chain (Slibrary 2) applied to the data economy: value is created at every stage, activities are linked, and competitive advantage arises from performing one or more stages better or more cheaply than rivals (Curry, 2016; Miller and Mork, 2013).

WHO DEVELOPED IT

Introduced as the 'Big Data Value Chain' by Miller and Mork (2013) and refined by the European Big Data Value Association (Curry, 2016). It adapts Porter's firm-level value chain (Porter, 1985) to the specific sequence of activities that convert raw data into decisions — particularly relevant to platform businesses and data-driven firms.

WHY IT WAS DEVELOPED

Firms accumulating data often stored it without a clear view of where in the value chain it was being underutilised. The value chain exposes the specific stage that is blocking monetisation — whether it is collection completeness, storage cost, processing latency, analytical capability, or the handoff to action (Curry, 2016).

FIGURE 1 · THE FIVE-STAGE DATA CHAIN

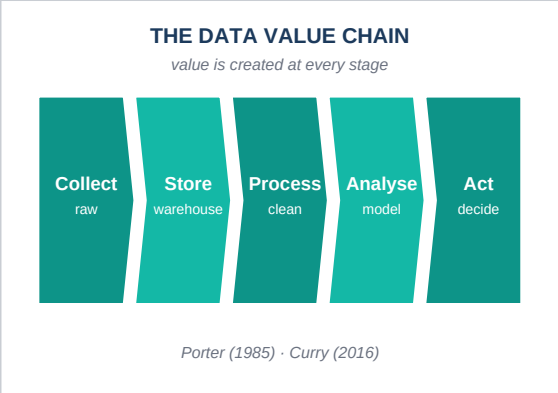

Figure 1. The data value chain as a chevron flow. Each stage adds value and can itself be a source of advantage; the chain ends in the action that produces business outcomes (Curry, 2016; Miller and Mork, 2013).

HOW IT IS USED

Firms map their data activities onto the chain, benchmark each stage against competitors and industry leaders, and identify the weakest link as the focus of next-year investment — whether instrumentation, warehousing, ML platform, or decision automation.

WHEN IT IS USED

Data strategy reviews, AI investment prioritisation, data platform architecture reviews, and M&A due diligence where the target firm's data assets must be assessed stage by stage.

BY WHOM IT IS USED

Chief Data Officers, Chief Digital Officers, data platform teams, strategy consultants, and investors assessing data-driven company valuations.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the generic DIKW pyramid (Data → Information → Knowledge → Wisdom), the value chain is operational: each stage corresponds to concrete engineering and organisational capabilities. Compared with Porter's physical value chain, it highlights the unique economics of data — infinite reproducibility, increasing returns, and compounding network effects (Curry, 2016).

RELATED FRAMEWORKS IN THIS COURSE

Porter's Value Chain · *Slibrary 2* | **Data Moat** · *Slibrary 7* | **Feature Engineering** · *Slibrary 7* | **CRISP-DM Cycle** · *Slibrary 3* | **Data Readiness Ladder** · *Slibrary 3*

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Data Moat Flywheel

Users → Data → Better Model → Better Product → More Users

PROFESSIONAL DEFINITION

The Data Moat Flywheel is a self-reinforcing cycle in which more users generate more data, which trains a better model, which produces a better product, which attracts more users. Each turn of the flywheel widens the performance gap between the incumbent and any entrant, creating a defensible competitive moat — as long as the loop is engineered and measurable (Moore, 2018; Hagiu and Wright, 2023).

WHO DEVELOPED IT

The flywheel metaphor was developed by Jim Collins in Good to Great (Collins, 2001) and applied to data businesses by Andrei Hagiu and Julian Wright (Hagiu and Wright, 2023). The 'data moat' language was popularised by Geoffrey Moore (Moore, 2018) and by venture capitalists evaluating SaaS and platform businesses (Bessemer, 2022).

WHY IT WAS DEVELOPED

Venture investors and corporate strategists needed a framework to distinguish genuine data advantages — where more data measurably improves the product — from stored-data vanity claims. The flywheel specifies the four concrete transitions that must all be operational for a true data moat to exist (Hagiu and Wright, 2023).

FIGURE 1 · THE SELF-REINFORCING LOOP

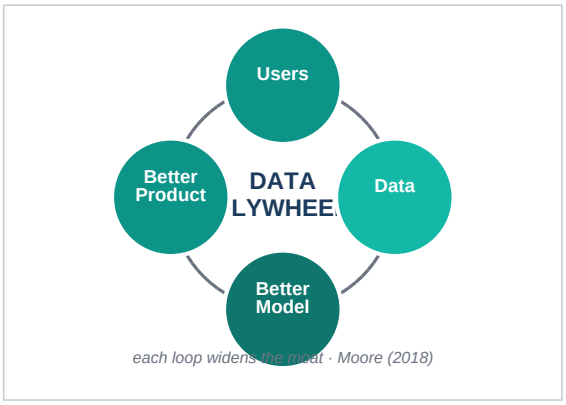

Figure 1. The data flywheel. Each transition — users to data, data to model, model to product, product to more users — must work for the loop to compound; a broken link flatlines the advantage (Moore, 2018; Hagiu and Wright, 2023).

HOW IT IS USED

Strategists map their product to the four transitions, identify which are functioning and which are broken, and invest in the weakest link. The health of the flywheel is measured in each transition's rate (data per user, model accuracy gain per data increment, etc.).

WHEN IT IS USED

Product strategy reviews, venture due diligence, competitive positioning analysis, and any time a firm claims a 'data advantage' and must prove it is real rather than asserted.

BY WHOM IT IS USED

Product leaders, Chief Data Officers, venture investors, corporate development teams, and business-school strategy courses.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with static 'data is the new oil' narratives, the flywheel specifies the mechanism — the dynamic loop — through which data creates value. Compared with the generic RBV (Slibrary 2), it identifies the specific resource structure (a feedback loop) that produces VRIN advantage in digital markets. Hagiu and Wright (2023) show empirically that most claimed data moats fail on at least one of the four transitions.

RELATED FRAMEWORKS IN THIS COURSE

Data Value Chain · *Slibrary 7* | **Resource-Based View** · *Slibrary 2* | **VRIO Framework** · *Slibrary 2* | **Data Network Effects** · *Slibrary 7* | **First vs Third Party Data** · *Slibrary 7*

SOURCES (HARVARD REFERENCING STYLE)

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First-, Second-, and Third-Party Data

SLIBRARY 07 · SHEET 3 / 5

Three sourcing regimes · three competitive profiles

PROFESSIONAL DEFINITION

The First / Second / Third Party taxonomy classifies data by its source relative to the using firm. First-party data is generated by the firm's own customer interactions; second-party is a partner's first-party data shared via contract; third-party is data aggregated and sold by external providers. As privacy regulation tightens and third-party cookies are deprecated, first-party data has become the dominant source of durable AI advantage (Marr, 2019; Boston Consulting Group, 2024).

WHO DEVELOPED IT

The taxonomy emerged from digital-marketing practice in the 2010s (IAB, 2015; Winterberry Group, 2018) and was extended to AI training data by BCG (2024) and McKinsey (2024). It has particular strategic significance after GDPR (2018) and Apple's ATT framework (2021) restricted third-party data flows.

WHY IT WAS DEVELOPED

Firms spent large sums on third-party data without realising that competitors were buying the same feeds — producing no differentiation. The taxonomy made the distinction explicit and clarified why first-party data, being uniquely owned, is the only source that can produce durable competitive advantage.

FIGURE 1 · THREE DATA SOURCING REGIMES

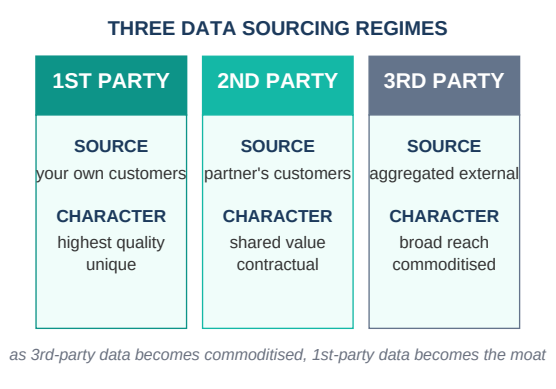

Figure 1. The three data parties and their characteristics. First-party is unique and high-quality but limited to owned channels; third-party is broad but commoditised; second-party sits in between (Marr, 2019; BCG, 2024).

HOW IT IS USED

Firms audit their AI training data by party, calculate the share of their model performance that relies on each party, and invest in first-party collection (product telemetry, customer consent flows) where third-party feeds are at risk.

WHEN IT IS USED

Data strategy reviews, privacy compliance audits, marketing-technology stack decisions, and any AI programme that depends on external data feeds.

BY WHOM IT IS USED

Chief Marketing Officers, Chief Data Officers, privacy officers, CIOs, and investors assessing the defensibility of a firm's AI capability.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with a simple 'data is data' view, the party taxonomy reveals the strategic heterogeneity of data assets. Compared with pure cost accounting, it prices the fact that some data sources (first-party) produce moats while others (third-party) merely buy parity. In the post-GDPR era, the taxonomy is increasingly the starting point for any data strategy discussion.

RELATED FRAMEWORKS IN THIS COURSE

Data Value Chain · *Slibrary 7* | **Data Moat Flywheel** · *Slibrary 7* | **Data Network Effects** · *Slibrary 7* | **VRIO Framework** · *Slibrary 2* | **EU AI Act Risk Pyramid** · *Slibrary 1*

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Feature Engineering Pipeline

From raw columns to model-ready features — the craft at the heart of applied ML

PROFESSIONAL DEFINITION

The Feature Engineering Pipeline is the sequence of transformations that converts raw data columns into the features a machine-learning model actually consumes: cleaning and imputing missing values, encoding categoricals, constructing new features from combinations, and selecting the most predictive subset. For tabular data, the quality of this pipeline typically matters more than the choice of algorithm (Kuhn and Johnson, 2019; Zheng and Casari, 2018).

WHO DEVELOPED IT

Feature engineering has no single author — it is the accumulated craft of applied statistics. Modern codification appears in Kuhn and Johnson's Feature Engineering and Selection (Kuhn and Johnson, 2019) and Zheng and Casari's Feature Engineering for Machine Learning (Zheng and Casari, 2018). Automated feature engineering (featuretools, autofeat, H2O.ai) has since extended the discipline.

WHY IT WAS DEVELOPED

A well-engineered feature can turn a mediocre algorithm into a strong predictor, while poor features can make even the best algorithm useless. Codifying the pipeline as a sequence forces teams to approach feature work systematically rather than improvising, and enables reproducible experimentation (Kuhn and Johnson, 2019).

FIGURE 1 · FIVE-STEP PIPELINE

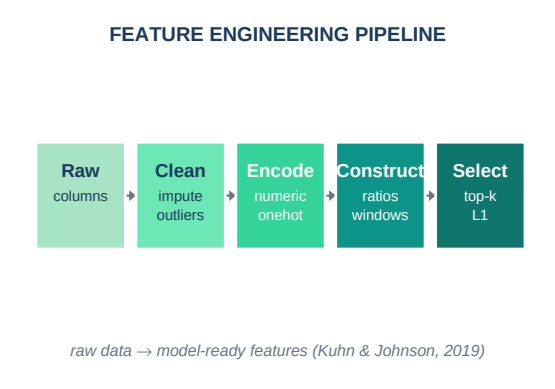

Figure 1. The feature engineering pipeline. Each step is a defined transformation with inputs and outputs; the ordered sequence is reproducible and auditable (Kuhn and Johnson, 2019).

HOW IT IS USED

Data scientists build the pipeline as a reusable set of transformation objects (e.g. scikit-learn Pipeline) that can be fit on training data and re-applied identically at inference time, preventing train-serve skew.

WHEN IT IS USED

On every tabular ML project; especially critical for regulated models where the pipeline must be auditable, and for production models where train-serve consistency is a correctness requirement.

BY WHOM IT IS USED

Data scientists, feature engineers (a role emerging in larger firms), ML engineers, and auditors verifying that production features match training features.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with ad-hoc data munging, the pipeline discipline makes feature work reproducible, testable, and versionable. Compared with end-to-end deep learning, explicit features remain dominant for tabular business problems (Grinsztajn, Oyallon and Varoquaux, 2022). Automated feature engineering is narrowing but not yet eliminating the value of human craft at this stage.

RELATED FRAMEWORKS IN THIS COURSE

Data Value Chain · *Slibrary 7* | **Data Readiness Ladder** · *Slibrary 3* | **CRISP-DM Cycle** · *Slibrary 3* | **Model Selection Tree** · *Slibrary 6* | **MLOps Loop** · *Slibrary 6*

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Data Network Effects

When more users make the product better, and better products attract more users

PROFESSIONAL DEFINITION

Data Network Effects are a specific form of indirect network effect in which each additional user's data improves the product for all users — typically by training a better model. Unlike classical network effects (where users value other users directly), data network effects operate through an intermediating algorithm. The result is super-linear value growth with scale, creating durable incumbency advantages (Hagiu and Wright, 2023; Evans and Schmalensee, 2016).

WHO DEVELOPED IT

The concept emerged from platform economics (Rochet and Tirole, 2003; Parker, Van Alstyne and Choudary, 2016) and was specialised to data-driven products by Andrei Hagiu and Julian Wright (Hagiu and Wright, 2023). It has particular relevance to search engines, recommender systems, translation services, and any AI product where model quality scales with training data volume.

WHY IT WAS DEVELOPED

Strategists needed to distinguish firms that genuinely benefited from data scale from firms whose scale merely produced cost reductions. Data network effects specify the precise mechanism — user-to-model-to-product feedback — that creates a defensible moat in digital businesses (Hagiu and Wright, 2023).

FIGURE 1 · SUPER-LINEAR VALUE GROWTH

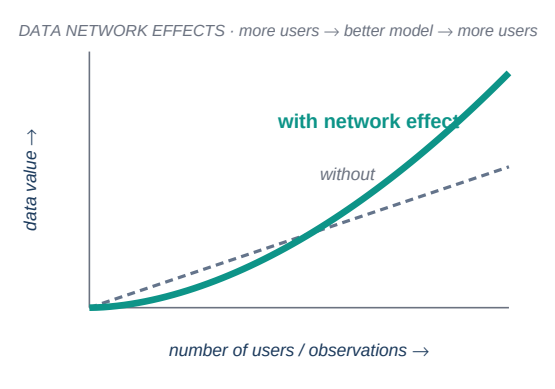

Figure 1. With data network effects, value grows super-linearly in users (teal); without them, value grows linearly or sublinearly (grey). The gap is the moat (Hagiu and Wright, 2023).

HOW IT IS USED

Strategists test whether a product has real data network effects by measuring whether an additional user's data measurably improves the product for other users — a high bar that many claimed 'data moats' fail to meet.

WHEN IT IS USED

Product strategy design, venture investment due diligence, board-level discussions of platform strategy, and regulatory analysis of market dominance in AI-driven markets.

BY WHOM IT IS USED

Product strategists, venture capitalists, platform-business executives, competition regulators, and antitrust economists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with classical network effects, data network effects can operate even when users do not interact with each other directly — a user's data benefits the shared model, which benefits future users. Compared with scale economies, data network effects compound over time rather than only reducing cost. The distinction is central to how regulators assess dominance in platform and AI markets (European Commission, 2022).

RELATED FRAMEWORKS IN THIS COURSE

Data Moat Flywheel · *Slibrary 7* | **Data Value Chain** · *Slibrary 7* | **First vs Third Party Data** · *Slibrary 7* | **Disruption S-Curves** · *Slibrary 2* | **Porter's Five Forces** · *Slibrary 2*

SOURCES (HARVARD REFERENCING STYLE)

Evans, D.S. and Schmalensee, R. (2016) Matchmakers: The New Economics of Multisided Platforms. Boston, MA: Harvard Business Review Press.
Hagiu, A. and Wright, J. (2023) 'When Data Creates Competitive Advantage', Harvard Business Review, January–February 2023, pp. 94–101.
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Rochet, J.-C. and Tirole, J. (2003) 'Platform Competition in Two-Sided Markets', Journal of the European Economic Association, 1(4), pp. 990–1029.

Confusion Matrix

Four outcomes · four cost profiles — the foundation of classifier evaluation

PROFESSIONAL DEFINITION

The Confusion Matrix is a 2×2 table that cross-tabulates a classifier's predictions against the true labels, producing four outcomes: True Positive, False Positive, True Negative, False Negative. Every classification metric — accuracy, precision, recall, F1, specificity — is a ratio of these four counts. Understanding the matrix is the prerequisite for choosing the right metric for the business problem (Fawcett, 2006; Kuhn and Johnson, 2013).

WHO DEVELOPED IT

The matrix has roots in classical statistics (Yerushalmy, 1947) and became standard in the machine-learning literature through Fawcett's widely-cited ROC review (Fawcett, 2006). Its business framing — mapping the four cells to asymmetric cost profiles — is taught in Kuhn and Johnson (2013) and Provost and Fawcett (2013).

WHY IT WAS DEVELOPED

Reporting a single accuracy number hides the failure pattern: a spam filter that blocks every email is '99% accurate' if spam is 1% of traffic, but catastrophic. The confusion matrix forces explicit reasoning about which errors the business can tolerate — false positives or false negatives — before a single metric is chosen.

FIGURE 1 · THE FOUR OUTCOMES

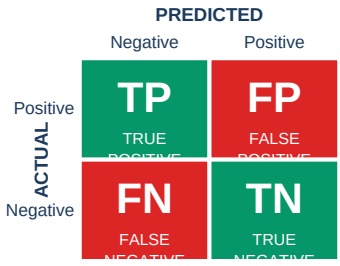

Figure 1. The confusion matrix. Green cells are correct predictions; red cells are errors. Precision = TP/(TP+FP); Recall = TP/(TP+FN); F1 combines both (Fawcett, 2006).

HOW IT IS USED

Before selecting a metric, the team writes the business consequence for each of the four cells — the cost of a False Positive vs a False Negative — then picks the metric that best captures the actual business tradeoff.

WHEN IT IS USED

At metric-selection stage of every classification project; during model monitoring when reported accuracy looks suspicious; and during external audits of regulated AI systems where cell-by-cell costs must be made explicit.

BY WHOM IT IS USED

Data scientists, product managers, business sponsors, model risk officers, and regulators reviewing AI classification systems.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-number accuracy, the confusion matrix exposes the full trade-off surface. Compared with classical statistical tests, it is interpretable to non-statisticians — the four cells map directly to business consequences. The matrix is the universal starting point for any serious conversation about classifier performance.

RELATED FRAMEWORKS IN THIS COURSE

ROC / AUC Curve · *Slibrary 8* | **Model Explainability** · *Slibrary 8* | **Bias-Variance Tradeoff** · *Slibrary 5* | **Cross-Validation** · *Slibrary 6* | **Fairness Metrics** · *Slibrary 8*

SOURCES (HARVARD REFERENCING STYLE)

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Provost, F. and Fawcett, T. (2013) Data Science for Business. Sebastopol, CA: O'Reilly Media.
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ROC / AUC Curve

The threshold-independent measure of classifier quality

PROFESSIONAL DEFINITION

The Receiver Operating Characteristic (ROC) curve plots a classifier's True Positive Rate against its False Positive Rate at every possible decision threshold. The Area Under the Curve (AUC) summarises the curve in a single number between 0.5 (random) and 1.0 (perfect). AUC has the intuitive interpretation of the probability that the classifier ranks a random positive example above a random negative example (Fawcett, 2006; Hanley and McNeil, 1982).

WHO DEVELOPED IT

ROC curves originate from signal detection theory during World War II (Peterson, Birdsall and Fox, 1954) and were adapted to diagnostic medicine by Hanley and McNeil (1982). Fawcett (2006) introduced the technique to the ML community; it is now the default performance plot for binary classifiers in scikit-learn and most ML tooling.

WHY IT WAS DEVELOPED

Classifier performance depends on the threshold chosen to convert probabilities into decisions. ROC separates the classifier's intrinsic ranking ability from the threshold choice — letting teams compare classifiers without committing to a threshold up front (Fawcett, 2006).

FIGURE 1 · ROC AND AUC

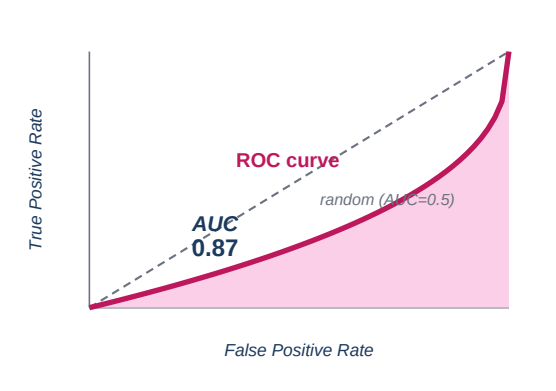

Figure 1. The ROC curve (pink) and the shaded AUC. A curve bending toward the top-left corner indicates better discrimination; the diagonal is random guessing (Fawcett, 2006).

HOW IT IS USED

Teams plot ROC for each candidate classifier, compare AUCs, and choose the best model; the operating threshold is then set based on the business cost of FPs vs FNs, independent of model selection.

WHEN IT IS USED

Standard first-line evaluation for binary classification; less informative when classes are highly imbalanced (use Precision-Recall curves instead) or when calibration matters more than ranking.

BY WHOM IT IS USED

Data scientists, model risk managers, and peer reviewers; medical-AI validation teams where ROC has been the standard since the 1980s.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with accuracy, AUC is threshold-independent. Compared with precision or recall at a single threshold, ROC shows the full trade-off surface. Compared with F1, AUC is more stable under class imbalance. The curve is the single most cited classifier-evaluation plot in scientific ML papers.

RELATED FRAMEWORKS IN THIS COURSE

Confusion Matrix · *Slibrary 8* | **Cross-Validation** · *Slibrary 6* | **Fairness Metrics** · *Slibrary 8* | **Model Explainability** · *Slibrary 8* | **Model Drift Monitoring** · *Slibrary 8*

SOURCES (HARVARD REFERENCING STYLE)

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Model Explainability (SHAP / LIME)

SLIBRARY 08 · SHEET 3 / 5

Turning black-box predictions into audit-ready explanations

PROFESSIONAL DEFINITION

Model Explainability is the set of techniques that decompose an opaque model's prediction into human-interpretable feature contributions. The dominant methods — SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) — attribute a score to each feature showing how much it pushed the prediction up or down for a given instance or dataset (Lundberg and Lee, 2017; Ribeiro, Singh and Guestrin, 2016).

WHO DEVELOPED IT

LIME was introduced by Ribeiro, Singh and Guestrin (2016); SHAP was developed by Lundberg and Lee (2017), drawing on Shapley values from cooperative game theory (Shapley, 1953). Christoph Molnar's Interpretable Machine Learning (Molnar, 2022) is the standard reference. Both techniques are implemented in open-source libraries (shap, lime) widely used in industry.

WHY IT WAS DEVELOPED

Regulators, customers, and internal auditors increasingly require explanations for AI decisions — why was this loan denied? why was this claim flagged? Black-box models cannot answer those questions directly. SHAP and LIME provide a standardised, mathematically-grounded way to generate such explanations post-hoc.

FIGURE 1 · FEATURE IMPORTANCE (SHAP)

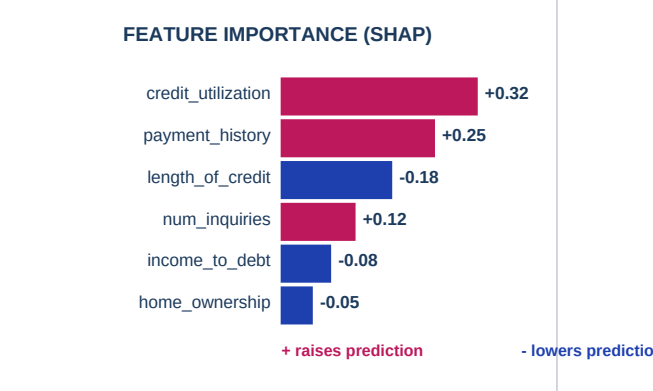

Figure 1. A SHAP feature-importance plot for a single prediction. Bars show each feature's signed contribution; + pushes the prediction toward the positive class, - pushes toward negative (Lundberg and Lee, 2017).

HOW IT IS USED

Teams compute SHAP values on a representative sample of predictions, surface the top features driving each decision in the user interface, and log the feature contributions for audit. Global summaries (feature importance across all predictions) drive model debugging.

WHEN IT IS USED

For any model used in customer-facing decisions, regulated industries, or where stakeholder trust is a deployment blocker. Also essential for debugging unexpected model behaviour.

BY WHOM IT IS USED

Data scientists, model risk officers, compliance teams, customer-service leaders needing explainable outputs, and auditors validating model behaviour.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with opaque predictions, SHAP/LIME convert black boxes into audit-able decisions. Compared with interpretable models (Slibrary 5, Sheet 5), they allow the firm to retain high-accuracy complex models while satisfying explanation requirements. Critics (Rudin, 2019) caution that post-hoc explanations can themselves be misleading — SHAP/LIME are an addition to, not a substitute for, interpretable-by-design models in high-stakes contexts.

RELATED FRAMEWORKS IN THIS COURSE

Confusion Matrix · *Slibrary 8* | **Accuracy-Interpretability** · *Slibrary 5* | **Fairness Metrics** · *Slibrary 8* | **NIST AI RMF** · *Slibrary 1* | **EU AI Act Risk Pyramid** · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

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Model Drift Monitoring

Detecting when a production model's assumptions have expired

SLIBRARY 08 · SHEET 4 / 5

PROFESSIONAL DEFINITION

Model Drift Monitoring is the continuous measurement of a deployed model's performance and input distributions to detect when they have drifted from training-time conditions. Three types of drift are distinguished: concept drift (the relationship between inputs and outputs changes), data drift (the input distribution shifts), and prior probability shift (class proportions change). Undetected drift causes silent accuracy loss, which in financial or medical contexts can be catastrophic (Gama et al., 2014; Huyen, 2022).

WHO DEVELOPED IT

The statistical theory of drift detection developed in the data-stream-mining literature (Gama et al., 2014; Žliobait■, 2010). Practical industrial frameworks are documented in Huyen's Designing Machine Learning Systems (Huyen, 2022), Microsoft's MLOps Maturity Model (Microsoft, 2022), and specialised tooling such as Evidently AI and Fiddler.

WHY IT WAS DEVELOPED

Static models silently degrade as the world changes — customer behaviour shifts, new product categories appear, regulatory regimes change. Without monitoring, the first signal of drift is usually a business problem (complaints, losses). Monitoring catches drift before it causes harm and triggers retraining.

FIGURE 1 · ACCURACY DRIFT AND RETRAINING

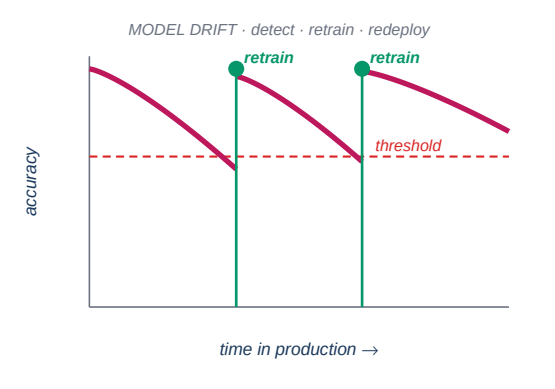

Figure 1. A model's accuracy degrades over time (pink); when it approaches the threshold (red dashed), a retrain (green) restores performance. The cycle repeats (Gama et al., 2014).

HOW IT IS USED

Teams instrument both output metrics (accuracy, AUC) and input distribution statistics (PSI, KL divergence) on live traffic, set alert thresholds, and trigger retraining or rollback when thresholds are breached.

WHEN IT IS USED

For every production ML system — especially those in fast-changing domains (marketing, fraud, finance, demand forecasting); regulatory compliance increasingly requires documented drift monitoring.

BY WHOM IT IS USED

MLOps engineers, SREs supporting ML, model risk officers, internal auditors, and operations leaders responsible for ML system uptime.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with 'deploy once, forget' static systems, drift monitoring converts ML into an observable discipline with measurable SLAs. Compared with periodic retraining on a schedule, drift-triggered retraining is more responsive and cost-efficient. The discipline is a prerequisite for the HITL Feedback Loop (Slibrary 4) and for EU AI Act post-market monitoring obligations.

RELATED FRAMEWORKS IN THIS COURSE

MLOps Loop · *Slibrary 6* | **HITL Feedback Loop** · *Slibrary 4* | **ROC / AUC Curve** · *Slibrary 8* | **Model Explainability** · *Slibrary 8* | **NIST AI RMF** · *Slibrary 1*

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Fairness Metrics

Four incompatible definitions · one design choice

PROFESSIONAL DEFINITION

Fairness Metrics quantify whether an ML model treats different demographic groups equitably. Four are most commonly used: Demographic Parity (same prediction rate across groups), Equal Opportunity (same TPR), Equalised Odds (same TPR and FPR), and Predictive Parity (same precision). Kleinberg, Mullainathan and Raghavan (2017) proved an impossibility theorem: except in trivial cases, the metrics are mathematically incompatible — satisfying all simultaneously is impossible (Kleinberg et al., 2017; Barocas, Hardt and Narayanan, 2019).

WHO DEVELOPED IT

Formal definitions emerged from the fair-ML research community: Demographic Parity (Dwork et al., 2012); Equalised Odds (Hardt, Price and Srebro, 2016); Predictive Parity (Chouldechova, 2017). The impossibility theorem is due to Kleinberg, Mullainathan and Raghavan (2017) and Chouldechova (2017) independently. Barocas, Hardt and Narayanan's Fairness and Machine Learning (2019) is the standard textbook.

WHY IT WAS DEVELOPED

High-profile cases — the COMPAS recidivism algorithm, biased hiring systems, discriminatory credit scoring — revealed that ML systems were encoding and amplifying historical inequities. The metrics give teams a vocabulary to measure fairness rather than argue about it qualitatively, and to make the inevitable trade-offs explicit (Barocas, Hardt and Narayanan, 2019).

FIGURE 1 · FOUR METRIC FAMILIES

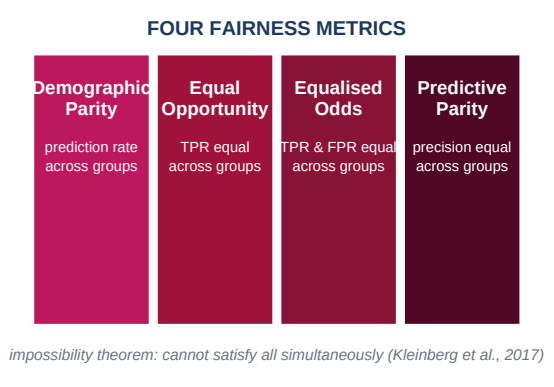

Figure 1. Four common fairness metrics. The impossibility theorem (Kleinberg et al., 2017) means that firms must choose which fairness criterion is most relevant to their context — satisfying all is mathematically impossible in realistic settings.

HOW IT IS USED

Teams compute the relevant metric across protected attribute groups, set acceptable gap thresholds, and mitigate violations through reweighting, post-processing, or in-model constraints; the choice of metric is a policy decision, not a technical one.

WHEN IT IS USED

Any AI system with impact on individuals — hiring, lending, healthcare, policing, insurance, education; increasingly required by regulators under EU AI Act and emerging national AI laws.

BY WHOM IT IS USED

AI ethics committees, model risk officers, legal and compliance teams, data scientists, and civil-society organisations auditing deployed AI.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with informal 'bias audits', the formal metrics allow reproducible, defensible measurements. Compared with the assumption that fairness has a single definition, the impossibility theorem forces an explicit policy choice about which inequality is most important to correct. The theorem has reshaped how serious firms approach AI governance — the question shifted from 'is the model fair?' to 'fair according to which metric, and why that one?'.

RELATED FRAMEWORKS IN THIS COURSE

Confusion Matrix · *Slibrary 8* | **Model Explainability** · *Slibrary 8* | **NIST AI RMF** · *Slibrary 1* | **EU AI Act Risk Pyramid** · *Slibrary 1* | **Responsible AI Scorecard** · *Slibrary 16*

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Marketing Evolution 1.0 → 6.0

Six generations of marketing thought from Product to Metaverse + AI

PROFESSIONAL DEFINITION

The Marketing 1.0–6.0 evolution traces six successive paradigms in the Kotler school of thought: Marketing 1.0 (product-centric), 2.0 (customer-centric), 3.0 (human-centric and values-driven), 4.0 (digital convergence), 5.0 (human-technology collaboration), and 6.0 (immersive metaverse experiences powered by generative AI). Each generation adds a layer rather than replacing the previous; firms operate across several simultaneously (Kotler, Kartajaya and Setiawan, 2017; 2021; 2024).

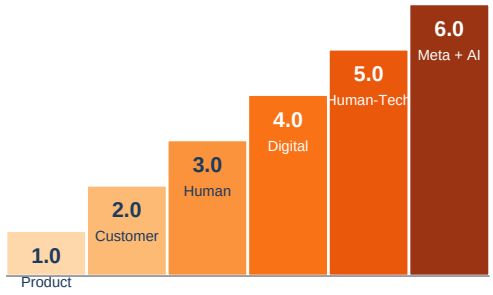
WHO DEVELOPED IT

Introduced and progressively extended by Philip Kotler and co-authors Hermawan Kartajaya and Iwan Setiawan across four books: Marketing 3.0 (2010), 4.0 (2017), 5.0 (2021), and 6.0 (2024). Kotler is the founding author of modern marketing theory (Kotler, 1967). Each new generation was named in response to observable market shifts rather than predicting them.

WHY IT WAS DEVELOPED

Marketing practice was outpacing marketing theory — mobile, social, AI, and metaverse technologies reshaped customer relationships faster than academic frameworks could track. The generational labelling gave practitioners a shared vocabulary to discuss where they were and where they were heading.

FIGURE 1 · SIX GENERATIONS



Kotler et al. (1967–2024) · six generations of marketing thought

Figure 1. The six marketing generations as an ascending staircase. Each adds a capability layer to the one below; Marketing 6.0 integrates all previous generations into immersive, AI-augmented, metaverse-ready experiences (Kotler, Kartajaya and Setiawan, 2024).

HOW IT IS USED

Firms diagnose which generation their current marketing operating model reflects, then build a roadmap toward the capabilities of the next generation — typically without discarding the foundations of the previous generations.

WHEN IT IS USED

Marketing strategy reviews, transformation programme design, executive education, and when justifying investment in newer channels (AR/VR, GenAI content systems) against legacy spend.

BY WHOM IT IS USED

CMOs, marketing transformation leaders, brand strategists, and agency partners advising on next-horizon capabilities.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the single-channel taxonomies (the 4 Ps, the 7 Ps), the generational view captures how marketing has fundamentally shifted unit of analysis over time — from products to customers to human beings to digitally-augmented communities. Compared with pure technology hype cycles, Kotler's framing keeps the human at the centre: Marketing 6.0 is not about replacing humans with AI but about using AI to serve richer human experiences (Kotler et al., 2024).

RELATED FRAMEWORKS IN THIS COURSE

Personalisation Hierarchy · *Slibrary 10* | **Customer Journey Map** · *Slibrary 11* | **Trust Equation** · *Slibrary 12*

AI Maturity Staircase · *Slibrary 1* | **Perez Techno-Economic S-Curve** · *Slibrary 1*

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The Metaverse Marketing Stack

SLIBRARY 09 · SHEET 2 / 5

Physical · Extended Reality · Virtual — three layers of customer immersion

PROFESSIONAL DEFINITION

The Metaverse Marketing Stack describes three vertically integrated layers in which customer experiences are increasingly staged: the Physical layer (stores, events, products), the Extended Reality (XR) layer (AR overlays, VR environments), and the fully Virtual layer (game-like immersive worlds, avatars, digital goods). Marketing 6.0 programmes run across all three layers simultaneously, moving customers between them as the journey demands (Kotler, Kartajaya and Setiawan, 2024; Ball, 2022).

WHO DEVELOPED IT

The tripartite stack was articulated by Kotler, Kartajaya and Setiawan in Marketing 6.0 (2024), synthesising Matthew Ball's influential metaverse framing (Ball, 2022) and Meta's commercial metaverse strategy papers (Meta, 2022). It complements earlier O2O (online-to-offline) concepts by adding the XR middle layer.

WHY IT WAS DEVELOPED

Marketers needed a unifying taxonomy for an explosion of new channels — AR filters, VR showrooms, Roblox activations, NFT drops — that did not fit neatly into physical/digital binaries. The stack gives a coherent architecture for thinking about customer immersion as a spectrum rather than a switch.

FIGURE 1 · THREE LAYERS OF IMMERSION

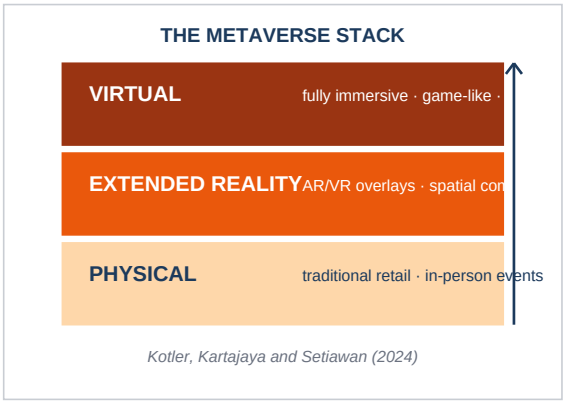

Figure 1. The metaverse stack. Moving upward adds immersion, richness, and data depth; moving downward adds tangibility and mass reach. Marketing 6.0 programmes span all three (Kotler et al., 2024; Ball, 2022).

HOW IT IS USED

Brand teams map each campaign touchpoint to one of the three layers, identify white-space in their current portfolio, and pilot XR or virtual experiences where the target segment has genuine engagement — avoiding novelty-for-novelty's-sake.

WHEN IT IS USED

Campaign architecture reviews, experiential-marketing investment decisions, and brand refresh programmes targeting Gen Z and Gen Alpha consumers where virtual-native experiences are baseline expectations.

BY WHOM IT IS USED

Brand strategists, experiential marketing leads, creative agencies, CMOs, and digital transformation teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with O2O (online-to-offline) framings, the stack admits the XR middle layer — where most consumer activity actually sits in 2025 (filters, AR try-on, hybrid events). Compared with the hype of 'full metaverse' discourse, the stack grounds investment decisions in observable engagement by layer and segment (Kotler et al., 2024).

RELATED FRAMEWORKS IN THIS COURSE

Marketing Evolution 1.0–6.0 · *Slibrary 9*
GenAI in Marketing · *Slibrary 9*
Physical-Digital Convergence · *Slibrary 9*
Omnichannel Orchestration · *Slibrary 11*
Customer Journey Map · *Slibrary 11*

SOURCES (HARVARD REFERENCING STYLE)

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Physical-Digital Convergence

Phygital experiences · seamless flow across channels

PROFESSIONAL DEFINITION

Physical-Digital Convergence — also called 'phygital' — is the merging of online and offline customer experiences into a single continuous journey in which the customer moves freely between channels without friction. Data flows in both directions: in-store behaviour informs digital personalisation, and digital history informs in-store service. The aim is for the customer to experience the brand, not the channels (Kotler, Kartajaya and Setiawan, 2021).

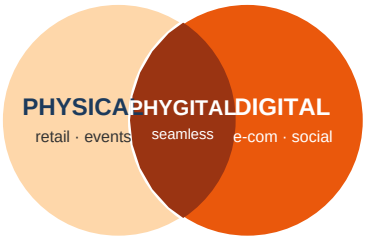
WHO DEVELOPED IT

The 'phygital' term was coined by the Australian agency Momentum (2007) and systematised in the Marketing 4.0 framework (Kotler, Kartajaya and Setiawan, 2017). Walmart, Nike, and Starbucks are the most-cited exemplars of integrated phygital strategies (Harvard Business School cases, 2022–2024).

WHY IT WAS DEVELOPED

Siloed channel organisations produced fragmented customer experiences — data from one channel was invisible to another, and promotions contradicted each other. Phygital convergence forces cross-channel data unification and operational integration, allowing coherent customer treatment regardless of touchpoint.

FIGURE 1 · THE PHYGITAL OVERLAP



convergence of physical and digital customer experience

Figure 1. The convergence zone. Physical and digital are not alternatives but complements; the overlap is where modern customer experience happens (Kotler, Kartajaya and Setiawan, 2021).

HOW IT IS USED

Firms audit each customer touchpoint for its position on the physical-digital spectrum, identify handoff failures (e.g. customer switches devices mid-journey and loses context), and invest in unified CDPs (customer data platforms) that restore continuity.

WHEN IT IS USED

Retail transformation, omnichannel programme design, loyalty programme redesign, and customer-experience benchmarking against digital-native and physical-native competitors.

BY WHOM IT IS USED

CMOs, Chief Digital Officers, customer experience officers, retail operations leaders, and CDP implementation teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with multichannel marketing (parallel channels), phygital emphasises continuity — the same customer state persists across channels. Compared with the metaverse discourse, phygital is immediately actionable and does not require customers to adopt new devices or platforms. It is the most mature of the Marketing 6.0 capability layers.

RELATED FRAMEWORKS IN THIS COURSE

Marketing Evolution 1.0–6.0 · *Slibrary 9*
Metaverse Marketing Stack · *Slibrary 9*
Omnichannel Orchestration · *Slibrary 11*
Personalisation Hierarchy · *Slibrary 10*
Customer Journey Map · *Slibrary 11*

SOURCES (HARVARD REFERENCING STYLE)

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Generative AI in Marketing

Four use-case families · content · ads · email · conversational

PROFESSIONAL DEFINITION

Generative AI in Marketing is the application of large language models and generative-image models to four core marketing functions: Content creation (copy, images, video), Advertising (creative variants and A/B at scale), Email (hyper-personalised send), and Conversational interfaces (chatbots and virtual agents). McKinsey estimates that ~40% of marketing labour hours are now directly augmentable by generative AI (McKinsey, 2023; 2024).

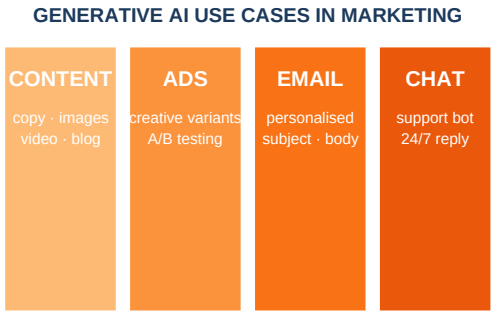
WHO DEVELOPED IT

The four-use-case taxonomy is synthesised from McKinsey's GenAI marketing studies (McKinsey, 2023; 2024), Gartner's martech hype cycle (Gartner, 2024), and HubSpot's State of Marketing report (HubSpot, 2024). Academic grounding comes from Eloundou et al.'s task-level labour market analysis (Eloundou et al., 2023) and Davenport and Mittal (2022).

WHY IT WAS DEVELOPED

Marketing leaders faced a proliferation of GenAI tools without a structured way to prioritise investment. The four-use-case taxonomy organises the space into recognisable function families, each with measurable business metrics and each with different governance considerations.

FIGURE 1 · FOUR GENAI USE CASES



McKinsey (2024) · 40% of marketing time now touched by GenAI

Figure 1. The four core GenAI marketing use cases. Each has distinct cost profiles, risk surfaces, and required governance controls (McKinsey, 2024; HubSpot, 2024).

HOW IT IS USED

Marketing teams audit each of the four use cases for current adoption maturity and gap vs peers, then sequence investments based on expected value and risk — typically starting with content creation before moving to high-stakes conversational applications.

WHEN IT IS USED

Annual marketing budget allocation, martech RFPs, creative agency briefs, and governance reviews of AI-generated content for brand-safety compliance.

BY WHOM IT IS USED

CMOs, marketing operations leaders, creative directors, and compliance officers auditing generative content for regulatory and brand risk.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with treating GenAI as a single 'tool', the four-use-case view surfaces that each family has distinct data, governance, and performance characteristics. Compared with open-ended experimentation, it imposes a portfolio logic that makes budgets and outcomes comparable. The framework is now standard in major consultancies' marketing-AI playbooks (McKinsey, 2024; BCG, 2024).

RELATED FRAMEWORKS IN THIS COURSE

Marketing Evolution 1.0–6.0 · *Slibrary 9* | **Metaverse Marketing Stack** · *Slibrary 9* | **Personalisation Hierarchy** · *Slibrary 10*

Chatbot Design · *Slibrary 12* | **Copyright Framework** · *Slibrary 17*

SOURCES (HARVARD REFERENCING STYLE)

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Human-Centric Marketing

Human at the centre · experience around them · tech at the edge

PROFESSIONAL DEFINITION

Human-Centric Marketing is the principle that even in the AI-saturated Marketing 6.0 era, the organising unit of marketing strategy must remain the human being — their aspirations, anxieties, and values — rather than the technology deployed to reach them. Technology is the outermost ring; experience wraps around the human at the centre (Kotler, Kartajaya and Setiawan, 2021; 2024).

WHO DEVELOPED IT

Introduced as a corrective to tech-driven Marketing 4.0 by Kotler, Kartajaya and Setiawan in Marketing 3.0 (2010), reaffirmed in Marketing 5.0 (2021), and restated as Marketing 6.0's organising premise (Kotler et al., 2024). Academic grounding draws on humanistic management (Pirson, 2017) and stakeholder theory (Freeman, 1984).

WHY IT WAS DEVELOPED

As generative AI enabled ever-more-powerful automated content and personalisation, the risk grew that marketing would optimise for algorithmic engagement metrics at the expense of genuine human benefit.
Human-centric marketing reasserts that customer outcomes — not engagement per se — remain the ultimate goal.

FIGURE 1 · CONCENTRIC PRIORITIES

human at the centre · tech serves experience (Kotler, 2024)

Figure 1. Concentric rings: Human at the centre, Experience around them, Technology at the edge. Technology serves experience; experience serves the human (Kotler, Kartajaya and Setiawan, 2024).

HOW IT IS USED

Marketing teams apply the rings as a review filter: does this campaign's technology choice serve a better customer experience? Does that experience serve a genuine human need? Campaigns failing the filter are redesigned or halted.

WHEN IT IS USED

Campaign approvals, martech procurement reviews, AI ethics assessments of marketing programmes, and leadership alignment exercises during transformation programmes.

BY WHOM IT IS USED

CMOs, brand stewards, marketing ethics boards, and agency account teams translating technology capabilities into brand-appropriate activations.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with technology-first strategies (Marketing 4.0's digital imperative), human-centric framing restores a normative anchor. Compared with ESG-driven marketing, human-centric marketing is narrower and more operational — it applies at the campaign level, not only at corporate strategy. The framework is increasingly reflected in industry bodies' AI-marketing guidelines (WFA, 2024).

RELATED FRAMEWORKS IN THIS COURSE

Marketing Evolution 1.0–6.0 · *Slibrary 9* | **Trust Equation** · *Slibrary 12* | **GenAI in Marketing** · *Slibrary 9* | **Fairness Metrics** · *Slibrary 8* | **Responsible AI Scorecard** · *Slibrary 16*

SOURCES (HARVARD REFERENCING STYLE)

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Personalisation Hierarchy

SLIBRARY 10 · SHEET 1 / 5

Mass → Segment → Micro-segment → 1-to-1 — four levels of targeting fidelity

PROFESSIONAL DEFINITION

The Personalisation Hierarchy is a four-tier ladder describing how precisely a marketing message or experience can be tailored: Mass (one-size-fits-all), Segment (a handful of broad personas), Micro-segment (hundreds of fine-grained clusters), and 1-to-1 (true individual-level personalisation). Each ascending tier demands more data, better models, and more operational sophistication, and delivers higher conversion at higher cost (Kumar, 2018; McKinsey, 2021).

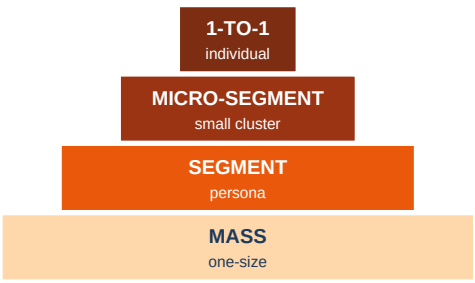
WHO DEVELOPED IT

The hierarchy derives from direct-marketing segmentation theory (Wedel and Kamakura, 2000) and was made AI-specific by V. Kumar (Kumar, 2018) and practitioner research at McKinsey (McKinsey, 2021; 2024). Adobe's and Salesforce's personalisation maturity models are operational variants of the same scheme.

WHY IT WAS DEVELOPED

Marketing leaders were told to 'personalise' without clarity on what level was achievable or desirable for their context. The hierarchy maps each tier to its data prerequisites and expected lift, letting executives decide on a target tier rather than chasing a vague aspiration.

FIGURE 1 · THE FOUR-TIER LADDER



from mass → one-to-one · Kumar (2018) · McKinsey (2021)

Figure 1. The four tiers of personalisation. McKinsey (2021) finds that firms moving from Segment to Micro-segment typically see 10–15% revenue lift; 1-to-1 adds a further 5–10% but at sharply higher operational cost.

HOW IT IS USED

Marketing teams audit current personalisation by channel and customer journey stage, plot the distribution across the four tiers, and select a target tier for each touchpoint based on cost-value analysis.

WHEN IT IS USED

Personalisation strategy reviews, martech RFPs, customer-experience roadmaps, and board-level conversations on AI marketing value.

BY WHOM IT IS USED

CMOs, personalisation leads, customer analytics teams, and martech architects.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with binary 'personalised vs not' framing, the hierarchy acknowledges gradations and matches investment to achievable gains. Compared with pure technology hype, it links each tier to concrete data and operational prerequisites. Kumar (2018) shows empirically that most firms claim 1-to-1 but operate at Micro-segment — a gap the hierarchy makes visible.

RELATED FRAMEWORKS IN THIS COURSE

Recommendation Engine Types · *Slibrary 10* | **Real-time vs Batch** · *Slibrary 10* | **Privacy-Personalisation** · *Slibrary 10* | **Marketing Evolution 1.0–6.0** · *Slibrary 9* | **Customer Journey Map** · *Slibrary 11*

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Recommendation Engine Types

SLIBRARY 10 · SHEET 2 / 5

Content-based · Collaborative filtering · Hybrid — three engines, three data regimes

PROFESSIONAL DEFINITION

Recommendation Engines are the AI systems that personalise product, content, or experience surfacing. Three families dominate: Content-based (recommends items similar to those the user already liked, using item attributes); Collaborative Filtering (recommends items liked by users similar to the current user); and Hybrid (combines both signals). Each family has distinct data prerequisites, cold-start behaviour, and failure modes (Ricci, Rokach and Shapira, 2022).

WHO DEVELOPED IT

Content-based recommendation is rooted in information retrieval (Salton and McGill, 1983). Collaborative filtering was pioneered by Goldberg et al. (1992) and systematised by Koren, Bell and Volinsky's Netflix Prize work (Koren, Bell and Volinsky, 2009). Hybrid methods are comprehensively reviewed in Ricci, Rokach and Shapira (2022).

WHY IT WAS DEVELOPED

Different business contexts have different data profiles: a new streaming service has rich item metadata but no usage history (content-based wins); a large e-commerce site has vast behavioural data but too many items for manual metadata (collaborative wins); mature platforms combine both (hybrid).

FIGURE 1 · THREE ENGINE FAMILIES

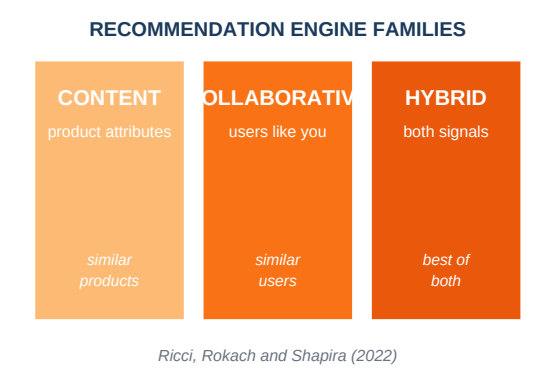

Figure 1. The three recommendation engine families. Amazon, Netflix, and Spotify all operate hybrid systems in production; pure content or pure collaborative systems are now rare in mature digital businesses (Ricci et al., 2022).

HOW IT IS USED

Product teams pick the starting approach based on available data — content-based for cold-start, collaborative after sufficient user history, hybrid as the long-term architecture — then evolve the stack as data accumulates.

WHEN IT IS USED

Recommendation-system design reviews, new-product launch planning (cold-start handling), and vendor selection for personalisation platforms.

BY WHOM IT IS USED

Product managers, ML engineers, data scientists, and personalisation platform teams at retailers, streaming services, and content platforms.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with rule-based merchandising, algorithmic recommendation scales personally to each user. Compared with pure collaborative or content approaches, the hybrid architecture captures both data modalities and is robust to missing signals. Modern deep-learning recommenders (e.g. two-tower models, BERT4Rec) are architectural extensions of the hybrid paradigm.

RELATED FRAMEWORKS IN THIS COURSE

Personalisation Hierarchy · *Slibrary 10* | **Real-time vs Batch** · *Slibrary 10* | **Next-Best-Action Engine** · *Slibrary 11* | **Data Value Chain** · *Slibrary 7* | **Three ML Paradigms** · *Slibrary 5*

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Real-time vs Batch Personalisation

SLIBRARY 10 · SHEET 3 / 5

Two latency regimes · two cost profiles · two operating models

PROFESSIONAL DEFINITION

Real-time vs Batch Personalisation is the architectural decision of whether to compute personalised content on demand as each user interaction happens (real-time) or to pre-compute recommendations on a schedule and cache them (batch). Real-time is responsive to fresh context but expensive per request; batch is cheap and deterministic but cannot react to the current session. The right choice depends on the customer's decision window (Huyen, 2022).

WHO DEVELOPED IT

The distinction is standard in ML systems literature (Huyen, 2022; Kleppmann, 2017). It operationalises what marketers call 'freshness vs reach' and maps directly to the deployment patterns in Slibrary 6. Major martech vendors (Adobe, Salesforce, Bloomreach) explicitly architect around this distinction.

WHY IT WAS DEVELOPED

Firms that default to real-time frequently over-spend on infrastructure where batch would suffice; firms that default to batch miss high-conversion real-time opportunities (cart abandonment, in-session intent). Making the choice explicit aligns architecture to customer-journey stage.

FIGURE 1 · SIDE-BY-SIDE COMPARISON

	BATCH	REAL-TIME
latency	hours	ms
cost	low	high
freshness	yesterday	now
use	email	web UI

pattern choice driven by customer's decision window

Figure 1. Batch vs real-time trade-offs across four dimensions. Batch dominates email and push notifications; real-time dominates on-site recommendations and search relevance (Huyen, 2022).

HOW IT IS USED

Teams classify each personalisation touchpoint by its customer decision window. Windows measured in hours or days go to batch; windows measured in seconds or minutes go to real-time. Hybrid architectures use batch for base scores and real-time for session reranking.

WHEN IT IS USED

Martech architecture reviews, cost-optimisation programmes for personalisation platforms, and new-channel launches where the appropriate latency model is not yet decided.

BY WHOM IT IS USED

Martech architects, platform engineers, product managers, FinOps teams monitoring personalisation platform costs.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with a single-latency default, the explicit choice matches infrastructure cost to business value. Compared with pure batch legacy systems, real-time enables in-session personalisation that can lift conversion materially. Modern systems increasingly use hybrid approaches — pre-computed candidates reranked in real time — as the cost-efficient sweet spot.

RELATED FRAMEWORKS IN THIS COURSE

Deployment Patterns · *Slibrary 6* | **Personalisation Hierarchy** · *Slibrary 10* | **Recommendation Engine Types** · *Slibrary 10* | **Data Value Chain** · *Slibrary 7* | **TCO Framework** · *Slibrary 14*

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Privacy-Personalisation Tradeoff

The creepy cliff · where personalisation stops helping and starts harming

PROFESSIONAL DEFINITION

The Privacy-Personalisation Tradeoff describes the non-linear relationship between how much personal data a firm uses for personalisation and the customer response: up to a point, more personalisation increases value; past an inflection point (the 'creepy cliff'), it triggers backlash, churn, and regulatory risk. The shape of the curve depends on context, trust, and consent salience (Martin and Murphy, 2017; Aguirre et al., 2015).

WHO DEVELOPED IT

The concept was formalised in marketing ethics literature by Martin and Murphy (2017) and Aguirre et al. (2015); the 'creepy' framing was popularised by John Tehranian and later mainstream journalism.
Post-GDPR and post-ATT, the tradeoff has become operational rather than theoretical (European Commission, 2022).

WHY IT WAS DEVELOPED

Firms optimising for engagement metrics sometimes crossed the creepy cliff without realising it — triggering regulatory investigations, press crises, and customer churn. The tradeoff gives a principled framework for when to collect and use data and when to stop.

FIGURE 1 · THE CREEPY CLIFF

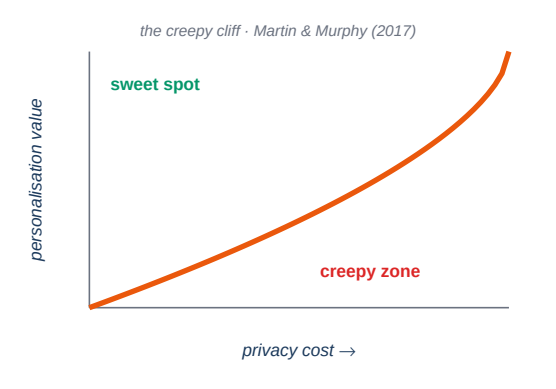

Figure 1. The tradeoff. Personalisation value rises with privacy cost up to a point, then plunges sharply. The sweet spot sits left of the inflection (Martin and Murphy, 2017).

HOW IT IS USED

Personalisation teams stress-test each use case by asking 'would the average customer be delighted or alarmed to know we did this?' Use cases where alarm is likely are either redesigned, moved to opt-in, or dropped.

WHEN IT IS USED

Personalisation campaign reviews, data-use approvals, privacy impact assessments, and any time behavioural targeting is being added to a new data source.

BY WHOM IT IS USED

Privacy officers, marketing ethics committees, CMOs, legal teams, and regulators auditing personalisation programmes under GDPR and CCPA.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with a pure compliance view (is it legal?), the tradeoff introduces the customer perception dimension (does it feel right?). Compared with unconstrained personalisation, it acknowledges that data maximisation can destroy value. Post-GDPR and post-ATT, the tradeoff is the central question in any serious personalisation strategy discussion.

RELATED FRAMEWORKS IN THIS COURSE

Personalisation Hierarchy · *Slibrary 10*
First vs Third Party Data · *Slibrary 7*
Trust Equation · *Slibrary 12*
Responsible AI Scorecard · *Slibrary 16*
EU AI Act Risk Pyramid · *Slibrary 1*

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Uplift Modeling

Four customer types · one who matters — targeting the Persuadables

PROFESSIONAL DEFINITION

Uplift Modeling is a causal-inference technique that estimates, for each customer, the incremental effect of a marketing treatment — in contrast to traditional response modelling, which estimates the absolute probability of response. Customers are classified into four behavioural types: Persuadables (respond because treated), Sure Things (respond regardless), Lost Causes (do not respond), and Do Not Disturbs (respond negatively to treatment). Only Persuadables should receive the treatment (Radcliffe and Surry, 2011).

WHO DEVELOPED IT

Uplift modelling was developed by Nicholas Radcliffe at Stochastic Solutions (Radcliffe, 2007; Radcliffe and Surry, 2011), building on differential response analysis (Lo, 2002) and treatment-effect estimation (Rubin, 1974). It is now standard in sophisticated direct-marketing programmes and political campaigning.

WHY IT WAS DEVELOPED

Traditional response models identify customers likely to respond — but cannot distinguish those who would have responded anyway (Sure Things) from those genuinely swayed by the marketing treatment (Persuadables). This leads to waste and, worse, to annoying Do-Not-Disturbs and triggering their churn.

FIGURE 1 · THE FOUR QUADRANTS

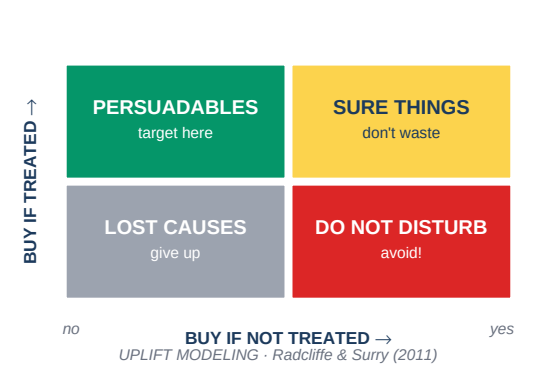

Figure 1. The uplift 2×2. Persuadables — respond if treated, do not respond if not — are the only quadrant where marketing spend earns incremental value (Radcliffe and Surry, 2011).

HOW IT IS USED

Teams run randomised A/B tests, fit uplift models on the treatment and control groups, predict each customer's uplift score, and target only those above a threshold — typically delivering more revenue at lower spend than response-based targeting.

WHEN IT IS USED

Direct marketing campaigns, retention and win-back programmes, political campaigning, and any context where the cost of contacting a Do-Not-Disturb is high (e.g. cold outreach).

BY WHOM IT IS USED

Marketing scientists, direct-marketing analysts, political campaign data teams, and customer-value-optimisation groups at large retailers and telcos.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with response modelling, uplift targets the causal effect — the only source of incremental value. Compared with broad segmentation, it identifies the individual customers for whom marketing actually works. Empirically, uplift-based targeting consistently delivers 10–30% more revenue at 20–50% less spend than response-based targeting (Radcliffe and Surry, 2011).

RELATED FRAMEWORKS IN THIS COURSE

Personalisation Hierarchy · *Slibrary 10*
Recommendation Engine Types · *Slibrary 10*
Next-Best-Action Engine · *Slibrary 11*
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Customer Journey Map with AI Touchpoints

Awareness · Consideration · Decision · Retention · Advocacy

PROFESSIONAL DEFINITION

The Customer Journey Map is a visual representation of the end-to-end sequence of interactions a customer has with a brand, from Awareness through Consideration, Decision, Retention, and Advocacy. In the AI-augmented version, each stage is mapped to specific AI capabilities — programmatic advertising at Awareness, recommendation engines at Consideration, conversational commerce at Decision, next-best-action at Retention, and net-promoter analytics at Advocacy (Lemon and Verhoef, 2016; Kotler, Kartajaya and Setiawan, 2021).

WHO DEVELOPED IT

The journey concept was popularised by Chan Kim and Mauborgne (2005) and operationalised as a mapping tool by design-thinking practitioners at IDEO and Adaptive Path (2002). The AI-augmented version appears in Kotler, Kartajaya and Setiawan's Marketing 5.0 and 6.0 (Kotler et al., 2021; 2024) and in Lemon and Verhoef's influential review (Lemon and Verhoef, 2016).

WHY IT WAS DEVELOPED

Siloed marketing functions optimised individual stages without visibility of the full journey — causing friction at handoffs and missed cross-stage opportunities. The journey map creates a shared artefact that the whole organisation can use to orchestrate AI investments coherently across the full customer lifecycle.

FIGURE 1 · FIVE STAGES · FIVE AI TOUCHPOINTS

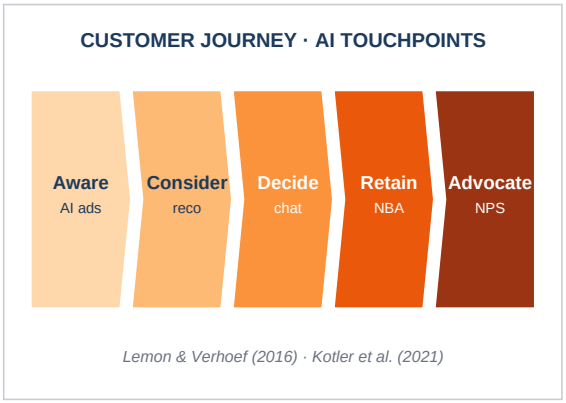

Figure 1. The five-stage customer journey with representative AI capabilities at each stage. NBA = Next-Best-Action; NPS = Net Promoter Score analytics (Lemon and Verhoef, 2016).

HOW IT IS USED

Cross-functional teams map their current touchpoints onto the five stages, identify gaps and friction points, and prioritise AI investments against customer-experience pain rather than departmental KPIs.

WHEN IT IS USED

Annual marketing strategy, CX transformation programmes, martech roadmap reviews, and post-mortem investigations of customer churn or NPS drops.

BY WHOM IT IS USED

CMOs, Chief Customer Officers, CX leaders, design researchers, and transformation consultants.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with funnel metaphors (AIDA, awareness to purchase), the journey includes post-purchase stages (retention, advocacy) where loyalty and LTV are created. Compared with channel-by-channel marketing plans, the journey forces cross-channel coherence. AI adds the ability to personalise treatment at every stage based on current journey state.

RELATED FRAMEWORKS IN THIS COURSE

Next-Best-Action Engine · *Slibrary 11* | **Omnichannel Orchestration** · *Slibrary 11* | **Personalisation Hierarchy** · *Slibrary 10* | **Marketing Evolution 1.0–6.0** · *Slibrary 9* | **Sentiment Pipeline** · *Slibrary 11*

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Next-Best-Action Engine

The orchestration brain that picks one action per customer per moment

PROFESSIONAL DEFINITION

A Next-Best-Action (NBA) engine is an AI system that, given a customer's current state and context, selects the single most valuable action to take at this moment — an email, a push notification, a call-centre prompt, an offer. It combines uplift scores, business constraints (contact frequency, offer costs), and priors (customer preferences, regulatory rules) into a ranked recommendation for each customer (Rust and Huang, 2014; Pega, 2022).

WHO DEVELOPED IT

The NBA concept has roots in operations research bandit-arm problems (Gittins, 1979) and in marketing decision-support systems at Pegasystems (Pega, 2010). Academic grounding in real-time customer analytics is due to Rust and Huang (2014) and Kumar et al. (2019). Major CRM platforms (Salesforce, Pega, Adobe) embed NBA engines as core offerings.

WHY IT WAS DEVELOPED

Marketing teams firing every available treatment at every eligible customer produced noise, fatigue, and opt-outs. The NBA engine forces the one-best-action discipline: each customer, each moment, one message — chosen algorithmically rather than by channel ownership.

FIGURE 1 · INPUTS → ENGINE → ACTION

Figure 1. The NBA engine ingests customer behaviour, context, and profile data; applies uplift scores and constraint logic; and emits a single ranked action (Rust and Huang, 2014; Pega, 2022).

HOW IT IS USED

Firms consolidate all candidate actions across channels, train uplift models for each, and deploy the NBA engine as the orchestration layer that decides which action wins in each micro-moment — replacing channel-silo-based planning.

WHEN IT IS USED

Loyalty programme design, retention campaign orchestration, contact-centre next-best-offer routing, and any context where too many candidate actions compete for the same customer attention budget.

BY WHOM IT IS USED

Customer-value-management teams, CRM platform owners, contact-centre operations, and marketing orchestration leads at large B2C firms.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with channel-siloed campaigns, NBA enforces cross-channel optimisation. Compared with static rules engines, ML-driven NBA adapts to observed uplift. Compared with pure uplift modelling, NBA adds business constraints (budget, contact caps, regulation) that make the recommendation operationally deployable.

RELATED FRAMEWORKS IN THIS COURSE

Customer Journey Map · *Slibrary 11* | **Uplift Modeling** · *Slibrary 10* | **Omnichannel Orchestration** · *Slibrary 11*

Recommendation Engine Types · *Slibrary 10* | **HITL Feedback Loop** · *Slibrary 4*

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Omnichannel Orchestration

One customer · one state · every channel coherent

PROFESSIONAL DEFINITION

Omnichannel Orchestration is the operating-model in which every customer channel — web, app, email, SMS, store, call-centre, social, push — shares a unified real-time view of each customer's state, and actions taken on any channel update that state for all others. The customer experiences a single brand, not a federation of channels (Verhoef, Kannan and Inman, 2015; Kotler et al., 2021).

WHO DEVELOPED IT

The omnichannel concept was formalised by Verhoef, Kannan and Inman (2015) and popularised in retail by case studies of Apple, Disney, and Nordstrom. The AI-driven orchestration layer is described in Kotler, Kartajaya and Setiawan (2021) and in Harvard Business Review's customer-experience research (HBR, 2023).

WHY IT WAS DEVELOPED

Multichannel — parallel channels without shared state — produced contradictory messaging, duplicated contacts, and broken handoffs (the customer sees an offer on email that is invalid in-store). Omnichannel resolves these gaps by architecting around a unified customer-data platform as the source of truth.

FIGURE 1 · HUB-AND-SPOKE ARCHITECTURE

Figure 1. The customer sits at the centre; every channel is a spoke. State changes on any spoke propagate instantly to all others (Verhoef et al., 2015; Kotler et al., 2021).

HOW IT IS USED

Firms consolidate customer data into a single CDP (customer data platform), define a canonical state model, and require every channel system to read from and write to it. The NBA engine (Sheet 2) then runs on top of the unified state.

WHEN IT IS USED

Retail transformation programmes, banking customer-experience overhauls, telecom churn-reduction initiatives, and any context where channel silos are destroying value.

BY WHOM IT IS USED

Chief Customer Officers, Chief Digital Officers, CDP implementation teams, and customer experience strategists.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with multichannel marketing, omnichannel adds cross-channel continuity as a hard requirement. Compared with per-channel analytics, it produces unified attribution and identity resolution. The move from multi- to omni-channel is typically the largest single CX investment firms make in a decade.

RELATED FRAMEWORKS IN THIS COURSE

Customer Journey Map · *Slibrary 11*
Next-Best-Action Engine · *Slibrary 11*
Physical-Digital Convergence · *Slibrary 9*
First vs Third Party Data · *Slibrary 7*
Data Value Chain · *Slibrary 7*

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Sentiment Analysis Pipeline

Voice · Extract · Classify · Act — voice-of-customer at scale

PROFESSIONAL DEFINITION

A Sentiment Analysis Pipeline is an AI system that ingests unstructured customer feedback — reviews, call transcripts, social posts, chat logs — and produces structured signals about customer sentiment that feed operational action. The four-stage pipeline (Voice → Extract → Classify → Act) converts qualitative noise into the voice of customer at a scale impossible with manual review (Liu, 2015; Pang and Lee, 2008).

WHO DEVELOPED IT

Sentiment analysis as a field was established by Pang and Lee (2008) and Liu (2015). The four-stage pipeline view synthesises industrial practice at Medallia, Qualtrics, and Clarabridge (now Qualtrics XM). Modern variants use large language models for extraction and classification in a single step (Devlin et al., 2019; OpenAI, 2023).

WHY IT WAS DEVELOPED

Manual review of customer feedback cannot keep pace with the volume of digital channels. The pipeline automates classification while preserving the qualitative richness needed to diagnose root causes — turning feedback from a cost centre into an operational feedback loop.

FIGURE 1 · FOUR-STAGE PIPELINE

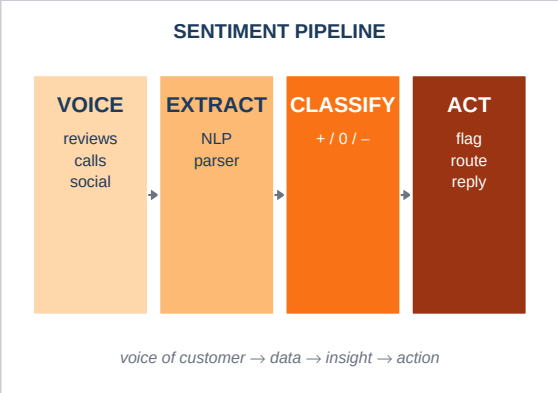

Figure 1. The sentiment pipeline. Voice ingests raw customer expressions; Extract parses entities and aspects; Classify scores sentiment; Act routes the signal to the correct operational owner (Liu, 2015).

HOW IT IS USED

Teams connect the pipeline to all customer-touching unstructured channels, set routing rules for negative sentiment (complaints → service escalation; product issues → product team), and measure time-to-resolution as a closed-loop metric.

WHEN IT IS USED

Customer-service operations, product quality monitoring, brand reputation management, and regulatory-complaint tracking in financial services.

BY WHOM IT IS USED

Customer experience teams, product managers, brand managers, and compliance officers monitoring complaints patterns.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with manual review of a sample, the pipeline processes 100% of feedback. Compared with lexicon-based sentiment of the 2010s, modern LLM-based extraction captures nuance, sarcasm, and aspect-level sentiment (this feature is bad, that feature is great). The pipeline increasingly feeds real-time operational dashboards rather than quarterly reports.

RELATED FRAMEWORKS IN THIS COURSE

Customer Journey Map · *Slibrary 11* | **Next-Best-Action Engine** · *Slibrary 11* | **Chatbot Design** · *Slibrary 12*

GenAI in Marketing · *Slibrary 9* | **HITL Feedback Loop** · *Slibrary 4*

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Customer Lifetime Value

The one metric that reconciles acquisition, retention, and pricing

PROFESSIONAL DEFINITION

Customer Lifetime Value (CLV) is the present value of the future cash flow attributable to a customer across their relationship with the firm. Classical formulations multiply margin per period by retention probability, discounted by time. AI refines every component: margin is predicted from purchase pattern; retention is estimated by churn models; individual CLV distributions replace segment averages (Gupta et al., 2006; Fader and Hardie, 2020).

WHO DEVELOPED IT

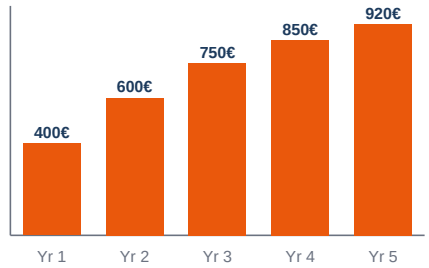
Formal CLV emerged from direct-marketing analytics (Dwyer, 1989; Gupta et al., 2006). The probabilistic AI variants were developed by Fader and Hardie (BG/NBD, Gamma-Gamma models; Fader and Hardie, 2005; 2020). Modern deep-learning CLV models (Gabel and Timoshenko, 2022) use sequence models to predict future purchase trajectories.

WHY IT WAS DEVELOPED

Firms over-investing in acquisition without considering retention burned cash chasing low-LTV customers. CLV gives the common currency in which acquisition spend, retention investment, and pricing decisions can be compared, forcing coherent resource allocation.

FIGURE 1 · THE MULTI-YEAR REVENUE STREAM

CLV = Σ (Margin × Retention rate) / (1+r)■



AI refines each component: margin, retention, churn (Gupta et al., 2006)

Figure 1. CLV as the sum of discounted annual contributions. AI refines the estimates for each year's margin and retention probability (Gupta et al., 2006; Fader and Hardie, 2020).

HOW IT IS USED

Firms score each customer's CLV and segment investment accordingly: aggressive retention for high-CLV, efficient service for mid-CLV, automated handling for low-CLV. Acquisition spend is capped at a fraction of expected CLV.

WHEN IT IS USED

Pricing strategy, acquisition-budget allocation, loyalty programme design, customer-value-management programmes, and investor due diligence on SaaS and subscription businesses.

BY WHOM IT IS USED

Chief Customer Officers, CFOs, customer-value-management teams, and investors evaluating customer-economics-driven business models.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with segment-average CLV, AI-driven individual CLV enables one-to-one investment decisions. Compared with static CLV from cohort averages, dynamic CLV updates with each customer interaction, feeding the NBA engine. CLV is the central metric in modern SaaS valuation (the 'rule of 40' and related heuristics) and increasingly in retail and financial services.

RELATED FRAMEWORKS IN THIS COURSE

Uplift Modeling · *Slibrary 10* | **Next-Best-Action Engine** · *Slibrary 11* | **Cost-Benefit Framework** · *Slibrary 13* | **TCO Framework** · *Slibrary 14* | **Personalisation Hierarchy** · *Slibrary 10*

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The Trust Equation

Credibility + Reliability + Intimacy, divided by Self-Orientation

PROFESSIONAL DEFINITION

The Trust Equation defines trustworthiness as a ratio: the sum of three components (Credibility, Reliability, Intimacy) in the numerator, divided by Self-Orientation in the denominator. Because Self-Orientation multiplies the whole, an organisation that appears self-serving destroys trust far faster than new capability adds to it. The equation applies equally to individual advisors, brands, and AI systems (Maister, Green and Galford, 2000).

WHO DEVELOPED IT

Formulated by David Maister, Charles Green and Robert Galford in The Trusted Advisor (Maister, Green and Galford, 2000), drawing on organisational trust research (Mayer, Davis and Schoorman, 1995). Green later extended the framework specifically to organisational and brand trust (Green, 2006; Reynolds, 2021).

WHY IT WAS DEVELOPED

Trust is widely cited as an outcome but rarely decomposed into actionable components. The equation gives executives four specific levers — demonstrate competence, follow through, engage personally, and act unselfishly — and specifies that the fourth lever (self-orientation) has the largest impact.

FIGURE 1 · THE RATIO FORM

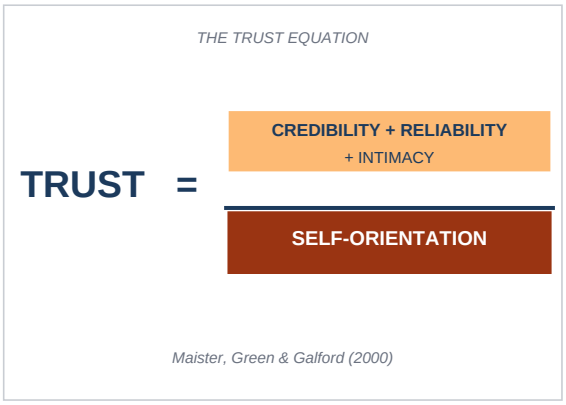

Figure 1. The Trust Equation. Because self-orientation is the denominator, a 10% increase in self-serving behaviour reduces trust more than a 10% improvement in any numerator component (Maister, Green and Galford, 2000).

HOW IT IS USED

Leaders audit their brand's and their AI systems' behaviour against the four components, identify the weakest element, and prioritise interventions — typically reducing self-orientation yields the largest trust gains.

WHEN IT IS USED

Brand crisis response, customer experience reviews, AI deployment ethics assessments, and executive coaching programmes for client-facing leaders.

BY WHOM IT IS USED

CMOs, brand stewards, customer experience officers, AI ethics committees, and professional-services leaders managing advisory relationships.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with vague 'build trust' aspirations, the equation specifies four observable dimensions with distinct interventions. Compared with cognitive trust models, the self-orientation denominator captures the outsized impact of perceived self-interest. For AI, the framework diagnoses why some AI brands (e.g. medical AI) demand far higher trust thresholds than others.

RELATED FRAMEWORKS IN THIS COURSE

Brand Personality AI · *Slibrary 12* | **Chatbot Design** · *Slibrary 12* | **Privacy-Personalisation** · *Slibrary 10*
Responsible AI Scorecard · *Slibrary 16* | **Human-Centric Marketing** · *Slibrary 9*

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AI-Measured Brand Personality

Aaker's five dimensions · quantified from unstructured customer data

PROFESSIONAL DEFINITION

Brand Personality is the set of human characteristics attributed to a brand, organised by Jennifer Aaker (1997) into five dimensions: Sincerity, Excitement, Competence, Sophistication, and Ruggedness. Modern AI systems measure these dimensions automatically by applying NLP to customer reviews, social media, and call transcripts — converting qualitative brand perceptions into quantitative, tracked metrics (Aaker, 1997; Keller, 2020).

WHO DEVELOPED IT

The five-dimension framework is due to Jennifer Aaker's 1997 Journal of Marketing Research paper (Aaker, 1997). The AI-measurement adaptation comes from computational marketing research (Ordenes, Ludwig, et al., 2017; Culotta and Cutler, 2016) and practitioner tools such as Brandwatch and YouGov BrandIndex.

WHY IT WAS DEVELOPED

Traditional brand-tracking surveys are expensive, lag reality by months, and suffer from response bias. Large-scale NLP extraction of personality dimensions from real customer language is cheaper, more current, and more honest — it captures what customers actually say, not what they say to researchers.

FIGURE 1 · FIVE-DIMENSION PROFILE

Figure 1. A sample AI-measured brand personality profile. Values are normalised between 0 and 1 relative to a category benchmark (Aaker, 1997; Culotta and Cutler, 2016).

HOW IT IS USED

Brand teams benchmark their AI-measured profile against target positioning and competitor profiles, identify gaps, and brief creative or activation programmes to close the most important gaps — tracking the profile monthly as a leading indicator of brand equity.

WHEN IT IS USED

Brand equity reviews, creative campaign briefs, post-launch brand tracking, M&A due diligence assessing acquired-brand personality, and competitor analysis.

BY WHOM IT IS USED

Brand managers, insights teams, strategic planners at agencies, and investors assessing brand as an intangible asset.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with survey-based trackers, AI measurement is continuous and closer to reality. Compared with sentiment analysis (Slibrary 11), personality measurement captures qualitative character beyond positive/negative. The technique is increasingly integrated into brand-equity valuation methodologies (Interbrand, Kantar BrandZ, 2024).

RELATED FRAMEWORKS IN THIS COURSE

Trust Equation · *Slibrary 12* | **Sentiment Pipeline** · *Slibrary 11* | **GenAI in Marketing** · *Slibrary 9* | **Human-Centric Marketing** · *Slibrary 9* | **Customer Lifetime Value** · *Slibrary 11*

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Chatbot & Virtual Assistant Design

Conversation design · escalation · transparency · trust

PROFESSIONAL DEFINITION

Chatbot & Virtual Assistant Design is the discipline of engineering conversational AI interfaces that serve customers effectively and build, rather than erode, trust. Good design covers scope (what the bot is allowed to do), transparency (disclosing AI status), escalation (clean handoff to human agents when needed), and personality (tone appropriate to brand and context). Best practice is now codified in human-AI interaction guidelines (Amershi et al., 2019; Google, 2022).

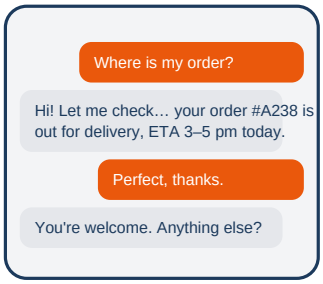
WHO DEVELOPED IT

Foundational guidelines come from Microsoft's Human-AI Interaction research (Amershi et al., 2019), Google's People + AI Guidebook (Google, 2022), and Gartner's Conversational AI playbooks. Academic grounding in conversational UX is due to Amershi, Weld and Vorvoreanu (2019) and McTear (2022).

WHY IT WAS DEVELOPED

Poorly-designed chatbots are a leading source of customer frustration and brand damage — bots that pretend to be human, cannot escalate, or loop indefinitely erode the Trust Equation's self-orientation dimension. Design guidelines turn chatbots from a risk into a measurable customer-experience improvement.

FIGURE 1 · CHATBOT UX ESSENTIALS



design for handoff · escalate early · no fake empathy

Figure 1. A well-designed exchange: clear bot identity, specific answer, graceful closure. Escalation paths and transparency about AI status are non-negotiable (Amershi et al., 2019; Google, 2022).

HOW IT IS USED

Design teams apply the 18 guidelines from Amershi et al. (2019) to each conversational flow, run usability testing with real customers, and measure containment rate, escalation success rate, and CSAT as the core KPIs.

WHEN IT IS USED

Chatbot launch reviews, IVR modernisation programmes, customer-service digital transformation, and generative AI-powered assistant deployments where hallucination risk adds to the design burden.

BY WHOM IT IS USED

Conversation designers, UX researchers, customer-experience teams, and AI ethics reviewers auditing deployed conversational systems.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with rule-based IVR systems, modern LLM-backed assistants handle nuance and open-ended requests. Compared with early chatbots, best-practice designs now require transparency and escalation as non-negotiable. The EU AI Act's transparency provisions (Article 50) make the 'disclose AI identity' guideline legally binding in the EU.

RELATED FRAMEWORKS IN THIS COURSE

Trust Equation · *Slibrary 12* | **Sentiment Pipeline** · *Slibrary 11* | **GenAI in Marketing** · *Slibrary 9* | **HITL Feedback Loop** · *Slibrary 4* | **EU AI Act Risk Pyramid** · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

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Deepfake & Synthetic Media Risk

SLIBRARY 12 · SHEET 4 / 5

Three escalation tiers · scams · impersonation · synthetic executives

PROFESSIONAL DEFINITION

Deepfake & Synthetic Media Risk is the exposure that brands and executives face from AI-generated audio, video, or image content created in their name without authorisation. Three escalating tiers of threat are commonly distinguished: financial scams (voice cloning of family members or executives), brand impersonation (fake advertising or endorsement), and synthetic executive fraud (CFO video calls, authorised wire transfers). All three tiers have been observed in real incidents by 2024–2026 (FBI, 2024; Deloitte, 2024).

WHO DEVELOPED IT

Risk frameworks come from the FBI's 2022 and 2024 advisories on synthetic media fraud, Deloitte's Center for Financial Services reports (Deloitte, 2024), and academic research on deepfake detection (Verdoliva, 2020). The 'synthetic CEO' variant was highlighted by the 2024 Hong Kong firm $25m incident (HKPF, 2024).

WHY IT WAS DEVELOPED

Firms underestimated the attack surface that generative-AI opened: publicly-available executive video and voice became training material for deepfakes. The three-tier framework gives security and brand teams a shared vocabulary to prioritise monitoring and response investments.

FIGURE 1 · THREAT ESCALATION CURVE

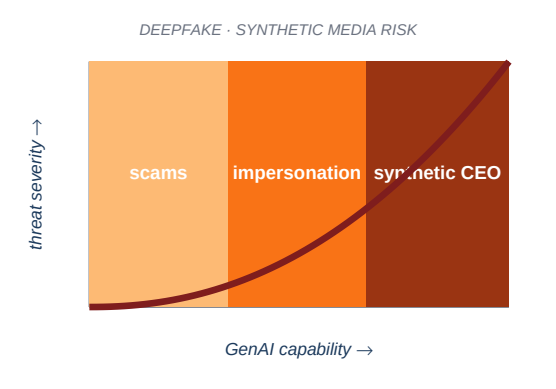

Figure 1. As generative capability rises, threat severity rises super-linearly. The 'synthetic CEO' tier represents existential financial risk to mid-size firms (FBI, 2024; Deloitte, 2024).

HOW IT IS USED

Firms run tabletop exercises for each tier, implement verification protocols (call-back policies for wire transfers, passphrases for executive requests), and monitor public channels for brand impersonation with specialised detection tools.

WHEN IT IS USED

Cybersecurity strategy reviews, brand reputation risk assessments, anti-fraud control design, and executive protection programmes for C-suite and board members.

BY WHOM IT IS USED

CISOs, CFOs (for financial fraud vectors), brand protection teams, legal counsel, and board risk committees.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional identity fraud, deepfake fraud bypasses voice and video verification — historically the strongest 'prove you're you' signals. Compared with text-based phishing, synthetic media attacks are far more convincing and scale cheaply. The EU AI Act (Article 50) now requires providers to label synthetic media, a regulatory counter-measure to the threat.

RELATED FRAMEWORKS IN THIS COURSE

Trust Equation · *Slibrary 12* | **Chatbot Design** · *Slibrary 12* | **EU AI Act Risk Pyramid** · *Slibrary 1* | **NIST AI RMF** · *Slibrary 1* | **Responsible AI Scorecard** · *Slibrary 16*

SOURCES (HARVARD REFERENCING STYLE)

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Loyalty Programme Analytics

New · Regular · VIP — AI-driven tier progression and reward design

PROFESSIONAL DEFINITION

Loyalty Programme Analytics is the AI-driven measurement and optimisation of loyalty programme performance: predicting tier progression, forecasting redemption liability, personalising reward offers, and identifying at-risk high-value customers. Modern programmes move beyond fixed point-based schemes to dynamic, per-customer value propositions calibrated by CLV and AI-estimated price elasticity (Kumar and Reinartz, 2018; McKinsey, 2022).

WHO DEVELOPED IT

Analytics-driven loyalty theory is developed in Kumar and Reinartz (2018) and Fader and Hardie's customer-centric work (Fader and Hardie, 2020). Practitioner frameworks come from McKinsey's retail loyalty reports (McKinsey, 2022), Bain & Company (Reichheld and Markey, 2011), and Harvard Business Review case studies of Starbucks Rewards, Sephora, and Tesco Clubcard.

WHY IT WAS DEVELOPED

Legacy loyalty programmes offered the same reward mix to every member, producing liability on the balance sheet without corresponding retention lift. AI analytics personalise rewards, retire dead points, and redirect spend to the customers most influenced by the programme — converting loyalty from a cost centre to a measured value driver.

FIGURE 1 · THREE-TIER SEGMENTATION

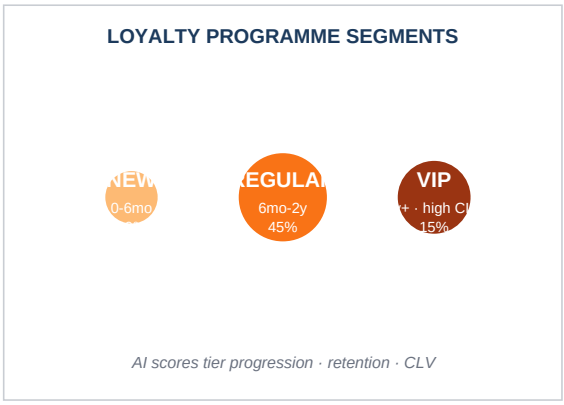

Figure 1. A typical three-tier loyalty distribution. VIP members (~15%) typically generate 40–60% of programme revenue; retention of the VIP tier is the single highest-value analytic task (McKinsey, 2022).

HOW IT IS USED

Teams score each member's next-tier transition probability and CLV, personalise reward offers to accelerate tier progression for customers who respond, and redesign liability accruals based on realistic redemption forecasts.

WHEN IT IS USED

Loyalty programme redesign, retention strategy reviews, balance-sheet liability forecasting, and cross-functional coordination between finance, marketing, and customer service.

BY WHOM IT IS USED

Loyalty programme managers, customer-analytics teams, finance teams forecasting liability, and CMOs reviewing customer economics.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with fixed-point schemes, personalised loyalty increases redemption-per-dollar and reduces slush. Compared with segment-based programmes, individual-level analytics identify the specific customers for whom the programme drives genuine incremental behaviour (Kumar and Reinartz, 2018). Starbucks Rewards and Sephora Beauty Insider are the most-cited examples of fully-analytic loyalty programmes.

RELATED FRAMEWORKS IN THIS COURSE

Customer Lifetime Value · *Slibrary 11* | **Uplift Modeling** · *Slibrary 10* | **Next-Best-Action Engine** · *Slibrary 11* | **Personalisation Hierarchy** · *Slibrary 10* | **Trust Equation** · *Slibrary 12*

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AI Cost-Benefit Waterfall

Base → Revenue → Savings → Costs → NPV — the investment-case bridge

PROFESSIONAL DEFINITION

The AI Cost-Benefit Waterfall is the visual decomposition of an AI investment's net present value (NPV) into its constituent benefit and cost components, ordered sequentially from the business-as-usual baseline through each positive and negative contribution to the bottom-line NPV. It is the standard deliverable presented to investment committees, boards, and CFOs evaluating AI proposals (Brealey, Myers and Allen, 2022; McKinsey, 2023).

WHO DEVELOPED IT

The waterfall format originated in McKinsey management reporting (McKinsey, 1990s) and became standard in investment-case presentation across the Big Four consultancies and investment banks. Its application to AI projects is codified in McKinsey's 'State of AI' reports (McKinsey, 2023; 2024), Deloitte's AI adoption surveys, and Gartner's AI Value Pyramid.

WHY IT WAS DEVELOPED

AI investment cases too often presented gross benefits without offsetting costs, or discussed costs without placing them against quantified benefits. The waterfall forces a disciplined, auditable bridge from baseline to NPV, making every assumption visible and contestable.

FIGURE 1 · ILLUSTRATIVE AI BUSINESS CASE

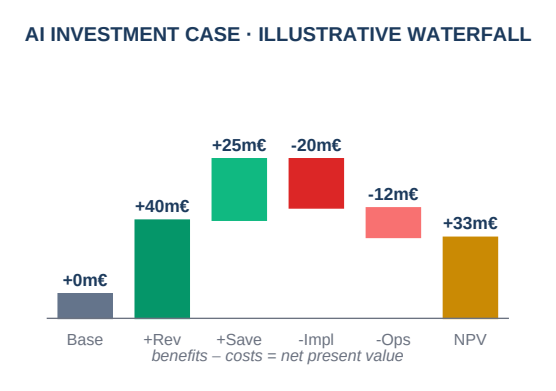

Figure 1. A six-bar waterfall showing how an AI investment translates into NPV. Green bars are positive contributions; red bars are costs; the final gold bar is the net position (Brealey, Myers and Allen, 2022).

HOW IT IS USED

Teams build the waterfall at three phases: first pass at business-case stage, refined after the pilot, and updated post-deployment with actuals — allowing board-level comparison of projected vs realised value.

WHEN IT IS USED

AI investment committee submissions, annual AI portfolio reviews, post-implementation reviews, and CFO-level conversations about AI ROI that require auditable defences of claimed value.

BY WHOM IT IS USED

CFOs, strategy and corporate development teams, AI programme leaders, and finance business-partners supporting AI initiatives.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with summary ROI numbers, the waterfall makes every assumption visible. Compared with scenario analysis, it provides a single most-likely bridge that can be compared like-for-like across AI investments. The format forces the discipline of naming each cost and benefit — preventing hidden items and surprise post-implementation write-offs.

RELATED FRAMEWORKS IN THIS COURSE

ROI Payback Period · *Slibrary 13* | **AI Value Drivers** · *Slibrary 13* | **NPV Formula** · *Slibrary 13* | **TCO Framework** · *Slibrary 14* | **Build vs Buy** · *Slibrary 3*

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ROI Payback Period

When cumulative cash turns positive — the investor's first question

PROFESSIONAL DEFINITION

The Payback Period is the time elapsed until an AI investment's cumulative cash flow turns positive — i.e., until the project has returned its initial outlay. It is the simplest investment metric and the first question most executives ask: 'When do we break even?' Modern variants (discounted payback) apply a discount rate before summation to reflect time value of money (Brealey, Myers and Allen, 2022).

WHO DEVELOPED IT

Payback analysis is a foundational capital-budgeting technique documented in classical corporate finance texts (Brealey, Myers and Allen, 2022; Ross, Westerfield and Jaffe, 2022). Its application to AI investments is emphasised in Gartner AI ROI guidance (Gartner, 2024), McKinsey's AI financial-impact studies (McKinsey, 2023), and practitioner literature (Davenport, 2018).

WHY IT WAS DEVELOPED

NPV and IRR are more theoretically sound than payback but harder to communicate to non-financial executives. Payback's intuitive framing ('18 months to break even') complements NPV as an accessible first-screen metric and a liquidity risk indicator for capital-constrained firms.

FIGURE 1 · CUMULATIVE CASH FLOW CURVE

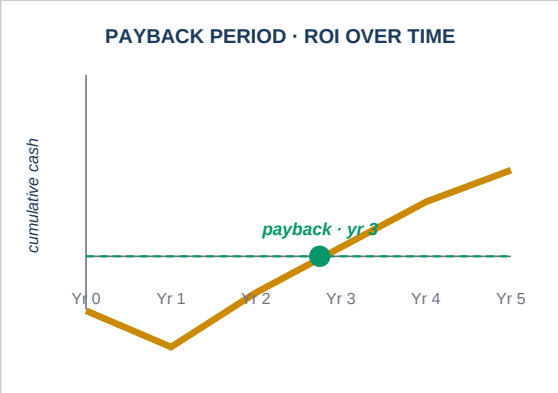

Figure 1. Cumulative cash flow over five years. The curve starts negative (initial investment), deepens during the build phase, then rises and crosses zero at the payback point (here, Year 3) (Brealey, Myers and Allen, 2022).

HOW IT IS USED

Teams model monthly or annual cash flows, plot the cumulative curve, and report payback alongside NPV and IRR. Projects with payback longer than the firm's tolerance are restructured or rejected.

WHEN IT IS USED

Early investment-screening, board-level approvals, investor presentations for AI-driven companies, and capital rationing discussions when multiple AI projects compete for limited funds.

BY WHOM IT IS USED

CFOs, investment committees, corporate-development teams, and investors evaluating AI-adoption claims by portfolio companies.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with NPV alone, payback surfaces liquidity risk explicitly — how long must the firm sustain the loss? Compared with simple ROI ratios, payback captures the time profile of returns. For AI specifically, payback helps surface the 'J-curve' pattern in which costs hit early (talent, compute, change management) and benefits arrive later (adoption, optimisation).

RELATED FRAMEWORKS IN THIS COURSE

Cost-Benefit Waterfall · *Slibrary 13* | **NPV Formula** · *Slibrary 13* | **AI Value Drivers** · *Slibrary 13* | **TCO Framework** · *Slibrary 14* | **Build vs Buy** · *Slibrary 3*

SOURCES (HARVARD REFERENCING STYLE)

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Four AI Value Drivers

Revenue · Cost · Risk · Experience — where AI creates measurable value

PROFESSIONAL DEFINITION

The Four AI Value Drivers is a taxonomy of the benefit categories into which AI-generated value flows: Revenue Growth (new products, conversion lift, price optimisation), Cost Reduction (automation, efficiency, compute optimisation), Risk Mitigation (fraud prevention, compliance automation, predictive maintenance), and Experience Uplift (NPS, retention, brand equity). McKinsey's AI adoption studies consistently show roughly 28/35/17/20 splits across large firms (McKinsey, 2023; 2024).

WHO DEVELOPED IT

The four-category taxonomy is synthesised across McKinsey's State of AI reports (McKinsey, 2023; 2024), BCG's AI value-capture research (BCG, 2024), and Harvard Business Review's applied-AI case studies. Academic grounding on categorising AI value is in Davenport (2018) and Agrawal, Gans and Goldfarb (2018).

WHY IT WAS DEVELOPED

AI business cases historically focused narrowly on cost reduction, leaving revenue, risk, and experience benefits unquantified and therefore invisible to approval committees. The four-driver taxonomy forces teams to consider the full value landscape — and typically reveals benefits 2–3× larger than the cost-only case.

FIGURE 1 · BENEFIT SPLIT BY DRIVER

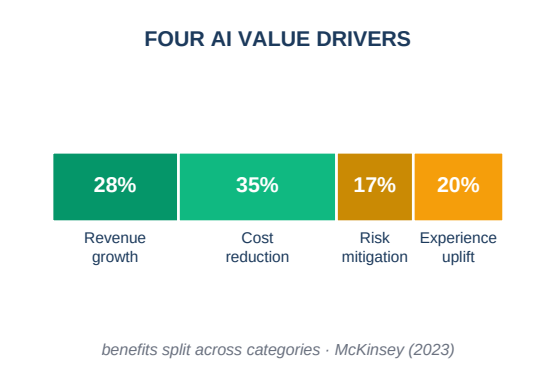

Figure 1. Typical split of AI benefits across the four drivers at McKinsey 'AI-leading' firms. Cost reduction remains the largest single category but all four contribute materially (McKinsey, 2023).

HOW IT IS USED

Teams mapping each AI use-case to its dominant value driver build a portfolio view of AI benefits across the four categories, then ensure that investment isn't skewed entirely to one category (typically cost) when material opportunity exists in others.

WHEN IT IS USED

AI portfolio reviews, annual strategy off-sites, investor communications on AI value, and CFO-level discussions about expected AI contribution to enterprise value.

BY WHOM IT IS USED

AI programme leaders, CFOs, strategy teams, corporate development, and investor relations teams communicating AI benefits externally.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with one-dimensional 'productivity' framing of AI, the four-driver view captures revenue and risk dimensions that are equally important. Compared with pure financial ROI, the experience dimension captures value that is real but flows indirectly through brand equity and CLV. The taxonomy is used in most large-firm AI value-tracking dashboards post-2023.

RELATED FRAMEWORKS IN THIS COURSE

Cost-Benefit Waterfall · *Slibrary 13* | **ROI Payback Period** · *Slibrary 13* | **NPV Formula** · *Slibrary 13* | **Cost Breakdown** · *Slibrary 13* | **Responsible AI Scorecard** · *Slibrary 16*

SOURCES (HARVARD REFERENCING STYLE)

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NPV Formula for AI

Discounted cash flows minus initial investment — the single most important equation in corporate finance

PROFESSIONAL DEFINITION

Net Present Value (NPV) is the sum of all future cash flows from an AI investment, each discounted back to today's money using a risk-adjusted discount rate r, minus the initial investment I0. If NPV > 0, the project creates shareholder value; if NPV < 0, it destroys value. NPV is the theoretically correct capital-budgeting metric against which all others (IRR, payback) should be reconciled (Brealey, Myers and Allen, 2022).

WHO DEVELOPED IT

Discounted cash flow analysis was formalised by Irving Fisher (Fisher, 1930) and John Burr Williams (Williams, 1938), and established as the gold standard of investment appraisal by Joel Dean (Dean, 1951). Its central position in modern corporate finance is documented in Brealey, Myers and Allen (2022) and every standard MBA finance textbook.

WHY IT WAS DEVELOPED

Undiscounted project evaluation overstates value by ignoring the opportunity cost of capital; simple ROI ratios obscure timing. NPV unifies both — the time value of money is captured by the discount rate, and the total value is a single summable number. For AI specifically, NPV surfaces the risk premium in the discount rate — frontier-AI projects demand higher r than proven ones.

FIGURE 1 · THE NPV FORMULA

NET PRESENT VALUE · Brealey et al. (2022)

$$NPV = \sum \frac{CF_t}{(1+r)^t} - I_0$$

CF cash flow r discount rate I initial invest t time period

Figure 1. NPV decomposed. Each future cash flow CFt is divided by (1+r)^t, summed, and reduced by the initial investment I0. The discount rate r embeds time value of money and project-specific risk (Brealey, Myers and Allen, 2022).

HOW IT IS USED

For each AI project, teams estimate annual cash flows, select a discount rate (weighted-average cost of capital plus a project-risk premium), compute NPV, and compare against other uses of capital — approving projects with positive NPV subject to portfolio and strategic constraints.

WHEN IT IS USED

All investment committee approvals, annual portfolio reviews, M&A valuation of AI-driven businesses, and regulatory assessments where financial projections must be defensible to auditors.

BY WHOM IT IS USED

CFOs, corporate development teams, investment committees, investment analysts, and any approver evaluating AI capital allocation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with payback or simple ROI, NPV is mathematically correct and comparable across projects with different time profiles. Compared with sensitivity-only analyses, NPV provides a single summary number. For AI, the key innovation is treating the discount rate as the lever through which technology risk (model performance, adoption, regulatory) enters the valuation — rather than haircutting cash flows directly.

RELATED FRAMEWORKS IN THIS COURSE

Cost-Benefit Waterfall · *Slibrary 13* | **ROI Payback Period** · *Slibrary 13* | **AI Value Drivers** · *Slibrary 13* | **TCO Framework** · *Slibrary 14* | **Portfolio AI (BlackRock)** · *Slibrary 22*

SOURCES (HARVARD REFERENCING STYLE)

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AI Cost Breakdown

Talent · Compute · Data · Software · Change — where every AI dollar goes

PROFESSIONAL DEFINITION

The AI Cost Breakdown is the decomposition of total AI programme spend into five recognisable cost categories: Talent (ML engineers, data scientists, AI product managers), Compute (cloud infrastructure, GPU clusters), Data (acquisition, labelling, platforms), Software (vendor tools and licences), and Change Management (training, organisational redesign). Gartner reports that talent typically represents the largest single share (~35%), followed by compute (~25%) (Gartner, 2024).

WHO DEVELOPED IT

The five-category taxonomy is synthesised from Gartner's AI TCO models (Gartner, 2023; 2024), Deloitte's 'State of AI' annual surveys (Deloitte, 2024), and McKinsey's AI cost-structure research (McKinsey, 2023). It is now standard in enterprise AI budget planning.

WHY IT WAS DEVELOPED

Firms frequently budgeted AI as 'technology' spend and under-provisioned the non-technology categories (talent, change) that in practice consume most of the actual cost. The breakdown forces balanced budgeting and surfaces hidden costs that otherwise produce programme overruns.

FIGURE 1 · FIVE-CATEGORY SPEND SPLIT

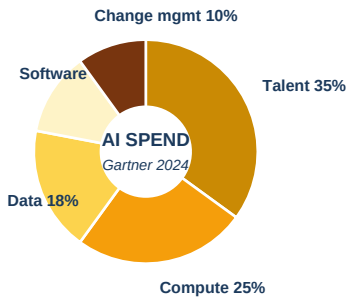

Figure 1. Typical enterprise AI cost breakdown. Talent dominates; compute is growing fastest; change management remains systematically under-budgeted despite being the category that most determines adoption success (Gartner, 2024).

HOW IT IS USED

AI programme leaders budget across all five categories, benchmark their mix against industry norms, and flag programmes where one category (typically technology) dominates at the expense of others (typically change management) — a common failure pattern.

WHEN IT IS USED

Annual AI budget cycles, TCO modelling, vendor negotiations, and post-implementation reviews analysing why an AI programme over- or under-ran budget.

BY WHOM IT IS USED

AI programme leaders, CFOs, CIOs, procurement teams, and transformation programme managers tracking full programme economics.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with IT-only cost views, the breakdown makes talent and change management explicit — both typically larger than software licensing. Compared with pure TCO accounting, the category view supports capability-level benchmarking. Gartner (2024) finds that high-performing AI programmes show a more balanced mix — compute plus software together around 35–40%, not 60%+.

RELATED FRAMEWORKS IN THIS COURSE

Cost-Benefit Waterfall · *Slibrary 13* | **AI Value Drivers** · *Slibrary 13* | **TCO Framework** · *Slibrary 14* | **Build vs Buy** · *Slibrary 3* | **BCG People Investment** · *Slibrary 4*

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TCO Iceberg for AI

Visible ~20 per cent · hidden ~80 per cent — where the real spend lies

PROFESSIONAL DEFINITION

The Total Cost of Ownership (TCO) Iceberg is a visualisation of the AI cost base in which the visible costs that reach the investment committee — licence fees, GPU compute, vendor subscriptions — form only the tip of an iceberg whose larger submerged mass consists of integration, data engineering, retraining, monitoring, compliance, and change-management costs. Gartner (2024) estimates visible costs at roughly 20 per cent of programme TCO, with 80 per cent below the waterline.

WHO DEVELOPED IT

The iceberg metaphor for TCO originated in manufacturing and IT infrastructure (Ellram, 1995) and was applied to enterprise software TCO by Gartner in the 1990s (Gartner, 1998). Its AI-specific application was codified in Gartner (2023; 2024), Deloitte's State of AI in the Enterprise surveys (Deloitte, 2024), and BCG's AI value-capture research (BCG, 2024).

WHY IT WAS DEVELOPED

AI business cases routinely capture licence and compute costs but omit the larger costs of integrating, operating, and governing models in production. This produces systematic under-estimates and programme overruns. The iceberg forces every submerged category to be named, quantified, and included in the TCO model before approval (Gartner, 2024).

FIGURE 1 · VISIBLE VS HIDDEN TCO

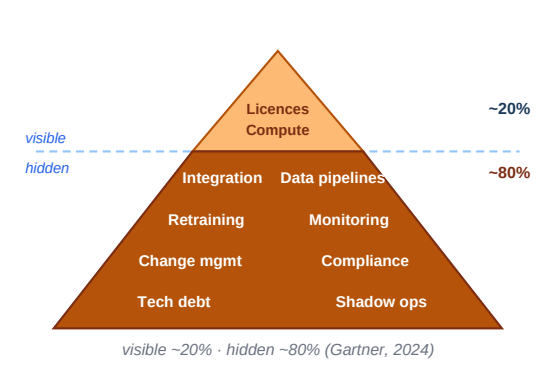

Figure 1. Visible AI costs (~20%) above the waterline — licences and compute — versus hidden TCO (~80%) below the line, covering integration, data, retraining, monitoring, compliance, change, technical debt, and shadow operations (Gartner, 2024).

HOW IT IS USED

Teams inventory every submerged category, estimate costs for each, and add them to the TCO model before business-case approval. Post-deployment, actuals are tracked against the estimate to surface the most underestimated categories for future proposals.

WHEN IT IS USED

AI business-case preparation, portfolio TCO modelling, vendor-comparison exercises, and post-implementation reviews where programmes overran budget and the question is which cost category was mis-estimated.

BY WHOM IT IS USED

CFOs, AI programme leaders, procurement teams, CIOs, and cost-transformation consultants modelling the full economic footprint of an AI initiative.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with licence-and-compute views, the iceberg quantifies categories that typically consume 3–4× more budget than the visible ones. Compared with generic IT TCO models, it foregrounds categories specific to AI — retraining, drift monitoring, and model governance — that do not appear in traditional software TCO (Gartner, 2024; Deloitte, 2024).

RELATED FRAMEWORKS IN THIS COURSE

Cloud vs On-Prem TCO · *Slibrary 14* | **GPU Cost Curve** · *Slibrary 14* | **Hidden Cost Categories** · *Slibrary 14* | **FinOps Loop** · *Slibrary 14* | **Cost-Benefit Waterfall** · *Slibrary 13*

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Cloud vs On-Premise TCO

OPEX that scales versus CAPEX that sinks — the deployment-mode choice

PROFESSIONAL DEFINITION

The Cloud vs On-Premise TCO comparison is a decision framework that contrasts the economics of renting AI infrastructure (cloud: usage-based OPEX, fast provisioning, data-egress risk) against owning it (on-premise: fixed CAPEX, fully-loaded cost, control over data gravity). The comparison is never static — optimal choice depends on utilisation, data gravity, regulatory constraints, and programme duration (Andreessen Horowitz, 2021; McKinsey, 2024).

WHO DEVELOPED IT

The cloud-versus-on-premise debate was formalised in Sarah Wang and Martin Casado's Andreessen Horowitz analysis of cloud repatriation (Andreessen Horowitz, 2021) and extended to AI workloads in Gartner's AI infrastructure research (Gartner, 2024) and McKinsey's analysis of hyperscaler economics (McKinsey, 2024). The CAPEX-OPEX framing goes back to classical management accounting (Horngren et al., 2014).

WHY IT WAS DEVELOPED

AI training and inference consume compute at scales where the cloud-vs-owned choice produces order-of-magnitude TCO differences. Many firms over-indexed on cloud for burst convenience and discovered that at steady-state inference volumes, owned GPUs would have been 2–3× cheaper — the repatriation hypothesis. Conversely, small-volume firms that built on-premise over-invested (Andreessen Horowitz, 2021).

FIGURE 1 · STACKED TCO COMPOSITION

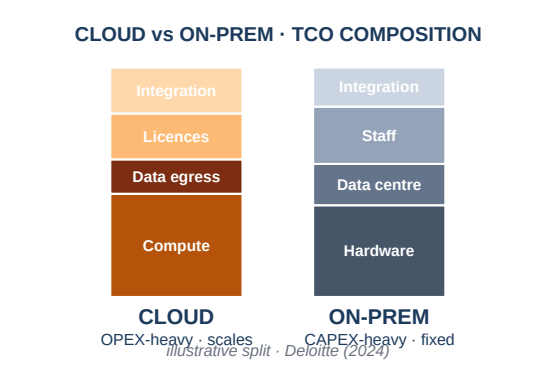

Figure 1. Illustrative TCO composition for equivalent AI workloads. Cloud is dominated by compute and egress charges; on-premise is dominated by hardware, data centre, and staff. Optimal choice depends on steady-state utilisation (Deloitte, 2024).

HOW IT IS USED

Teams model three-year TCO under each mode for their expected workload profile, apply sensitivity analysis on utilisation and data volume, and select the deployment mode — often hybrid — that minimises TCO while meeting non-financial constraints.

WHEN IT IS USED

Infrastructure decisions at programme launch, annual infrastructure review, when cloud spend hits organisational thresholds, and when regulatory or sovereignty constraints require reconsidering data residency.

BY WHOM IT IS USED

CTOs, CFOs, cloud architects, AI infrastructure leads, and procurement teams negotiating multi-year cloud commitments or capital expenditure on GPU clusters.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with simple cloud-first default thinking, this framework forces explicit modelling of steady-state costs. Compared with CAPEX-only IT planning, it surfaces cloud's hidden ratchet effects — egress lock-in, storage growth, reserved-instance commitments. The current repatriation wave (Meta, X, 37signals) validates the framework's predictive power.

RELATED FRAMEWORKS IN THIS COURSE

TCO Iceberg · *Slibrary 14* | **GPU Cost Curve** · *Slibrary 14* | **FinOps Loop** · *Slibrary 14* | **Hidden Cost Categories** · *Slibrary 14* | **Build vs Buy** · *Slibrary 3*

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GPU / Compute Cost Curve

~100× decline in cost per token — the deflationary force behind AI

SLIBRARY 14 · SHEET 3 / 5

PROFESSIONAL DEFINITION

The GPU Cost Curve charts the declining cost of AI training and inference per unit of compute. Between 2022 and 2025, the marginal cost of running large-language-model inference fell from roughly $20 per million tokens to $0.20 — a ~100× drop driven by newer silicon architectures, algorithmic improvements, and competitive pricing among providers. This deflation structurally reshapes the AI business case over time (Epoch AI, 2025; a16z, 2024).

WHO DEVELOPED IT

Empirical compute-cost tracking is maintained by Epoch AI (Epoch AI, 2025), Artificial Analysis (Artificial Analysis, 2025), and Andreessen Horowitz's State of AI reports (a16z, 2024). The theoretical framing of compute-driven deflation aligns with Wright's Law (Wright, 1936) and learning-curve economics.

WHY IT WAS DEVELOPED

Static business cases assume stable input costs, but AI compute prices are falling so fast that projects economically infeasible in year 1 become profitable by year 2. Explicit cost-curve modelling lets firms phase deployment with expected cost decline rather than rushing launches or deferring value indefinitely (Epoch AI, 2025).

FIGURE 1 · 2022–2025 COMPUTE COST DECLINE

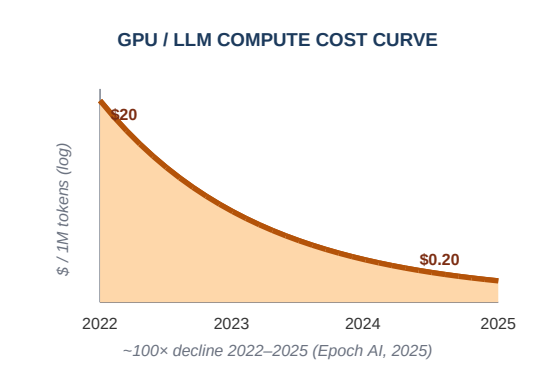

Figure 1. LLM inference cost per million tokens, logarithmic scale. ~100× decline over three years driven by newer silicon, algorithmic efficiency, and pricing competition (Epoch AI, 2025).

HOW IT IS USED

Teams model three scenarios for compute cost decline — conservative, base, aggressive — against use-case unit economics, identifying which use-cases become profitable first and planning a sequenced rollout tied to the cost curve.

WHEN IT IS USED

AI portfolio prioritisation, unit-economics modelling for AI-driven features, pricing decisions for AI-embedded products, and capacity planning for AI inference infrastructure.

BY WHOM IT IS USED

CFOs, AI product managers, pricing strategists, portfolio leaders, and investor relations teams modelling the economic trajectory of AI-enabled products.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with static unit-economics models, the cost-curve view captures a structural deflation that is consistently observed but rarely modelled explicitly. Compared with Moore's Law analogies, the curve is steeper and driven by a stack of factors (silicon, algorithms, competition) rather than a single trend. Firms that model it correctly time launches better.

RELATED FRAMEWORKS IN THIS COURSE

TCO Iceberg · *Slibrary 14* | **Cloud vs On-Prem** · *Slibrary 14* | **Perez S-Curve** · *Slibrary 1* | **Cost-Benefit Waterfall** · *Slibrary 13* | **AI Value Drivers** · *Slibrary 13*

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Hidden Cost Categories

Six under-estimated categories — integration to compliance

PROFESSIONAL DEFINITION

The Hidden Cost Categories framework names the six AI-programme cost lines most frequently underestimated in business cases: integration and APIs; data engineering; change management; retraining and drift; monitoring and operations; and compliance and audit. Deloitte's 2024 survey found 40–85 per cent of AI programmes report each category as 'significantly underestimated' at approval (Deloitte, 2024).

WHO DEVELOPED IT

Synthesised across Deloitte's State of AI annual surveys (Deloitte, 2024), Gartner's TCO research (Gartner, 2024), McKinsey's MLOps cost analyses (McKinsey, 2023), and BCG's AI value-capture research (BCG, 2024). Foundational MLOps cost literature is in Sculley et al. (2015)'s 'Hidden Technical Debt in Machine Learning Systems'.

WHY IT WAS DEVELOPED

Programmes consistently overrun not because compute costs more than expected, but because integration, change, and ops costs are 2–5× what was budgeted. Making each hidden category visible at approval time is the single most reliable way to bring AI programmes in on budget (Deloitte, 2024).

FIGURE 1 · UNDERESTIMATION RATES BY CATEGORY

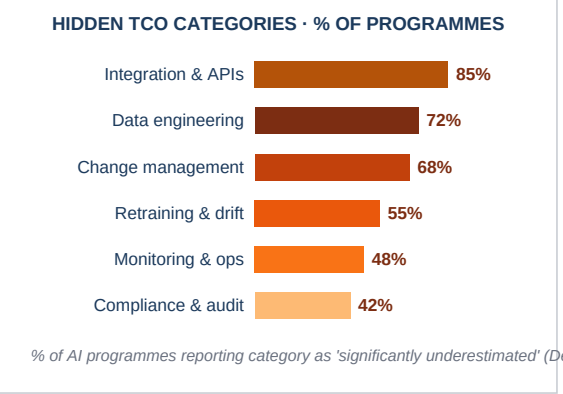

Figure 1. Per cent of AI programmes reporting each category as significantly underestimated at approval. Integration, data engineering, and change management top the list (Deloitte, 2024).

HOW IT IS USED

Business-case templates include explicit line items for each of the six categories, require a named owner and quantitative estimate for each, and check post-approval variance quarterly — converting the list into an active cost-management discipline.

WHEN IT IS USED

Business-case preparation, post-implementation reviews, portfolio health checks, and board-level AI investment oversight — wherever realistic cost estimation is material.

BY WHOM IT IS USED

CFOs, portfolio managers, AI programme leaders, PMO teams, and auditors reviewing AI business-case discipline.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-line 'AI spend' budgets, the six-category decomposition makes each cost a managed line item. Compared with Sculley et al.'s engineering view, this adds organisational categories (change, compliance) that dominate enterprise TCO. The framework turns cost discipline into a check-list rather than a cultural habit.

RELATED FRAMEWORKS IN THIS COURSE

TCO Iceberg · *Slibrary 14* | **Cloud vs On-Prem** · *Slibrary 14* | **FinOps Loop** · *Slibrary 14* | **AI Project Failure Funnel** · *Slibrary 3* | **Cost-Benefit Waterfall** · *Slibrary 13*

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FinOps Loop for AI

Inform · Optimise · Operate · Govern — continuous financial stewardship

PROFESSIONAL DEFINITION

The FinOps Loop is a four-phase operating model for continuous financial management of cloud and AI infrastructure: Inform (cost visibility and attribution), Optimise (rightsizing, reserved pricing, architectural efficiency), Operate (daily cost control in engineering practice), and Govern (policies, budgets, and forecasting). Applied to AI, FinOps adds specific disciplines around GPU utilisation, model efficiency, and inference cost attribution (FinOps Foundation, 2024; 2025).

WHO DEVELOPED IT

Codified by the FinOps Foundation under the Linux Foundation (FinOps Foundation, 2024), with industry adoption led by the hyperscaler cloud providers and major consultancies. The AI-specific adaptation is emerging in 2024–2025 practitioner literature (FinOps Foundation, 2025; McKinsey, 2024).

WHY IT WAS DEVELOPED

Cloud and AI infrastructure costs grow unpredictably and cannot be managed with traditional monthly finance cycles. FinOps embeds cost management in the engineering workflow — costs are visible daily, optimised continuously, and governed by policy rather than reacted to quarterly. Failing to run FinOps is the leading cause of cloud-cost surprises (FinOps Foundation, 2024).

FIGURE 1 · THE FOUR-PHASE CYCLE

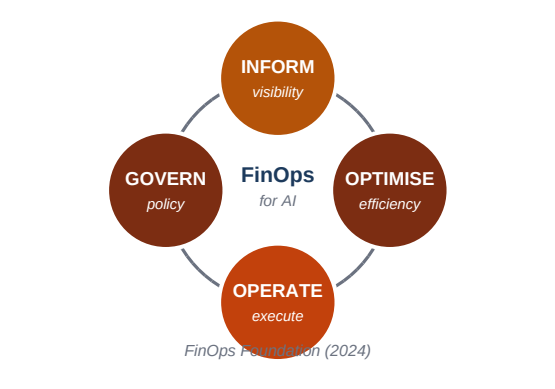

Figure 1. The FinOps loop — Inform, Optimise, Operate, Govern — cycles continuously. Each iteration generates new visibility that feeds the next round of optimisation (FinOps Foundation, 2024).

HOW IT IS USED

Organisations stand up a FinOps function with cross-functional representation, implement tagging and attribution, set monthly or weekly optimisation rhythms, and move AI teams from cost-oblivious to cost-aware engineering practice.

WHEN IT IS USED

When cloud spend exceeds a material threshold (typically >$1M/year), when AI programmes begin production deployment, and when the CFO can no longer explain month-over-month cost variance to the board.

BY WHOM IT IS USED

FinOps practitioners, cloud architects, CFOs, engineering leaders, and the newly emerging Head-of-AI-FinOps role in large enterprises.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with traditional IT budgeting, FinOps operates at the cadence of cloud itself — daily rather than annually. Compared with cost-cutting initiatives, it is continuous rather than episodic. For AI specifically, it is the discipline that closes the gap between engineering decisions (model size, inference frequency) and financial outcomes (the monthly cloud bill).

RELATED FRAMEWORKS IN THIS COURSE

TCO Iceberg · *Slibrary 14* | **Hidden Cost Categories** · *Slibrary 14* | **Cloud vs On-Prem** · *Slibrary 14* | **GPU Cost Curve** · *Slibrary 14* | **MLOps Loop** · *Slibrary 6*

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AI Risk Taxonomy

Four risk families — technical · operational · ethical · compliance

PROFESSIONAL DEFINITION

The AI Risk Taxonomy is a four-family classification of the risks AI systems introduce to an organisation: Technical (model accuracy, robustness, drift), Operational (process disruption, data quality, third-party dependency), Ethical (bias, fairness, privacy), and Compliance (EU AI Act, GDPR, sector-specific rules). It provides the shared vocabulary for risk registers, committee discussions, and audit workpapers across AI programmes (NIST, 2023; European Parliament and Council, 2024; ISO, 2023). Boxes in Figure 1 use abbreviated labels for layout; the full names are Technical, Operational, Ethical, and Compliance.

WHO DEVELOPED IT

Synthesised across the NIST AI Risk Management Framework (NIST, 2023), the EU AI Act risk classification (European Parliament and Council, 2024), ISO/IEC 42001 (ISO, 2023), and academic risk taxonomies from Amodei et al. (2016) on AI safety and Raji et al. (2020) on algorithmic audit.

WHY IT WAS DEVELOPED

Without a shared taxonomy, AI risks are discussed inconsistently — one team debates bias while another debates latency, and the risk committee cannot compare them. The taxonomy forces every risk to be classified, owned, measured, and reported using consistent categories, enabling portfolio-level aggregation (NIST, 2023).

FIGURE 1 · FOUR RISK FAMILIES

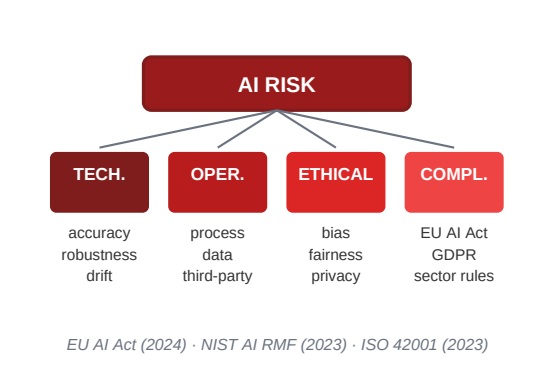

Figure 1. The four-family AI risk taxonomy, with representative sub-categories under each family. Every AI risk maps to exactly one of the four branches for consistent registration (NIST, 2023).

HOW IT IS USED

Risk registers are structured by the four families; each AI use-case is scored on each family at approval and quarterly; aggregated portfolio-level exposure is reported to the board's risk committee.

WHEN IT IS USED

AI use-case approval, ongoing AI portfolio oversight, third-party AI vendor due-diligence, annual risk reports, and regulatory audits under the EU AI Act and sector-specific supervisors.

BY WHOM IT IS USED

Chief Risk Officers, AI governance leads, compliance teams, internal auditors, and boards receiving AI portfolio risk reports.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-dimensional frameworks ('AI is an ethics issue' or 'AI is a technical issue'), the four-family taxonomy gives every stakeholder a seat at the table. Compared with existing ERM taxonomies, it adds the technical and ethical categories that standard risk management lacks vocabulary for. The taxonomy is regulator-compatible: the EU AI Act maps cleanly onto its compliance branch.

RELATED FRAMEWORKS IN THIS COURSE

SR 11-7 MRM · *Slibrary 15* | **Risk Heatmap** · *Slibrary 15* | **Three Lines of Defence** · *Slibrary 15* | **Incident Response** · *Slibrary 15* | **EU AI Act Risk Pyramid** · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

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SR 11-7 Model Risk Management

Development · Implementation · Validation — regulator-grade AI governance

PROFESSIONAL DEFINITION

SR 11-7 is the Federal Reserve Board's Supervisory Letter on Model Risk Management, issued in 2011 and applicable to banks using models for decisions. It defines three pillars — Development (rigorous design), Implementation (use with quality data and clear outcomes), and Validation (independent challenge) — under board oversight. Its 2020s application to AI is now the de-facto template for AI governance in regulated industries globally (Federal Reserve Board, 2011; OCC, 2011; EBA, 2023).

WHO DEVELOPED IT

Issued jointly by the Federal Reserve Board and Office of the Comptroller of the Currency (Federal Reserve Board, 2011; OCC, 2011) as SR 11-7 / OCC 2011-12. The framework has been extended by the European Banking Authority (EBA, 2023) and is cited in the UK PRA's model-risk principles (Bank of England, 2023).

WHY IT WAS DEVELOPED

Banks had deployed models for decades without consistent governance, and the 2008 financial crisis revealed material model failures (VaR, ratings models) that were not detected. SR 11-7 required board oversight of every material model, independent validation, and documented use — a regime now being applied to AI models in banking, insurance, and increasingly in other regulated sectors.

FIGURE 1 · THREE PILLARS UNDER BOARD OVERSIGHT

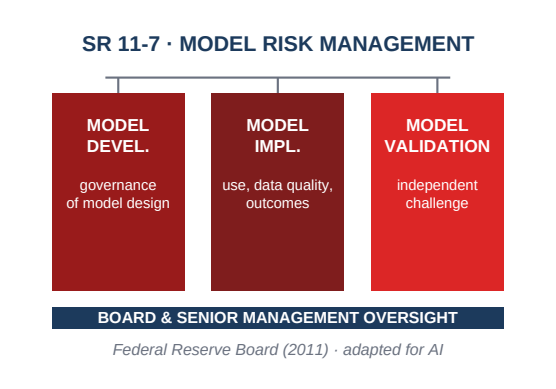

Figure 1. The SR 11-7 architecture. Three pillars — Development, Implementation, Validation — all rest on board and senior-management oversight. Independent validation is the central innovation (Federal Reserve Board, 2011).

HOW IT IS USED

Firms establish a model inventory, assign each AI model a materiality tier, run independent validation for high-tier models, and report to the board on model-risk exposure. The three pillars map directly onto the AI lifecycle (CRISP-DM, Slibrary 3).

WHEN IT IS USED

Banks and insurers deploying AI in credit, market risk, fraud, or capital models; increasingly in healthcare (FDA software-as-medical-device) and in regulated industries under the EU AI Act's high-risk-system obligations.

BY WHOM IT IS USED

Chief Risk Officers, Model Risk Management (MRM) functions, internal audit, and regulators assessing model-governance maturity.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with informal model-review practices, SR 11-7's requirement for independent validation creates a structural challenge mechanism that cannot be bypassed. Compared with governance frameworks that treat AI as purely technical, it embeds AI in a risk-and-board oversight regime. For non-banks adopting the framework, it is the most rigorous off-the-shelf AI governance template available.

RELATED FRAMEWORKS IN THIS COURSE

AI Risk Taxonomy · *Slibrary 15* | **Risk Heatmap** · *Slibrary 15* | **Three Lines of Defence** · *Slibrary 15* | **NIST AI RMF** · *Slibrary 1* | **EU AI Act Risk Pyramid** · *Slibrary 1*

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AI Risk Heatmap

SLIBRARY 15 · SHEET 3 / 5

Probability × impact — the board's single most-used risk visualisation

PROFESSIONAL DEFINITION

The AI Risk Heatmap is a two-dimensional grid that plots each identified risk on axes of probability and impact, colour-coded from low (green) through moderate (amber) to high (red). It is the standard visualisation used at board and risk-committee level to compare heterogeneous risks on a common canvas and to prioritise mitigation effort (COSO, 2017; Hubbard, 2020).

WHO DEVELOPED IT

The heatmap format originated in COSO's Enterprise Risk Management Integrated Framework (COSO, 2004; 2017) and was popularised by risk-consulting firms. Its AI application is covered in Hubbard (2020), NIST's AI RMF (NIST, 2023), and the IIA's Three Lines model for AI (IIA, 2020). Academic critiques by Hubbard (2020) caution against its mathematical limits.

WHY IT WAS DEVELOPED

Boards manage dozens of heterogeneous risks — cyber, regulatory, reputational, model — and need a consistent visual that communicates relative priority in one glance. The heatmap supplies that shared visual, enabling cross-domain comparison and steering the mitigation budget toward the top-right quadrant (high probability × high impact).

FIGURE 1 · 3×3 PROBABILITY × IMPACT GRID

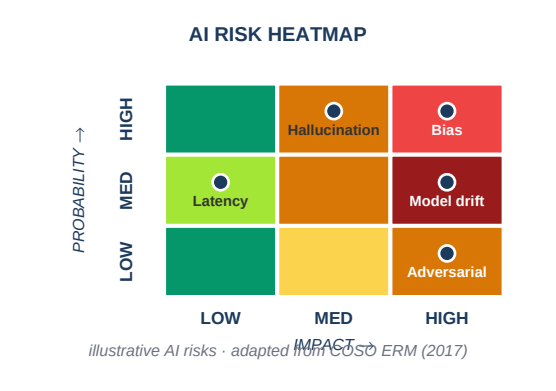

Figure 1. An illustrative AI risk heatmap. Five representative AI risks — bias, model drift, hallucination, latency, and adversarial attack — are placed on the probability × impact grid. Top-right quadrant risks receive priority mitigation (COSO, 2017).

HOW IT IS USED

Teams identify AI risks, score probability and impact on a 1–3 or 1–5 scale, plot the results, and run monthly or quarterly heatmap reviews with a risk committee. The heatmap is overlaid with mitigation status (treated, accepted, transferred) for a fuller view.

WHEN IT IS USED

Board and risk-committee meetings, annual risk reports, AI-portfolio oversight reviews, and regulatory submissions under the EU AI Act's high-risk-system documentation obligations.

BY WHOM IT IS USED

Chief Risk Officers, boards, audit committees, AI governance leads, and compliance teams — anyone who needs to communicate risk exposure to non-technical stakeholders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with risk registers as flat lists, the heatmap visualises priority instantly. Compared with purely quantitative risk models (VaR, loss distributions), it is accessible to boards without statistical training. Hubbard (2020) cautions that the visual can mask quantitative weakness — professional practice combines the heatmap with quantitative support underneath.

RELATED FRAMEWORKS IN THIS COURSE

AI Risk Taxonomy · *Slibrary 15* | **SR 11-7 MRM** · *Slibrary 15* | **Three Lines of Defence** · *Slibrary 15* | **Incident Response** · *Slibrary 15* | **EU AI Act Risk Pyramid** · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

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Three Lines of Defence for AI

Own · oversee · assure — who does what in AI governance

PROFESSIONAL DEFINITION

The Three Lines of Defence is an organisational model for risk governance: the First Line (operational management) owns and manages risk; the Second Line (risk and compliance functions) oversees and challenges; the Third Line (internal audit) provides independent assurance. All three lines operate under board oversight and external audit. Applied to AI, it clarifies who is accountable for what across technical, ethical, and compliance risk (IIA, 2020; BCBS, 2015).

WHO DEVELOPED IT

Codified by the Institute of Internal Auditors (IIA, 2013; updated 2020). Adopted by the Basel Committee on Banking Supervision (BCBS, 2015) and now a de-facto standard in most regulated industries. Applied to AI in the NIST AI RMF (NIST, 2023) and EU AI Act implementation guidance (European Commission, 2024).

WHY IT WAS DEVELOPED

Risk governance often fails because accountability is unclear — product teams think compliance will catch issues; compliance thinks audit will; audit assumes product teams have it covered. The Three Lines makes each role explicit and prevents gap-and-overlap problems in AI risk management (IIA, 2020).

FIGURE 1 · THREE LINES UNDER BOARD OVERSIGHT

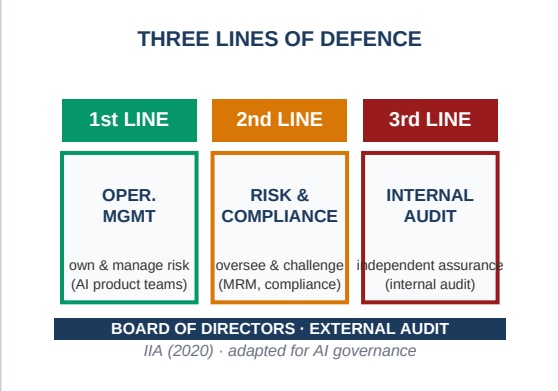

Figure 1. First Line (operational management) owns risk; Second Line (risk & compliance) oversees; Third Line (internal audit) assures. All three operate under board oversight and external audit (IIA, 2020).

HOW IT IS USED

Firms map every AI-risk activity to one of the three lines, check there are no gaps, and ensure the Second and Third Lines have sufficient AI expertise to challenge the First. The Second Line typically hosts the Responsible AI function.

WHEN IT IS USED

When standing up an AI governance operating model, during internal-audit reviews of AI controls, in regulatory examinations under the EU AI Act, and when boards ask the question 'who owns AI risk?'.

BY WHOM IT IS USED

Chief Risk Officers, Heads of Internal Audit, Chief Compliance Officers, AI governance leads, and boards — particularly audit and risk committees.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with informal risk-ownership structures, the Three Lines forces a structural separation between management of risk and assurance over risk. For AI specifically, it surfaces a common gap: the Second and Third Lines often lack the technical capability to challenge AI models meaningfully — addressing this is the current frontier in AI governance staffing.

RELATED FRAMEWORKS IN THIS COURSE

AI Risk Taxonomy · *Slibrary 15* | **SR 11-7 MRM** · *Slibrary 15* | **Risk Heatmap** · *Slibrary 15* | **Incident Response** · *Slibrary 15* | **Responsible AI Scorecard** · *Slibrary 16*

SOURCES (HARVARD REFERENCING STYLE)

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National Institute of Standards and Technology (2023) AI Risk Management Framework (AI RMF 1.0). Gaithersburg, MD: NIST.

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AI Incident Response Loop

Detect · triage · contain · eradicate · recover · learn

PROFESSIONAL DEFINITION

The AI Incident Response Loop adapts the six-phase cybersecurity incident-response cycle — Detect, Triage, Contain, Eradicate, Recover, Learn — to AI-specific failures such as model drift, harmful outputs, hallucinations, bias incidents, data leakage through prompts, and adversarial exploitation. The loop is now required for EU AI Act high-risk systems and is a core element of Responsible AI operations (NIST, 2012; AI Incident Database, 2024; European Parliament and Council, 2024).

WHO DEVELOPED IT

The six-phase structure comes from NIST SP 800-61 Computer Security Incident Handling Guide (NIST, 2012). Its AI-specific application is emerging through the AI Incident Database (AI Incident Database, 2024), Partnership on AI incident reports (PAI, 2024), and the EU AI Act's incident-reporting obligations (European Parliament and Council, 2024).

WHY IT WAS DEVELOPED

AI systems fail in modes unknown in traditional software — a model starts hallucinating, bias surfaces after deployment, an LLM leaks confidential data through a prompt. Incident response adapted from cybersecurity provides a structured, regulator-acceptable way to contain and learn from these incidents rather than reacting ad-hoc under time pressure.

FIGURE 1 · SIX-PHASE RESPONSE CYCLE

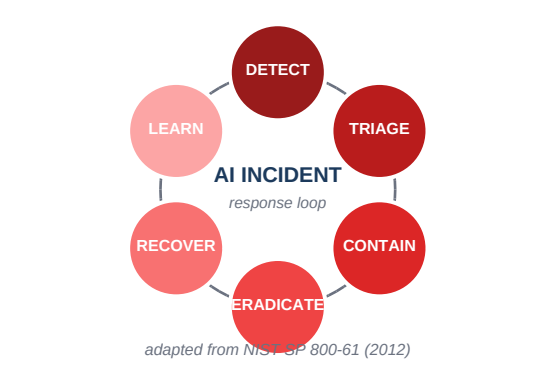

Figure 1. The six-phase AI incident response loop. Each incident cycles through detection, triage, containment, eradication, recovery, and learning, with the learning feeding improvements back into detection (NIST, 2012).

HOW IT IS USED

Firms operationalise an AI incident-response runbook, train response teams, run tabletop exercises, define detection signals per AI system, and establish the regulatory reporting path for serious incidents under the EU AI Act.

WHEN IT IS USED

When an AI system produces an unintended output that causes material harm — financial loss, reputational damage, regulatory breach, or safety issue — and as a preparation discipline before such incidents occur.

BY WHOM IT IS USED

AI operations teams, security incident-response teams (SIRT), Responsible AI officers, legal and communications teams, and regulators receiving mandatory incident reports under the EU AI Act.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with ad-hoc responses to AI failures, the loop provides structure under pressure. Compared with generic IT incident response, it captures AI-specific phases such as model rollback and training-data review. The 'Learn' phase is particularly important for AI: yesterday's incident becomes tomorrow's detection signal only if it is captured and shared across the organisation.

RELATED FRAMEWORKS IN THIS COURSE

AI Risk Taxonomy · *Slibrary 15* | **SR 11-7 MRM** · *Slibrary 15* | **Three Lines of Defence** · *Slibrary 15* | **Risk Heatmap** · *Slibrary 15* | **Model Drift Monitoring** · *Slibrary 8*

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Responsible AI Scorecard

Six dimensions on one radar — fairness to human oversight

PROFESSIONAL DEFINITION

The Responsible AI Scorecard is a six-dimension radar assessment covering fairness, transparency, privacy, security, accountability, and human oversight. Each dimension is scored on a standardised 0–100 scale against defined evidence criteria, and the resulting polygon shows the portfolio or system's ethical posture at a glance. The six dimensions align with the OECD AI Principles (2019) and NIST AI RMF Trustworthy AI Characteristics (NIST, 2023).

WHO DEVELOPED IT

The six-dimension structure is synthesised from the OECD AI Principles (OECD, 2019), the NIST AI Risk Management Framework (NIST, 2023), the EU's Ethics Guidelines for Trustworthy AI (European Commission, 2019), and Floridi et al.'s AI4People framework (Floridi et al., 2018). The radar visualisation format is adapted from BCG's Responsible AI Leader survey (BCG, 2024).

WHY IT WAS DEVELOPED

Responsible AI is often treated as a single concern — 'is this system ethical?' — which collapses distinct issues into one answer. The six-dimension scorecard separates them, enabling organisations to identify specific weaknesses and track improvement over time, with board-level visibility and regulator-compatible evidence (BCG, 2024).

FIGURE 1 · SIX-DIMENSION RADAR

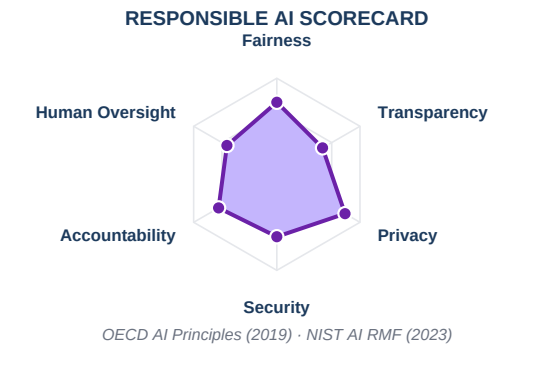

Figure 1. An illustrative Responsible AI scorecard. The purple polygon shows current maturity across the six dimensions; gaps between the polygon and the outer boundary indicate priorities for investment (BCG, 2024).

HOW IT IS USED

Teams score each AI system on the six dimensions using standardised evidence criteria, aggregate to portfolio level, and produce a quarterly scorecard for the AI/Risk Committee. The radar makes weaknesses instantly visible and becomes the anchor visual for board-level Responsible AI reporting.

WHEN IT IS USED

AI portfolio reviews, board-level Responsible AI reports, regulator submissions, investor ESG communication on AI practices, and customer due-diligence responses where AI governance is a procurement criterion.

BY WHOM IT IS USED

Chief Ethics Officers, Responsible AI leads, CDOs, board audit and risk committees, and procurement teams evaluating AI vendor maturity.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with checklist-based ethics reviews, the radar produces a continuous measurable score that can be tracked over time. Compared with narrative-only Responsible AI reporting, the six-dimension structure forces every organisation to address each dimension rather than emphasising only strong areas. It is the single most board-accessible Responsible AI artefact.

RELATED FRAMEWORKS IN THIS COURSE

Bias Audit Pipeline · *Slibrary 16* | **RAI Governance Maturity** · *Slibrary 16* | **Red-Team Loop** · *Slibrary 16* | **Board Oversight Chain** · *Slibrary 16* | **Fairness Metrics** · *Slibrary 8*

SOURCES (HARVARD REFERENCING STYLE)

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Bias Audit Pipeline

Define · measure · diagnose · mitigate · monitor — a five-step discipline

PROFESSIONAL DEFINITION

The Bias Audit Pipeline is a five-step operational discipline for detecting and mitigating unfair disparities in AI outputs across protected groups: Define the protected groups and fairness definition; Measure disparity metrics; Diagnose root cause (data, label, feature, model); Mitigate via data balancing, feature change, or model adjustment; and Monitor ongoing for drift. Each step has standard tooling and produces audit evidence (Raji et al., 2020; Mehrabi et al., 2021; Barocas et al., 2019).

WHO DEVELOPED IT

Formalised in Raji et al.'s Closing the AI Accountability Gap (Raji et al., 2020) and Mehrabi et al.'s Survey on Bias in Machine Learning (Mehrabi et al., 2021). The foundational fairness treatment is in Barocas, Hardt and Narayanan (Barocas et al., 2019). Tooling is emerging through IBM's AI Fairness 360 (IBM, 2018), Microsoft's Fairlearn, and Google's What-If Tool.

WHY IT WAS DEVELOPED

Bias in AI is often treated qualitatively — a concern rather than a measurement. Without a pipeline, teams cannot detect bias at scale, cannot audit mitigation effectiveness, and cannot demonstrate compliance under emerging regulation. The pipeline converts fairness into an engineering discipline with evidence (Raji et al., 2020).

FIGURE 1 · FIVE-STEP BIAS AUDIT FLOW

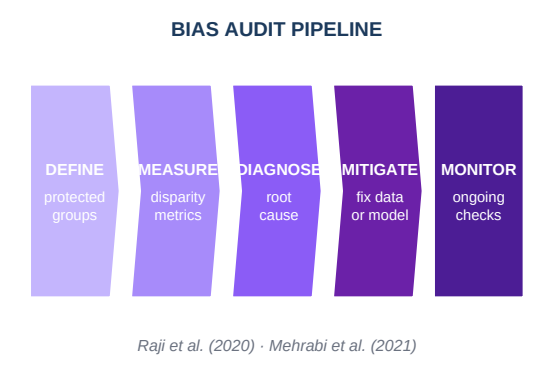

Figure 1. The bias audit pipeline. Each step produces audit evidence and must be completed before the next — Monitor loops back to Measure continuously (Raji et al., 2020).

HOW IT IS USED

Responsible AI teams run the pipeline at model approval and repeat quarterly in production, storing audit artefacts and producing bias-review reports for the Responsible AI function and, where required, external auditors or regulators.

WHEN IT IS USED

AI-system approval under the EU AI Act's high-risk regime, ongoing production monitoring, procurement of third-party AI, and civil rights compliance in employment, lending, housing, insurance, and similar regulated uses.

BY WHOM IT IS USED

Responsible AI engineers, ML engineers, compliance officers, civil rights counsel, and independent algorithmic auditors contracted for third-party review.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with ad-hoc fairness reviews, the pipeline produces repeatable quantitative evidence. Compared with purely academic fairness work, it is production-operable with standard tooling. The pipeline is now the expected practice under the EU AI Act's conformity-assessment regime for high-risk systems.

RELATED FRAMEWORKS IN THIS COURSE

Responsible AI Radar · *Slibrary 16* | **RAI Governance Maturity** · *Slibrary 16* | **Red-Team Loop** · *Slibrary 16* | **Board Oversight Chain** · *Slibrary 16* | **Fairness Metrics** · *Slibrary 8*

SOURCES (HARVARD REFERENCING STYLE)

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RAI Governance Maturity Matrix

SLIBRARY 16 · SHEET 3 / 5

Policy · People · Process · Platform — four dimensions × three levels

PROFESSIONAL DEFINITION

The RAI Governance Maturity Matrix assesses Responsible AI maturity on four dimensions — Policy (rules and principles), People (roles and skills), Process (operations and decision rights), and Platform (tooling and infrastructure) — each graded across three levels: Basic (ad hoc), Managed (consistent), and Optimised (continuously improved). The matrix reveals asymmetric maturity — e.g. strong policy but weak platform — and guides investment priorities (BCG, 2024; Gartner, 2024; McKinsey, 2024).

WHO DEVELOPED IT

Synthesised from BCG's AI Responsibility Index (BCG, 2024), Gartner's AI Governance Maturity Model (Gartner, 2024), and McKinsey's Responsible AI maturity work (McKinsey, 2024). The 4P (Policy-People-Process-Platform) structure is a well-established frame in transformation consulting, and builds on Capability Maturity Model principles (Paulk et al., 1993).

WHY IT WAS DEVELOPED

Responsible AI investment often over-indexes on one dimension — typically Policy (a published statement) — and under-invests in the others. The matrix forces balanced investment and surfaces the dimensions that are most likely to fail first. It also provides a common scale for benchmarking across business units.

FIGURE 1 · FOUR-BY-THREE MATURITY GRID

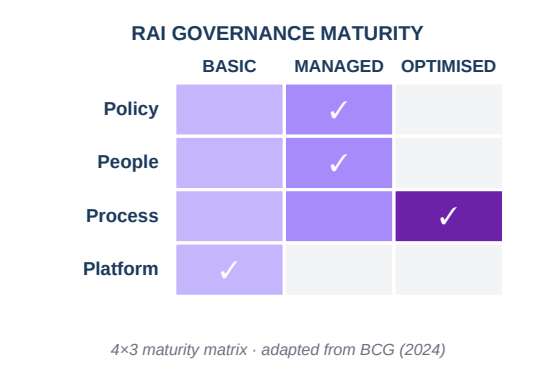

Figure 1. An illustrative maturity assessment. Check marks indicate current level on each dimension. Asymmetry — Process at Optimised while People at Basic — is the common pattern and the priority investment target (BCG, 2024).

HOW IT IS USED

Responsible AI leaders score each dimension using evidence, plot the matrix, and build a multi-year roadmap to raise the lowest-scoring dimension rather than advancing strong dimensions further.

WHEN IT IS USED

Annual Responsible AI strategy reviews, board-level maturity reports, M&A due-diligence on AI-heavy targets, and investor or regulator communication on AI governance strength.

BY WHOM IT IS USED

Responsible AI Officers, AI Governance Leads, Chief Ethics Officers, transformation leaders, and external auditors benchmarking AI maturity.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-score maturity rubrics, the matrix preserves asymmetry information — a 3.0 average can hide a 4/2/3/3 pattern where the 2 is the real risk. Compared with pure policy or pure tooling views, it forces the organisation to address governance as a coherent system rather than a technology deployment.

RELATED FRAMEWORKS IN THIS COURSE

Responsible AI Radar · *Slibrary 16* | **Bias Audit Pipeline** · *Slibrary 16* | **Red-Team Loop** · *Slibrary 16* | **Board Oversight Chain** · *Slibrary 16* | **AI Maturity Staircase** · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

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Red Team / Blue Team Loop

Adversarial probing → patching → re-probing — hardening AI systems

PROFESSIONAL DEFINITION

Red Teaming for AI is a systematic adversarial-probing exercise in which a dedicated Red Team attempts to elicit harmful outputs, bypass safety measures, or extract confidential data from an AI system; a Blue Team then patches the discovered weaknesses; the Red Team re-probes. The loop hardens systems against the real-world attackers who will probe them post-deployment (Perez et al., 2022; Anthropic, 2024; OpenAI, 2023).

WHO DEVELOPED IT

Adapted from cybersecurity red-team practice (Howard and Lipner, 2006) to AI by Perez et al. at DeepMind (Perez et al., 2022), Anthropic's frontier red-team programme (Anthropic, 2024), and OpenAI's Preparedness Framework (OpenAI, 2023). The practice is now an explicit requirement for frontier AI systems under major voluntary commitments and the EU AI Act's general-purpose model obligations.

WHY IT WAS DEVELOPED

Pre-deployment testing on expected inputs cannot find failures that occur only under adversarial prompting. Red teaming deliberately attacks the system with targeted probes — jailbreaks, prompt injection, data extraction — and uncovers failure modes that would otherwise be discovered by bad actors in production. The practice is one of the most cost-effective AI safety measures available.

FIGURE 1 · RED-TEAM / BLUE-TEAM CYCLE

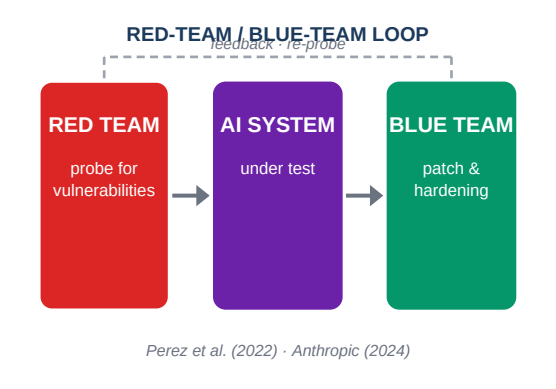

Figure 1. Red Team probes; the AI system either reveals a vulnerability or resists; Blue Team patches; the loop repeats. Feedback from each cycle informs detection for the next (Perez et al., 2022).

HOW IT IS USED

Responsible AI teams run red-team sprints before each major model deployment, document findings, prioritise patches by severity, and integrate the test cases into the model evaluation suite so the same vulnerability cannot regress.

WHEN IT IS USED

Pre-deployment for any user-facing AI system, after material model or prompt-template changes, under EU AI Act conformity assessment for high-risk systems, and as part of frontier-model safety commitments for general-purpose AI.

BY WHOM IT IS USED

Dedicated Red Team specialists, AI safety engineers, security researchers, ethics leads, and external red-team firms contracted for independent probing.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with pure pre-deployment testing, red-teaming attacks the system in the way real adversaries will — adversarial, persistent, and creative. Compared with post-deployment bug bounties, it catches issues before they harm users. The practice is becoming mandatory for frontier AI systems and is a leading indicator of serious AI governance.

RELATED FRAMEWORKS IN THIS COURSE

Responsible AI Radar · *Slibrary 16* | **Bias Audit Pipeline** · *Slibrary 16* | **RAI Governance Maturity** · *Slibrary 16* | **Board Oversight Chain** · *Slibrary 16* | **AI Incident Response** · *Slibrary 15*

SOURCES (HARVARD REFERENCING STYLE)

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Board-to-Operations Oversight

From Board tone to AI team execution — the four-tier chain

PROFESSIONAL DEFINITION

The Board-to-Operations Oversight Chain is a four-tier accountability structure connecting Board-level tone and oversight to operational AI execution: Board (strategy, tone, ultimate accountability) → AI / Risk Committee (policy and challenge) → AI Governance Office (framework, metrics, training) → AI Teams and Product (build and operate). Each tier has defined escalation rights and reporting cadence (OECD, 2019; European Parliament and Council, 2024; BCG, 2024).

WHO DEVELOPED IT

Synthesised from OECD AI Principles governance guidance (OECD, 2019), the EU AI Act's governance requirements (European Parliament and Council, 2024), the BCG AI Responsibility Index governance section (BCG, 2024), and the NACD's board guidance on AI (National Association of Corporate Directors, 2024).

WHY IT WAS DEVELOPED

Responsible AI efforts fail when Board-level statements and operational reality are disconnected — the Board commits to ethical AI but product teams ship without bias review. The oversight chain establishes a direct line from strategic commitment through to operational execution, with each tier accountable for specific outputs to the tier above.

FIGURE 1 · FOUR-TIER OVERSIGHT CHAIN

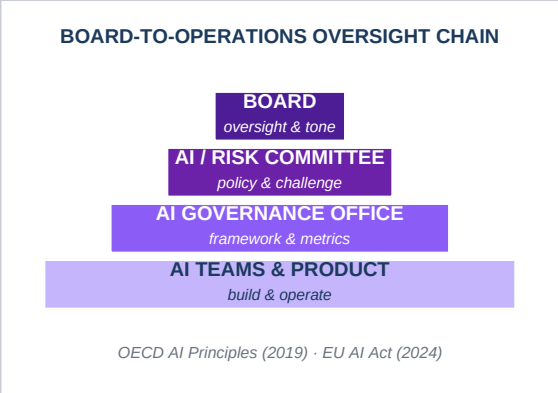

Figure 1. The four-tier oversight chain. Strategic tone flows downward from the Board; operational evidence flows upward through the AI Governance Office and AI / Risk Committee back to the Board (BCG, 2024).

HOW IT IS USED

Firms define membership, rights, and cadence for each tier, establish standard reports that flow upward, and run quarterly or biannual Board-level AI reviews with the Committee and Governance Office presenting a common scorecard.

WHEN IT IS USED

When standing up an AI governance operating model, during Board reviews of AI strategy, for regulator examinations under the EU AI Act, and in investor ESG and corporate governance communications.

BY WHOM IT IS USED

Corporate secretaries, Board members, AI Governance Officers, CROs, CEOs sponsoring AI transformation, and investors or regulators assessing governance quality.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with Board commitments that never translate into operational practice, the chain forces a direct line of accountability. Compared with purely operational AI governance, it embeds Board-level ownership. The structure is becoming explicit under the EU AI Act and is a leading indicator of regulator-ready AI governance.

RELATED FRAMEWORKS IN THIS COURSE

Responsible AI Radar · *Slibrary 16* | **Bias Audit Pipeline** · *Slibrary 16* | **RAI Governance Maturity** · *Slibrary 16* | **Red-Team Loop** · *Slibrary 16* | **Three Lines of Defence** · *Slibrary 15*

SOURCES (HARVARD REFERENCING STYLE)

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Copyright Ownership Decision Tree

SLIBRARY 17 · SHEET 1 / 5

Who owns AI-generated output — human, joint, public domain, or machine?

PROFESSIONAL DEFINITION

The Copyright Ownership Decision Tree maps the primary legal question raised by generative AI: who holds copyright in AI-generated output? The tree branches on the degree of human creative input — substantial human input produces a human-authored work; AI-generated-alone work is currently unprotectable in the US (Thaler v Perlmutter, 2023) and in flux elsewhere. Joint authorship and machine-rights regimes remain legally unsettled (US Copyright Office, 2023; WIPO, 2024).

WHO DEVELOPED IT

The US framing was articulated by the US Copyright Office in its 2023 guidance (US Copyright Office, 2023) and affirmed in Thaler v Perlmutter (2023). The WIPO Standing Committee on Copyright (WIPO, 2024) compiles the global divergence. Academic framing is in Samuelson (2023) and Abbott (2020).

WHY IT WAS DEVELOPED

Without a clear ownership rule, AI-generated content cannot be reliably licensed, transferred, or enforced. The decision tree gives counsel, commercial teams, and content creators a structured way to classify each AI output and to decide whether it is protectable, jointly held, or in the public domain.

FIGURE 1 · OWNERSHIP DECISION TREE

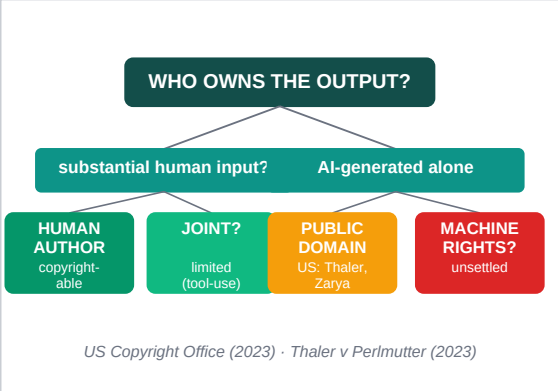

Figure 1. The ownership tree. Substantial human input → human author; tool-level input → joint or limited; AI-generated alone → public domain (US) or unsettled machine-rights regime (US Copyright Office, 2023).

HOW IT IS USED

Legal teams apply the tree to every new class of AI-generated output, document the classification, and choose the appropriate contractual protection — assignment, licence, or open release — based on the outcome.

WHEN IT IS USED

Content generation workflows, IP due-diligence on AI-heavy targets, publisher and broadcaster pipelines using generative AI, and policy submissions to national copyright offices conducting AI-specific consultations.

BY WHOM IT IS USED

General counsel, IP lawyers, content and product leaders, publishers, broadcasters, and policy analysts advising on AI and copyright.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with a single-rule 'AI cannot be an author' statement, the tree surfaces the gradient between pure-tool use and autonomous generation. Compared with purely policy-oriented framings, it is operational: a legal team can run each workflow through it and arrive at a documented decision. The tree also reveals where the law is unsettled — precisely the locations that require contractual safeguards rather than reliance on statute.

RELATED FRAMEWORKS IN THIS COURSE

Four Authorship Models · *Slibrary 17* | **Getty v Stability** · *Slibrary 17* | **Training Data Provenance** · *Slibrary 17* | **Moral Rights** · *Slibrary 17* | **AI Licensing Models** · *Slibrary 18*

SOURCES (HARVARD REFERENCING STYLE)

Abbott, R. (2020) The Reasonable Robot: Artificial Intelligence and the Law. Cambridge: Cambridge University Press.
Samuelson, P. (2023) 'Generative AI Meets Copyright', Science, 381(6654), pp. 158–161.
US Copyright Office (2023) Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence. Washington, DC: USCO.
World Intellectual Property Organization (2024) Intellectual Property and Generative Artificial Intelligence: Issues for Policymakers. Geneva: WIPO.

Four Authorship Models

Tool · Co-author · Autonomous AI · Creative Commons

PROFESSIONAL DEFINITION

The Four Authorship Models taxonomy names the distinct ways copyright authorship can be constituted around AI-generated work: the Tool Model (human uses AI as a tool, human owns the output), Co-author Model (shared contribution, joint but contested ownership), Autonomous AI Model (AI produces alone, unprotectable in the US), and Creative Commons Model (voluntary open licence on either inputs or outputs). Each carries different commercial and legal implications (US Copyright Office, 2023; WIPO, 2024; Creative Commons, 2023).

WHO DEVELOPED IT

Synthesised from the US Copyright Office's 2023 guidance (US Copyright Office, 2023), the WIPO Secretariat's 2024 comparative study (WIPO, 2024), Creative Commons' generative-AI policy statements (Creative Commons, 2023), and academic treatments in Abbott (2020) and Samuelson (2023).

WHY IT WAS DEVELOPED

Organisations deploying AI content pipelines need a vocabulary that distinguishes between the fundamentally different legal treatments of each scenario. The four models give commercial, legal, and product teams a shared language — an output under Tool Model is legally different from one under Autonomous AI Model, even if the user experience is identical.

FIGURE 1 · FOUR-QUADRANT MODEL MAP

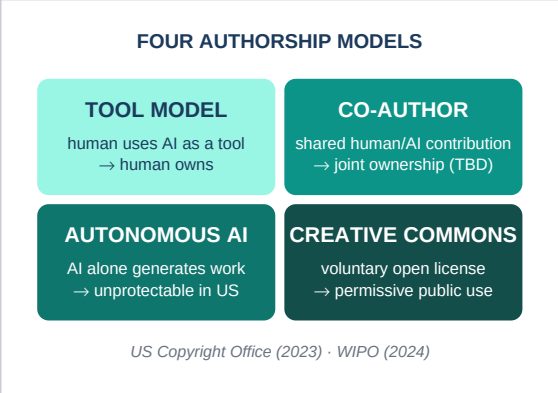

Figure 1. The four authorship models. Tool Model (top-left, human owns) is the dominant commercial pattern; Autonomous AI (bottom-left, unprotectable) is the rising edge case; Creative Commons (bottom-right) is the voluntary opt-out route (US Copyright Office, 2023).

HOW IT IS USED

Content and product teams classify each AI workflow against the four models, attach the correct terms-of-service language, and set downstream commercial policies — reuse, licensing, claims — consistent with the legal treatment.

WHEN IT IS USED

Terms-of-service drafting for AI products, content licensing negotiations, editor-and-journalist AI-use policies, and disclosure statements required under emerging AI transparency regulation.

BY WHOM IT IS USED

General counsel, content licensing teams, product and platform policy leaders, and editorial-standards managers at publishers using AI in production.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with binary 'AI-made or human-made' labelling, the four-model taxonomy covers the realistic spectrum of production workflows. Compared with case-by-case legal opinions, it produces consistent, policy-level classifications. For investors, it also clarifies what an AI-content asset actually is in portfolio terms.

RELATED FRAMEWORKS IN THIS COURSE

Copyright Ownership Tree · *Slibrary 17* | **Getty v Stability** · *Slibrary 17* | **Moral Rights** · *Slibrary 17* | **AI Licensing Models** · *Slibrary 18* | **Monetisation Models** · *Slibrary 20*

SOURCES (HARVARD REFERENCING STYLE)

Abbott, R. (2020) The Reasonable Robot: Artificial Intelligence and the Law. Cambridge: Cambridge University Press.
Creative Commons (2023) Understanding CC Licenses and Generative AI. San Francisco, CA: Creative Commons.
US Copyright Office (2023) Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence. Washington, DC: USCO.
World Intellectual Property Organization (2024) Intellectual Property and Generative Artificial Intelligence: Issues for Policymakers. Geneva: WIPO.

Getty Images v Stability AI

The industry's defining training-data copyright case

PROFESSIONAL DEFINITION

Getty Images v Stability AI is the landmark multi-jurisdiction copyright case brought by Getty in the US and UK (2023) alleging that Stability AI scraped approximately 12 million Getty-licensed images to train Stable Diffusion without authorisation, constituting copyright infringement and database-right violation. The case is a live test of the legal permissibility of training on copyrighted material at scale, with implications across the generative-AI industry (Getty Images v Stability AI Inc. [2025] EWHC 2863 (Ch)).

WHO DEVELOPED IT

Case filed by Getty Images Inc. in the US District Court for the District of Delaware (January 2023) and in the High Court of Justice (Chancery Division), England and Wales. Defended by Stability AI Ltd. Academic and industry coverage has been extensive, including analyses by the Stanford HAI Policy Center (Stanford HAI, 2024) and the Future of Life Institute (FLI, 2024).

WHY IT WAS DEVELOPED

The case tests the core legal question underlying all generative-AI training: is scraping copyrighted works to train a model 'fair use' (US) or within a relevant text-and-data-mining exception (UK / EU)? The answer determines the commercial viability of current training practices across every frontier model, and therefore the valuations and investment flows of the entire AI industry.

FIGURE 1 · CASE TIMELINE

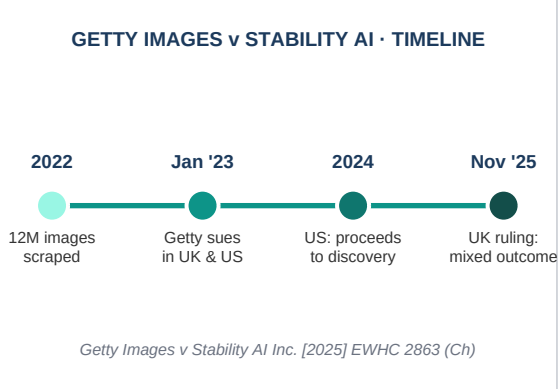

Figure 1. Key milestones in Getty v Stability: 2022 scraping; January 2023 filings in UK and US; 2024 discovery phase in US; November 2025 mixed ruling in the UK High Court (Getty Images v Stability AI Inc. [2025] EWHC 2863 (Ch)).

HOW IT IS USED

Investors, general counsel, and strategy leaders use the case as a reference point for modelling training-data legal risk, stress-testing AI business-case assumptions, and engaging in content-licensing negotiations that price the underlying legal uncertainty.

WHEN IT IS USED

AI investment due-diligence, content licensing negotiations between publishers and AI labs (New York Times v OpenAI, NYT v Microsoft), regulatory submissions on AI and copyright, and policy debates on text-and-data-mining exceptions.

BY WHOM IT IS USED

IP litigators, general counsel at AI labs and content companies, policy advisors, investors with exposure to training-data-sensitive investments, and regulators drafting the AI-and-copyright sections of new legislation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Getty v Stability is the first industry-scale case that forces a court to rule directly on training-data copyright — previous AI cases addressed output, not training. The case establishes evidentiary standards (what counts as proof of scraping) and clarifies the interplay between copyright and sui generis database rights in the EU/UK context.

RELATED FRAMEWORKS IN THIS COURSE

Copyright Ownership Tree · *Slibrary 17* | **Four Authorship Models** · *Slibrary 17* | **Training Data Provenance** · *Slibrary 17* | **Moral Rights** · *Slibrary 17* | **Data IP Protection** · *Slibrary 19*

SOURCES (HARVARD REFERENCING STYLE)

Future of Life Institute (2024) AI Copyright Litigation Tracker. Cambridge, MA: FLI.
Getty Images v Stability AI Inc. [2025] EWHC 2863 (Ch). London: High Court of Justice.
Stanford HAI (2024) The AI Copyright Landscape. Stanford, CA: Stanford Human-Centered AI Institute.
US Copyright Office (2023) Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence. Washington, DC: USCO.

Training Data Provenance Chain

SLIBRARY 17 · SHEET 4 / 5

Source → Collection → Training → Model → Output — the audit trail

PROFESSIONAL DEFINITION

The Training Data Provenance Chain traces the lineage of content from original source through collection, training, model weights, and finally generated output. Each stage requires documentation — source URL, collection terms, training-set inclusion flag, and output correlation — to enable rights-holder queries, licensing audits, and compliance under the EU AI Act's copyright-disclosure obligations (European Parliament and Council, 2024; C2PA, 2024).

WHO DEVELOPED IT

The Coalition for Content Provenance and Authenticity (C2PA, 2024) has codified content-provenance standards adopted by Adobe, Microsoft, BBC, and others. The EU AI Act's Article 53 (European Parliament and Council, 2024) obligates general-purpose-AI providers to publish training-data summaries. Academic grounding is in Gebru et al.'s Datasheets for Datasets (Gebru et al., 2021).

WHY IT WAS DEVELOPED

Without provenance, rights-holders cannot detect use of their work in training, firms cannot demonstrate compliance, and insurers cannot price training-data legal risk. The chain supplies the audit trail that connects rights claims to model behaviour, making licensing markets and enforcement actions operable (C2PA, 2024).

FIGURE 1 · FIVE-STAGE CHAIN WITH AUDIT TRAIL

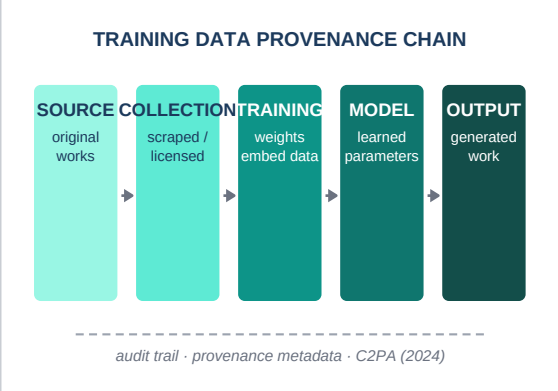

Figure 1. The provenance chain. Each stage produces metadata that feeds a continuous audit trail — the C2PA standard provides the technical format (C2PA, 2024).

HOW IT IS USED

AI labs implement provenance capture at each stage, publish training-data summaries under the EU AI Act, and respond to rights-holder queries with documented inclusion or exclusion. Content firms use the chain to verify that their licences cover downstream AI training.

WHEN IT IS USED

EU AI Act general-purpose-AI provider compliance, content licensing negotiations between AI labs and publishers, newsroom authentication of AI-assisted journalism, and insurance underwriting for training-data legal risk.

BY WHOM IT IS USED

ML engineers, data-engineering teams, general counsel, content rights officers at publishers and stock agencies, and regulatory teams preparing AI Act transparency reports.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with opaque training pipelines, provenance forces documentation that is auditable by rights-holders and regulators. Compared with after-the-fact disclosure, it builds the audit trail at capture time — when it is still cheap. C2PA turns a legal obligation into a technical standard that can be adopted across the supply chain.

RELATED FRAMEWORKS IN THIS COURSE

Copyright Ownership Tree · *Slibrary 17* | **Getty v Stability** · *Slibrary 17* | **Four Authorship Models** · *Slibrary 17* | **Data IP Protection** · *Slibrary 19* | **EU AI Act Risk Pyramid** · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

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European Parliament and Council (2024) Regulation (EU) 2024/1689 on AI — Article 53 Transparency Obligations. Brussels: European Union.
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National Institute of Standards and Technology (2023) AI Risk Management Framework (AI RMF 1.0). Gaithersburg, MD: NIST.

Moral Rights & Attribution

SLIBRARY 17 · SHEET 5 / 5

Attribution · Integrity · Disclosure · Withdrawal — the author's personal rights

PROFESSIONAL DEFINITION

Moral Rights are the non-economic, personal rights of an author under copyright: Attribution (right to be named as author), Integrity (right to prevent distortion or modification that harms reputation), Disclosure (right to decide when and how the work is published), and Withdrawal (right to remove the work from circulation). They are strong in civil-law jurisdictions (Germany, France, Spain), limited in common-law (US, UK), and newly relevant where AI systems reproduce or imitate named authors (Berne Convention Art. 6bis; WIPO, 2024).

WHO DEVELOPED IT

Moral rights were codified in the Berne Convention's Article 6bis (1928, revised 1971) and are implemented with varying strength across national copyright laws (WIPO, 1971/2024). The German Urhebergesetz treats them as inalienable; the US Visual Artists Rights Act (VARA, 1990) covers only visual art. Ginsburg (2020) and Rigamonti (2007) are authoritative comparative treatments.

WHY IT WAS DEVELOPED

AI systems now routinely reproduce recognisable styles of named authors — 'in the style of X' — raising attribution and integrity questions that are orthogonal to classical infringement analysis. Moral rights provide a framework distinct from economic rights, preserving authorial personality against AI-driven reproduction even when economic licensing is resolved.

FIGURE 1 · FOUR PILLARS OF MORAL RIGHTS

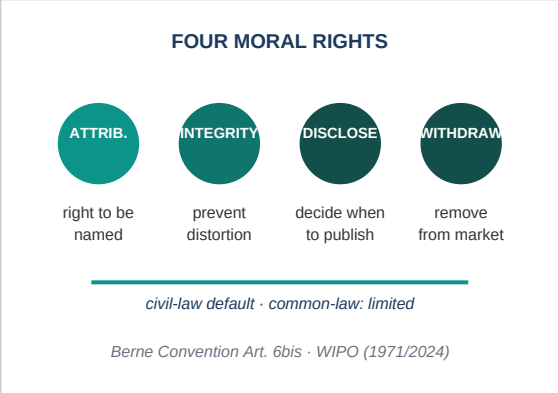

Figure 1. The four moral-rights pillars: Attribution, Integrity, Disclosure, and Withdrawal. All four are default rights in civil-law systems and weaker or absent in most common-law systems (Berne Convention Art. 6bis).

HOW IT IS USED

IP counsel advises authors and AI platforms on attribution and integrity claims, drafts terms-of-service that either respect or contractually waive moral rights (where jurisdictions permit), and runs style-imitation risk analyses for systems trained on named authors.

WHEN IT IS USED

Author disputes over AI-style reproduction, platform terms-of-service drafting, licensing negotiations with named creators, and compliance reviews for AI products deployed in civil-law jurisdictions where moral-rights enforcement is stronger.

BY WHOM IT IS USED

IP and media lawyers, creator rights organisations (SACEM, GEMA, BUMA/STEMRA), platform policy leaders, and AI-product managers designing style-imitation or voice-cloning features.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with economic-rights-only analysis, moral rights capture the personal harm caused by AI imitation of named creators. Compared with purely US-law framings, the Berne-Convention view surfaces the civil-law jurisdictions where AI firms face the strongest constraints. For global platforms, moral rights are the constraint that is hardest to contractually engineer around.

RELATED FRAMEWORKS IN THIS COURSE

Copyright Ownership Tree · *Slibrary 17* | **Four Authorship Models** · *Slibrary 17* | **Getty v Stability** · *Slibrary 17* | **Training Data Provenance** · *Slibrary 17* | **AI Licensing Models** · *Slibrary 18*

SOURCES (HARVARD REFERENCING STYLE)

Ginsburg, J.C. (2020) 'Moral Rights in the Age of Digital Distribution', in Copyright Law in an Age of Limitations and Exceptions. Cambridge: Cambridge University Press, pp. 256–288.

Rigamonti, C.P. (2007) 'The Conceptual Transformation of Moral Rights', American Journal of Comparative Law, 55(1), pp. 67–122.

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World Intellectual Property Organization (2024) Intellectual Property and Generative Artificial Intelligence: Issues for Policymakers. Geneva: WIPO.

Open-Closed AI Spectrum

From open-weights-and-data to fully closed — the strategic licensing choice

PROFESSIONAL DEFINITION

The Open-Closed AI Spectrum classifies AI systems along a continuum of openness: fully open (weights plus training data released), weights-available (weights only, no training data), API access (the model is callable but not downloadable), API with enterprise terms (API plus commercial controls), and fully closed (internal proprietary). Each position implies different licensing, commercial, and risk trade-offs (Widder et al., 2023; Open Source Initiative, 2024).

WHO DEVELOPED IT

The spectrum was articulated by Widder, West and Whittaker (Widder et al., 2023) at the AI Now Institute and extended by the Open Source Initiative's definition of Open Source AI (OSI, 2024). Industry perspectives are documented by Andreessen Horowitz (a16z, 2024) and the Mozilla Foundation (Mozilla, 2024).

WHY IT WAS DEVELOPED

Firms debating 'should we open-source our model?' tend to collapse the question into a binary, when in practice there are five distinct positions with materially different effects on adoption, competitive moats, and regulatory exposure. The spectrum forces explicit positioning and supports strategic consistency over time.

FIGURE 1 · FIVE POINTS ON THE SPECTRUM

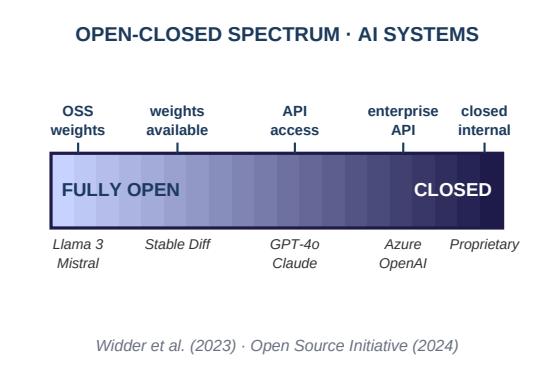

Figure 1. The open-closed spectrum with representative model examples. Fully open (Llama 3, Mistral) on the left; fully closed on the right. Enterprise-API with contractual controls is the dominant commercial pattern in 2025 (Widder et al., 2023).

HOW IT IS USED

Strategy teams position their model explicitly on the spectrum, check alignment with commercial objectives and competitive moats, and review the position annually against platform, community, and regulatory developments.

WHEN IT IS USED

Model-release decisions at AI labs, partner negotiations over model access, policy submissions on open-source AI exemptions under the EU AI Act, and investor communication on business-model defensibility.

BY WHOM IT IS USED

AI lab leadership, CTOs, product strategy leaders, policy advisors, and investors assessing AI business-model defensibility.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with binary 'open vs closed' framing, the spectrum captures the intermediate positions that dominate commercial practice. Compared with licence-text-first analysis, it foregrounds the strategic commitment; the licence is a consequence of position, not the position itself.

RELATED FRAMEWORKS IN THIS COURSE

AI Licence Types · *Slibrary 18* | **API Tiering** · *Slibrary 18* | **Revenue Share** · *Slibrary 18* | **IP Risk Allocation** · *Slibrary 18* | **Monetisation Canvas** · *Slibrary 20*

SOURCES (HARVARD REFERENCING STYLE)

Andreessen Horowitz (2024) State of Open-Source AI. Menlo Park, CA: a16z.
Mozilla Foundation (2024) Accelerating Progress Toward Trustworthy AI. San Francisco, CA: Mozilla.
Open Source Initiative (2024) The Open Source AI Definition 1.0. Palo Alto, CA: OSI.
Widder, D.G., West, S. and Whittaker, M. (2023) 'Open (For Business): Big Tech, Concentrated Power, and the Political Economy of Open AI', SSRN Working Paper.

AI Licence Types

Permissive · Copyleft · OpenRAIL · Proprietary — four commercial regimes

PROFESSIONAL DEFINITION

AI licences fall into four commercial regimes: Permissive (MIT, Apache-2.0, BSD — allow use, modification, and commercialisation with attribution), Copyleft (GPL, AGPL — require derivatives to use the same licence), OpenRAIL (open release with explicit behavioural use-restrictions), and Proprietary (EULA/ToS, licensor-defined, revocable). Selecting the right regime is a strategic question with multi-year consequences (OSI, 2024; BigScience, 2022).

WHO DEVELOPED IT

Permissive and copyleft originated in the OSS tradition (Raymond, 1999; Stallman, 2002) and are codified by the Open Source Initiative (OSI, 2024). OpenRAIL was developed by the BigScience workshop in 2022 (BigScience, 2022) for AI-specific ethical-use restrictions. Proprietary licensing is covered in Nimmer on Copyright and standard contract law.

WHY IT WAS DEVELOPED

Generic OSS licences were designed for software, not trained models, and do not address AI-specific concerns (use restrictions, training-data provenance, model weights distribution). OpenRAIL was designed to bridge that gap, and the four-licence taxonomy gives firms a structured choice between openness and control.

FIGURE 1 · FOUR LICENCE REGIMES

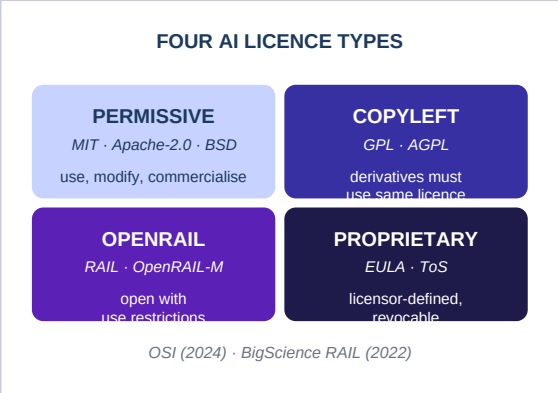

Figure 1. The four licence types. Permissive maximises adoption; Copyleft ensures derivative openness; OpenRAIL adds ethical use restrictions to openness; Proprietary maximises control (OSI, 2024; BigScience, 2022).

HOW IT IS USED

Legal and product teams match each AI asset — dataset, weights, fine-tune, wrapper service — to an appropriate licence type, maintain a portfolio view of licence commitments, and revisit choices when business model or regulatory environment changes.

WHEN IT IS USED

Model-release decisions, open-source contribution reviews, contract drafting for AI vendor agreements, and compliance checks on downstream use of licensed third-party models.

BY WHOM IT IS USED

Open-source programme offices, legal counsel, product leaders, CTOs, and developer-relations teams managing community engagement.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with applying OSS licences directly to AI artefacts, this taxonomy recognises that models are legally different from source code. Compared with purely proprietary ToS approaches, it maps the middle-ground licences (OpenRAIL) that are becoming standard for frontier-model releases (Llama, BLOOM, Falcon).

RELATED FRAMEWORKS IN THIS COURSE

Open-Closed Spectrum · *Slibrary 18* | **API Tiering** · *Slibrary 18* | **Revenue Share** · *Slibrary 18* | **IP Risk Allocation** · *Slibrary 18* | **Four Authorship Models** · *Slibrary 17*

SOURCES (HARVARD REFERENCING STYLE)

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Open Source Initiative (2024) The Open Source AI Definition 1.0. Palo Alto, CA: OSI.
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Stallman, R. (2002) Free Software, Free Society: Selected Essays of Richard M. Stallman. Boston, MA: GNU Press.

API Tiering Model

SLIBRARY 18 · SHEET 3 / 5

Free · Developer · Business · Enterprise — monetising API access

PROFESSIONAL DEFINITION

The API Tiering Model is the standard four-tier pricing structure used by frontier AI providers: Free (capped usage, rate-limited), Developer (pay-as-you-go for prototyping), Business (production SLAs and rate limits), and Enterprise (custom SLAs, dedicated capacity, data-residency controls). Each tier captures a different customer willingness-to-pay and serves a distinct funnel stage (OpenAI, 2024; Anthropic, 2024; a16z, 2024).

WHO DEVELOPED IT

The four-tier structure was effectively standardised by OpenAI (OpenAI, 2024), Anthropic (Anthropic, 2024), and Google (Google, 2024) during 2023–2024. Andreessen Horowitz documented the economics in their 2024 State-of-Generative-AI reports (a16z, 2024). The underlying SaaS logic draws on freemium pricing research (Kumar, 2014).

WHY IT WAS DEVELOPED

AI-inference workloads vary 1000× between casual developer exploration and large-scale production. A single pricing tier cannot serve both; differentiated tiering lets the provider capture value across the spectrum while using the free tier to drive developer adoption and the enterprise tier to protect margins.

FIGURE 1 · FOUR-TIER PYRAMID

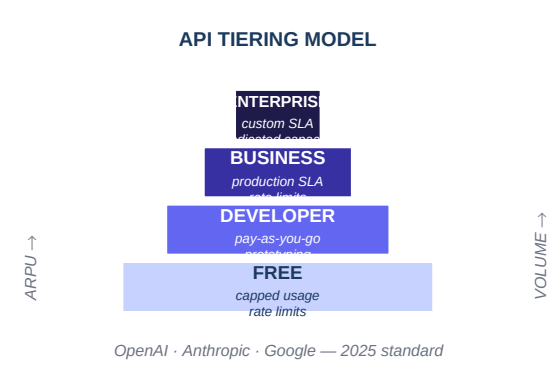

Figure 1. The four-tier pyramid. Volume (width) falls as you ascend; ARPU rises sharply. Free tier drives adoption; Enterprise tier drives profit (a16z, 2024).

HOW IT IS USED

Product leaders model customer flow through the tiers, tune conversion mechanics between Free→Developer and Developer→Business, and negotiate Enterprise contracts individually. Tier design is revisited quarterly as usage patterns evolve.

WHEN IT IS USED

When launching a new AI product or API, during pricing reviews, in partner negotiations, and in investor communications about the path to profitability for AI-inference businesses.

BY WHOM IT IS USED

AI product and pricing teams, revenue operations, enterprise sales leadership, and finance teams modelling AI-inference unit economics.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-price API access, the four-tier pyramid captures both volume and ARPU. Compared with traditional SaaS tiering, it accommodates the extreme usage variance and compute costs of AI workloads. The structure has become the implicit industry standard within two years of the generative-AI boom.

RELATED FRAMEWORKS IN THIS COURSE

Open-Closed Spectrum · *Slibrary 18* | **AI Licence Types** · *Slibrary 18* | **Revenue Share** · *Slibrary 18* | **IP Risk Allocation** · *Slibrary 18* | **Monetisation Canvas** · *Slibrary 20*

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OpenAI (2024) API Pricing and Tier Documentation. San Francisco, CA: OpenAI.

AI Revenue-Share Models

How value flows between platform, creator, and partner

PROFESSIONAL DEFINITION

AI Revenue-Share Models define how subscription, API, or per-use revenue is split among the platform (operator of the AI service), the creator (producer of content, data, or fine-tuned model), and the partner (distribution or ecosystem contributor). Typical splits are 50/30/20 but vary widely with bargaining power and category; the model determines incentives for participation in AI marketplaces (Andreessen Horowitz, 2024; Rochet and Tirole, 2003).

WHO DEVELOPED IT

Commercial practice by Apple (App Store 30/70), Epic Games, Shopify, and AWS Marketplace established the pattern; the AI-specific version is documented in Andreessen Horowitz's 2024 marketplace analyses (a16z, 2024). Theoretical grounding in two-sided market economics is in Rochet and Tirole (2003) and Parker, Van Alstyne and Choudary (Parker et al., 2016).

WHY IT WAS DEVELOPED

AI marketplaces — model hubs, agent stores, data exchanges — cannot function without a clear revenue-share rule. The platform wants to maximise network growth; the creator wants predictable earnings; the partner wants distribution economics. The model makes these trade-offs explicit and negotiable.

FIGURE 1 · THREE-WAY REVENUE SPLIT

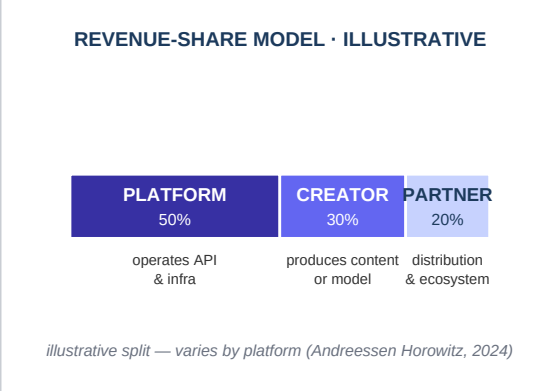

Figure 1. An illustrative 50/30/20 split among Platform (operates infrastructure), Creator (produces content or model), and Partner (distribution ecosystem). Actual splits vary by platform (a16z, 2024).

HOW IT IS USED

Platform strategists model revenue splits against creator-activation targets, benchmark against comparable platforms, and tune the split to sustain ecosystem growth while maintaining platform unit economics.

WHEN IT IS USED

AI marketplace launches, creator-partner negotiations, ecosystem strategy reviews, and regulatory submissions under Digital Markets Act self-preferencing rules.

BY WHOM IT IS USED

Platform strategy leaders, marketplace product managers, ecosystem partnership teams, and regulators examining AI marketplace competitive dynamics.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with uniform pricing, revenue-share models align incentives across three parties simultaneously. Compared with static splits, modern AI marketplaces use dynamic splits that adjust with scale — a pattern that Rochet-Tirole theory predicts for two-sided markets.

RELATED FRAMEWORKS IN THIS COURSE

Open-Closed Spectrum · *Slibrary 18* | **AI Licence Types** · *Slibrary 18* | **API Tiering** · *Slibrary 18* | **IP Risk Allocation** · *Slibrary 18* | **Monetisation Canvas** · *Slibrary 20*

SOURCES (HARVARD REFERENCING STYLE)

Andreessen Horowitz (2024) State of Generative AI: Enterprise. Menlo Park, CA: a16z.

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IP Risk Allocation in AI Contracts

Who bears training-data, output, trade-secret, and patent risk?

PROFESSIONAL DEFINITION

IP Risk Allocation is the contract-level decision about which party — vendor, customer, or both jointly — bears the legal cost of each category of AI-related IP risk: training-data copyright claims, output infringement, trade-secret leakage through prompts, and third-party patent exposure. Typical allocations are indemnities and caps that convert legal uncertainty into contractual risk transfer (IAPP, 2024; Gartner, 2024).

WHO DEVELOPED IT

Commercial practice is compiled by the International Association of Privacy Professionals (IAPP, 2024) and Gartner's contract-research team (Gartner, 2024). The underlying legal structures draw on classical IP indemnification clauses (Farnsworth, 2004) and vendor-contract risk allocation principles (Cummins and Venard, 2009).

WHY IT WAS DEVELOPED

AI contracts without explicit IP allocation leave the loss-bearing party undefined, producing disputes when real infringement claims arrive. Explicit allocation is the single most effective contractual tool for pricing and managing AI IP risk and is now standard practice in enterprise AI procurement.

FIGURE 1 · ILLUSTRATIVE ALLOCATION MATRIX

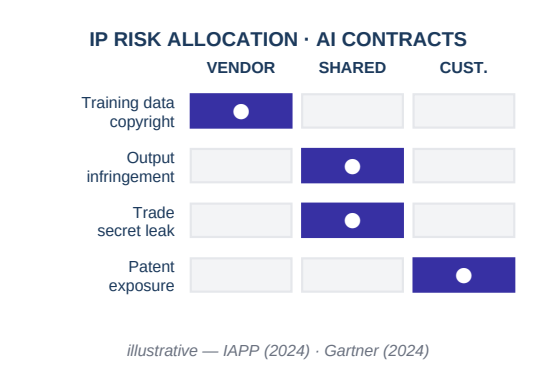

Figure 1. Illustrative allocation of four IP risk categories between vendor, shared, and customer. Market patterns show vendor bearing training-data risk, customer bearing output-infringement and patent risk, shared on trade secrets (IAPP, 2024).

HOW IT IS USED

General counsel and procurement negotiate explicit indemnities, caps, and warranties for each risk category, use standardised contract templates (e.g. Microsoft Copyright Commitment, Adobe Firefly indemnity) as benchmarks, and price the residual risk into the commercial terms.

WHEN IT IS USED

Enterprise AI procurement (Azure OpenAI, AWS Bedrock, Google Vertex), vendor contract renewals, and M&A due-diligence on AI-heavy targets where indemnity profiles determine valuation.

BY WHOM IT IS USED

General counsel, procurement teams, CIOs, insurance officers underwriting AI-IP risk, and M&A advisors evaluating AI business valuations.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with undifferentiated indemnity clauses, the matrix enforces explicit category-level allocation — recognising that the four IP risk types have fundamentally different underlying economics. Compared with purely insurance-based risk transfer, contractual allocation operates before any claim is filed.

RELATED FRAMEWORKS IN THIS COURSE

Open-Closed Spectrum · *Slibrary 18* | **AI Licence Types** · *Slibrary 18* | **API Tiering** · *Slibrary 18* | **Revenue Share** · *Slibrary 18* | **AI Risk Taxonomy** · *Slibrary 15*

SOURCES (HARVARD REFERENCING STYLE)

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Data Rights Stack

Ownership · Access · Use · Derived — the four layers of data rights

PROFESSIONAL DEFINITION

The Data Rights Stack is a four-layer decomposition of the rights associated with a dataset: Ownership (legal title), Access (who can see or query), Use (what can be done — analyse, train, commercialise), and Derived (rights in outputs produced from the data). Each layer can be allocated independently via contract, producing nuanced data-rights regimes that no single legal instrument cleanly captures (OECD, 2023; WIPO, 2024).

WHO DEVELOPED IT

Articulated in the OECD's Recommendation on Enhancing Access to and Sharing of Data (OECD, 2023) and in WIPO's IP-and-AI policy work (WIPO, 2024). The commercial practice is documented by the Data Governance Act implementation guidance (European Commission, 2024) and academic treatments by Purtova (2018) and Determann (2018) on data-as-asset classification.

WHY IT WAS DEVELOPED

Traditional IP categories (copyright, patent, trade secret) do not fit data cleanly — a dataset can be copyrightable as a selection, trade-secret-protectable in content, contractually restricted in access, and public-domain in individual records. The stack provides a structure for allocating these dimensions separately, which is essential for AI training-data contracts.

FIGURE 1 · FOUR-LAYER STACK

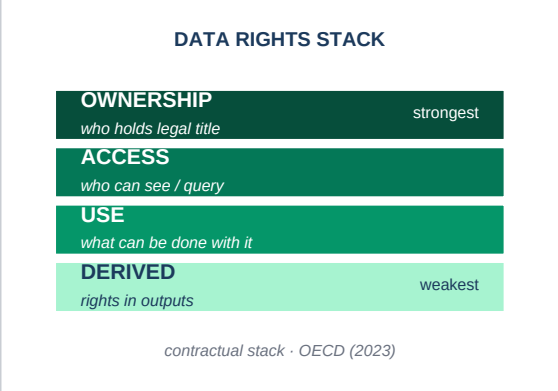

Figure 1. Ownership is the strongest right; Derived rights are the weakest. Each layer can be allocated independently via contract, producing fine-grained data-rights regimes (OECD, 2023).

HOW IT IS USED

Legal and commercial teams negotiate each layer separately in data-sharing contracts: who owns, who can access, for which uses, with what rights in derivatives. The stack makes it explicit where each layer sits and prevents ambiguity that would otherwise surface during AI training.

WHEN IT IS USED

Data-sharing agreements between enterprises and AI labs, publisher-AI licensing negotiations, M&A due-diligence on data-heavy targets, and regulatory submissions under the Data Governance Act or Data Act.

BY WHOM IT IS USED

Data licensing counsel, Chief Data Officers, commercial contract teams, procurement, and regulators evaluating data-sharing fairness.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-right data licences (which collapse all four layers into one), the stack reveals the negotiable structure beneath. Compared with generic IP categories, it acknowledges that data rights are assembled rather than granted. The framing underpins modern enterprise data-as-a-service contracts.

RELATED FRAMEWORKS IN THIS COURSE

Trade Secret vs Patent vs Copyright · *Slibrary 19*
EU Database Right · *Slibrary 19*
Data Licensing Taxonomy · *Slibrary 19*
Defensive Moats · *Slibrary 19*
Data Value Chain · *Slibrary 7*

SOURCES (HARVARD REFERENCING STYLE)

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Trade Secret vs Patent vs Copyright

Three IP regimes — which protects AI models and data best?

PROFESSIONAL DEFINITION

Three classical IP regimes offer distinct protection profiles for AI models and data: Trade Secret (indefinite duration, requires confidentiality, no filing), Patent (20-year duration, requires public disclosure and novelty, examination), and Copyright (70+ year duration, automatic at creation, protects expression not ideas). Each fits different AI asset types — model weights best protected as trade secret; novel architectures patentable; training data as compilations copyright-eligible (WIPO, 2024; Menell et al., 2020).

WHO DEVELOPED IT

Classical treatments in Menell, Lemley and Merges' Intellectual Property in the New Technological Age (Menell et al., 2020), WIPO's 2024 comparative study on AI and IP (WIPO, 2024), and the US PTO's guidance on AI inventorship (USPTO, 2024). Early academic framing in Samuelson (1994) on the IP protection of software, extended to AI in Abbott (2020).

WHY IT WAS DEVELOPED

AI firms ask whether to patent their architectures (exposes them via disclosure), rely on trade secret (limits inspection and enforcement), or use copyright (narrow protection for expression only). The comparison makes these trade-offs explicit and aligns protection choice with competitive strategy.

FIGURE 1 · THREE-REGIME COMPARISON

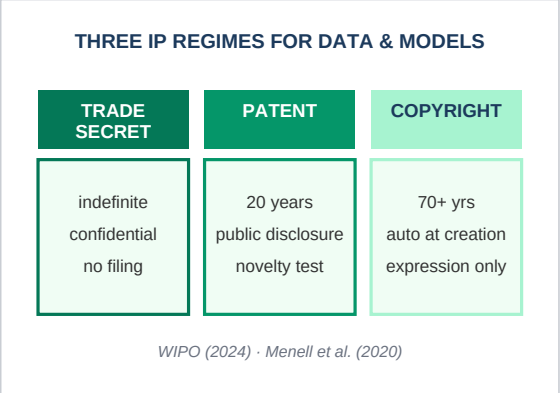

Figure 1. Trade Secret, Patent, and Copyright protection regimes compared on duration, disclosure requirements, and scope. Most frontier AI firms combine all three (WIPO, 2024).

HOW IT IS USED

IP strategy teams run each AI asset — model weights, training datasets, novel architectures, fine-tuning recipes — through the three-regime comparison and select the protection mode (or combination) that best matches the asset's economic half-life and competitive landscape.

WHEN IT IS USED

IP strategy reviews at AI labs, patent filing decisions, M&A IP due-diligence, and discussions about whether to publish frontier research or keep it confidential as trade secret.

BY WHOM IT IS USED

IP counsel, patent attorneys, Chief Research Officers, R&D leaders, and investors evaluating IP strength in AI company valuations.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with defaulting to patents, the three-regime comparison exposes cases where trade secret or copyright protection is stronger. For AI specifically, the dominant pattern — published architectures under trade-secret weights — is a direct consequence of this comparison rather than a compromise.

RELATED FRAMEWORKS IN THIS COURSE

Data Rights Stack · *Slibrary 19* | **EU Database Right** · *Slibrary 19* | **Data Licensing Taxonomy** · *Slibrary 19* | **Defensive Moats** · *Slibrary 19* | **AI Licence Types** · *Slibrary 18*

SOURCES (HARVARD REFERENCING STYLE)

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EU Sui Generis Database Right

SLIBRARY 19 · SHEET 3 / 5

The European protection that closes the gap left by copyright

PROFESSIONAL DEFINITION

The EU Sui Generis Database Right, established by Directive 96/9/EC, protects substantial investment in obtaining, verifying, or presenting the contents of a database against unauthorised extraction or re-utilisation, for 15 years from creation. It exists alongside copyright and applies where selection-and-arrangement copyright does not — the right closes the protection gap that copyright leaves in 'factual' databases, and is central to training-data legal disputes in Europe (European Parliament and Council, 1996; Getty Images v Stability AI [2025]).

WHO DEVELOPED IT

Created by the European Commission's Database Directive (96/9/EC) in 1996, transposed into EU and UK national law. Recent application to AI training is documented in Getty Images v Stability AI [2025] EWHC 2863 (Ch). Academic commentary by Derclaye (2008) is authoritative; Hugenholtz's 2024 commentary addresses AI-era application (Hugenholtz, 2024).

WHY IT WAS DEVELOPED

European policymakers observed that copyright protected only the selection-and-arrangement originality of databases, not the factual content. This left databases of comprehensive objective facts (phone directories, property records, image libraries) unprotected. The sui generis right fills this gap — and becomes one of the primary legal tools against AI training on European datasets.

FIGURE 1 · EU VS REST-OF-WORLD DIVIDE

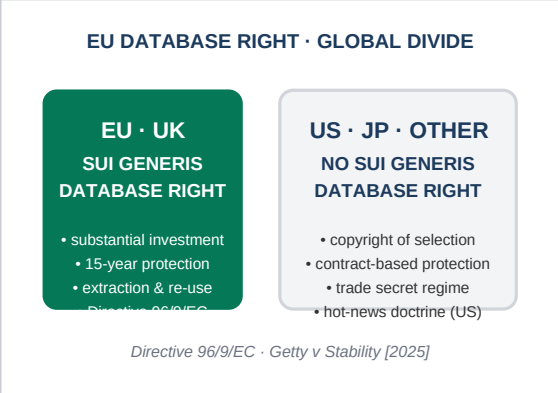

Figure 1. The EU/UK sui generis database right has no direct equivalent in US, Japan, or most other jurisdictions. European rights-holders have an extra legal tool against unauthorised training-data use (Directive 96/9/EC).

HOW IT IS USED

European rights-holders invoke the sui generis right alongside copyright in training-data disputes; US firms face weaker equivalents (contract, hot-news, trade secret). Firms operating across jurisdictions run dual analyses — copyright in the US, copyright plus sui generis in Europe.

WHEN IT IS USED

Training-data enforcement actions in Europe, licensing negotiations for European-sourced datasets, due-diligence on European data assets in cross-border M&A, and policy debates on AI text-and-data-mining exceptions.

BY WHOM IT IS USED

IP litigators in the EU and UK, general counsel at data businesses, regulators implementing Directive 96/9/EC, and policy advisors at the European Commission reviewing the Directive for AI-era updates.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with the US copyright-only regime, the sui generis right gives European rights-holders a distinct cause of action where copyright does not apply. For AI firms, the two-regime divide has direct strategic consequences for training-data acquisition — European data is legally different from US data.

RELATED FRAMEWORKS IN THIS COURSE

Data Rights Stack · *Slibrary 19* | **Trade Secret vs Patent vs Copyright** · *Slibrary 19* | **Data Licensing Taxonomy** · *Slibrary 19* | **Defensive Moats** · *Slibrary 19* | **Getty v Stability** · *Slibrary 17*

SOURCES (HARVARD REFERENCING STYLE)

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Hugenholtz, P.B. (2024) 'The New Database Right: Preserving the Free Flow of Information in the Age of Generative AI', Kluwer Copyright Blog.

Data Licensing Taxonomy

Scope × Duration × Exclusivity — the three-dimensional contract space

PROFESSIONAL DEFINITION

The Data Licensing Taxonomy is a three-dimensional classification of data licence terms: Scope (research only / internal / commercial / redistributable), Duration (one-time / time-limited / perpetual / revocable), and Exclusivity (non-exclusive / exclusive / first-right / assignable). Every practical data licence is a point in this three-dimensional space; the taxonomy enables consistent negotiation and comparison across heterogeneous agreements (Nimmer, 2024; OECD, 2023).

WHO DEVELOPED IT

Compiled from commercial data-licensing practice as documented in Nimmer on Copyright (Nimmer, 2024), OECD data-sharing guidance (OECD, 2023), and the standard template libraries maintained by the Open Data Institute (ODI, 2024) and data-exchange platforms (Snowflake, Databricks). Academic foundation in Lessig's Code and Other Laws of Cyberspace (Lessig, 2006).

WHY IT WAS DEVELOPED

Data licences are highly variable — a single firm can be subject to dozens of different licence terms simultaneously, and without a shared taxonomy, comparison and portfolio-level compliance become impossible. The three-dimensional taxonomy provides a consistent classification grid for any data licence.

FIGURE 1 · THREE DIMENSIONS WITH OPTIONS

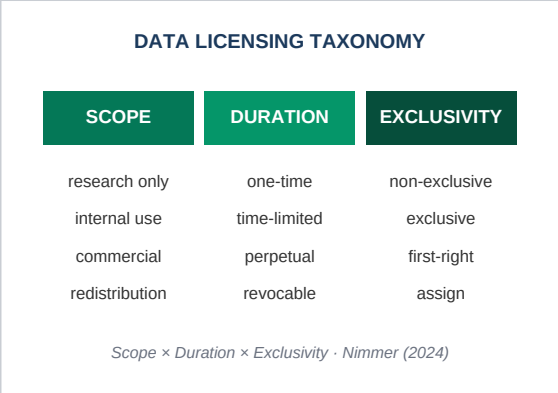

Figure 1. The three dimensions of data licence terms. Each licence is a point in this space; a portfolio of licences can be visualised as a scatter of points (Nimmer, 2024).

HOW IT IS USED

Data licensing teams classify every incoming licence against the taxonomy, maintain a portfolio view of licence commitments, and spot compliance risks where individual licences combine in conflicting ways.

WHEN IT IS USED

Data procurement reviews, licence portfolio audits, AI training-data sourcing due-diligence, and compliance reporting under contract-audit obligations.

BY WHOM IT IS USED

Data governance officers, procurement teams, data engineers responsible for training-data pipelines, and compliance auditors verifying licence adherence.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with reading each licence individually, the taxonomy enables portfolio-level comparison and compliance. Compared with generic contract classification, it foregrounds the three dimensions that most materially govern AI training-data use.

RELATED FRAMEWORKS IN THIS COURSE

Data Rights Stack · *Slibrary 19*
Trade Secret vs Patent vs Copyright · *Slibrary 19*
EU Database Right · *Slibrary 19*
Defensive Moats · *Slibrary 19*
AI Licence Types · *Slibrary 18*

SOURCES (HARVARD REFERENCING STYLE)

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Open Data Institute (2024) Data Licensing Template Library. London: ODI.

Defensive Moats for Data

Contractual · technical · trade-secret — compound protection

PROFESSIONAL DEFINITION

Defensive Moats for data are concentric layers of protection around a valuable dataset or model: contractual (NDAs, terms of service), technical (encryption, access controls, watermarking), and trade-secret (confidentiality regimes, employee covenants). Each layer protects against different failure modes; their combination — the compound moat — defines the practical defensibility of data and model assets (Barney, 1991; Porter, 1985; a16z, 2024).

WHO DEVELOPED IT

The moat metaphor in strategy comes from Buffett's writings and was formalised in the Resource-Based View (Barney, 1991) and Porter's competitive-strategy literature (Porter, 1985). Its application to AI/data defensibility is articulated in Andreessen Horowitz's 2024 State-of-AI reports (a16z, 2024) and Hamilton Helmer's 7 Powers (Helmer, 2016).

WHY IT WAS DEVELOPED

AI firms with strong models but weak data moats face rapid competitive erosion as rivals replicate capabilities. The concentric-moats framework shows that protection comes from combining multiple layers — no single layer is usually sufficient. Investors, in turn, price the depth of the moat into valuations.

FIGURE 1 · CONCENTRIC MOAT LAYERS

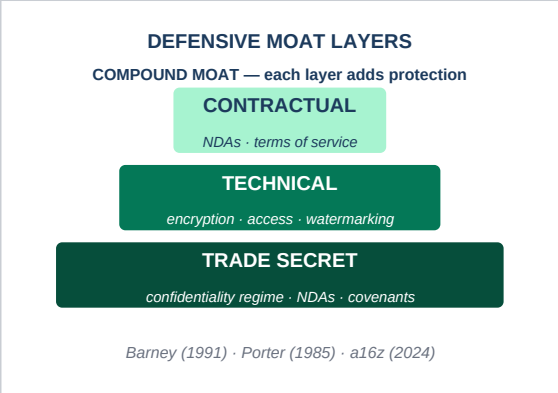

Figure 1. Three concentric layers of protection. Contractual (outer) is cheapest but thinnest; Trade Secret (inner) is strongest but requires active confidentiality management. A strong moat uses all three (a16z, 2024).

HOW IT IS USED

Strategy leaders assess each data or model asset against the three moat layers, identify gaps, and invest in strengthening the weakest layer. Portfolio-level moat strength informs IP protection strategy and investor communications.

WHEN IT IS USED

AI strategy reviews, IP protection investments, investor due-diligence on AI business defensibility, and competitive response planning when rivals replicate features.

BY WHOM IT IS USED

AI strategy leaders, CTOs, IP counsel, investor-relations teams, and investors evaluating AI business defensibility.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-moat thinking (contract-only, or patent-only), the concentric view captures the complementarity of layers. Compared with purely strategic 'moat narratives', it produces operational assessments that can be invested in and measured.

RELATED FRAMEWORKS IN THIS COURSE

Data Rights Stack · *Slibrary 19*
Trade Secret vs Patent vs Copyright · *Slibrary 19*
EU Database Right · *Slibrary 19*
Data Licensing Taxonomy · *Slibrary 19*
VRIO Framework · *Slibrary 2*

SOURCES (HARVARD REFERENCING STYLE)

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AI Monetisation Canvas

API · SaaS · Embedded · Licence — the four archetypes

SLIBRARY 20 · SHEET 1 / 5

PROFESSIONAL DEFINITION

The AI Monetisation Canvas is a four-archetype classification of how AI products capture economic value: API / Pay-per-use (OpenAI, Anthropic — billed by tokens or calls), SaaS / Subscription (Copilot, ChatGPT Plus — per-seat recurring), Embedded / OEM (Adobe Firefly inside Creative Cloud, Salesforce Einstein — AI as feature), and Licence / On-premise (Llama Enterprise, Mistral — model weights licensed for customer deployment). Each archetype implies distinct unit economics, go-to-market, and defensibility (a16z, 2024; Osterwalder and Pigneur, 2010).

WHO DEVELOPED IT

Synthesised from Andreessen Horowitz's commercial research (a16z, 2024), Business Model Canvas methodology (Osterwalder and Pigneur, 2010), and the observed commercial patterns of frontier AI providers in 2023–2025. Academic grounding in Teece's business-model-design literature (Teece, 2010).

WHY IT WAS DEVELOPED

Generative-AI firms in 2023 defaulted to API-pay-per-use, which frequently failed to capture value commensurate with perceived capability. The canvas forces explicit comparison of four archetypes, revealing that different AI workloads fit different archetypes — and that many successful firms combine two or more.

FIGURE 1 · FOUR-QUADRANT CANVAS

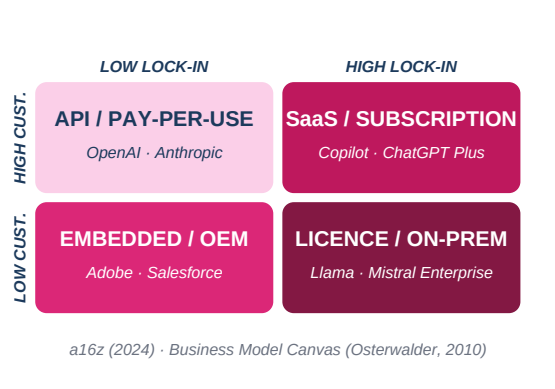

Figure 1. The four archetypes mapped on customisation × lock-in. API / pay-per-use is low-customisation, low-lock-in; Licence / on-prem is high-customisation, high-lock-in. Most firms occupy 1-2 quadrants (a16z, 2024).

HOW IT IS USED

Product and commercial teams classify each AI offering against the four archetypes, check fit with customer segments, and design explicit paths for customers to move between archetypes — e.g. API to SaaS upsell.

WHEN IT IS USED

AI product pricing and packaging decisions, go-to-market strategy reviews, M&A analysis of AI target business models, and investor pitches on monetisation strategy.

BY WHOM IT IS USED

AI product leaders, heads of go-to-market, pricing strategists, investors evaluating AI business-model scalability, and consultants advising on AI monetisation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with API-first default thinking, the canvas reveals where API is suboptimal. Compared with generic SaaS business-model frameworks, it captures the AI-specific archetypes (embedded and licence) that dominate enterprise deployments.

RELATED FRAMEWORKS IN THIS COURSE

AI Pricing Models · *Slibrary 20* | **Platform vs Product** · *Slibrary 20* | **Ecosystem Capture** · *Slibrary 20* | **Value Capture Curve** · *Slibrary 20* | **API Tiering** · *Slibrary 18*

SOURCES (HARVARD REFERENCING STYLE)

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AI Pricing Models

SLIBRARY 20 · SHEET 2 / 5

Per-token · per-seat · per-outcome · per-compute — the unit-economics choice

PROFESSIONAL DEFINITION

AI Pricing Models are the four dominant metered billing units: per-token (measuring model inputs and outputs — standard for LLM APIs); per-seat (monthly per-user — standard for AI-enhanced SaaS); per-outcome (billing for successful transactions — emerging for agents and automated workflows); and per-compute ($/GPU-hour — standard for infrastructure). Each aligns prices with a different value driver and has distinct margin and forecasting implications (a16z, 2024; Kumar, 2014).

WHO DEVELOPED IT

Commercial practice was shaped by OpenAI (per-token), Microsoft Copilot (per-seat), emerging agent startups (per-outcome), and AWS/Azure/Google (per-compute). The economics framing draws on Kumar's pricing literature (Kumar, 2014) and Andreessen Horowitz's 2024 generative-AI research (a16z, 2024). Academic treatment of value-based pricing in Hinterhuber (2008).

WHY IT WAS DEVELOPED

Per-token pricing captures compute cost but not value; per-seat captures ubiquity but not intensity; per-outcome captures value directly but requires attribution infrastructure. The typology exposes the trade-offs and the migration path — the industry is moving from per-token towards per-outcome as attribution technology matures (a16z, 2024).

FIGURE 1 · FOUR PRICING ARCHETYPES

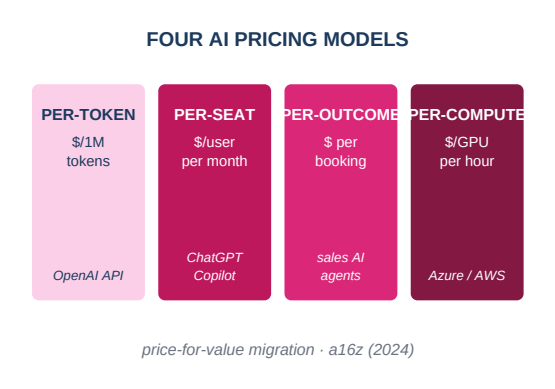

Figure 1. The four pricing archetypes, ordered approximately by value-alignment. Per-outcome is the emerging frontier; per-compute anchors the infrastructure layer (a16z, 2024).

HOW IT IS USED

Pricing teams match product to pricing archetype based on value attribution feasibility, customer willingness-to-pay, and competitive positioning. Hybrid pricing (per-seat with usage caps) is common during transitions.

WHEN IT IS USED

AI product pricing design, pricing reviews as product matures, enterprise contract negotiations, and competitive responses to rival pricing changes.

BY WHOM IT IS USED

Pricing strategists, product managers, revenue operations, CFOs modelling AI unit economics, and investors assessing AI business scalability.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with defaulting to per-token (infrastructure-aligned), the typology reveals that the leading commercial AI products are already migrating toward per-outcome as soon as attribution permits. The migration is the single biggest pricing transformation since the emergence of SaaS.

RELATED FRAMEWORKS IN THIS COURSE

Monetisation Canvas · *Slibrary 20*
Platform vs Product · *Slibrary 20*
Ecosystem Capture · *Slibrary 20*
Value Capture Curve · *Slibrary 20*
API Tiering · *Slibrary 18*

SOURCES (HARVARD REFERENCING STYLE)

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Platform vs Product Strategy

Ecosystem play or focused app — the fundamental AI go-to-market choice

PROFESSIONAL DEFINITION

The Platform vs Product Strategy is the fundamental strategic choice facing every AI company: become a Platform (horizontal infrastructure, APIs and developer ecosystem — e.g. OpenAI, AWS) or a Product (focused application for a specific use-case — e.g. Harvey for legal, Jasper for marketing). Platforms pursue network effects and winner-takes-most dynamics; Products pursue unit-economics and focused moats. Hybrid strategies exist but most firms cannot serve both well simultaneously (Parker, Van Alstyne and Choudary, 2016; a16z, 2024).

WHO DEVELOPED IT

The platform-vs-product framing draws on Parker, Van Alstyne and Choudary's Platform Revolution (Parker et al., 2016), the Rochet-Tirole two-sided markets literature (Rochet and Tirole, 2003), and Andreessen Horowitz's applied generative-AI analysis (a16z, 2024). Christensen's earlier disruption work (Christensen, 1997) is also relevant.

WHY IT WAS DEVELOPED

AI firms that try to be both platform and product typically fail at both — platform economics require broad neutrality, product economics require focused depth. The explicit choice clarifies strategy, focuses investment, and reduces the risk of diluted positioning as the firm scales.

FIGURE 1 · TWO STRATEGIC CHOICES

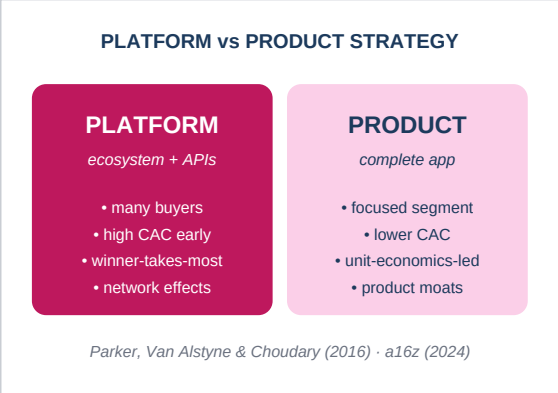

Figure 1. Platform and Product strategies differ on buyers, CAC, moat type, and examples. The choice is fundamental and difficult to reverse (Parker et al., 2016).

HOW IT IS USED

AI leaders make the platform-vs-product choice explicit, align investment and hiring to the chosen path, and review the position regularly as competition and scale reveal the fit — with migration from one to the other sometimes necessary but strategically costly.

WHEN IT IS USED

Company founding and Series A strategy, strategic pivots as product-market fit develops, investor pitches explaining go-to-market, and M&A decisions where acquirers must decide whether to absorb products into a platform.

BY WHOM IT IS USED

Founders, CEOs, boards, strategy leaders, investors in venture and growth-stage AI firms, and corporate-development teams evaluating acquisition integration paths.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with vague 'be both' positioning, the forced choice produces operational clarity. Compared with pure platform-economics framing, the AI-specific version recognises that high-quality vertical products can compete with horizontal platforms longer than in earlier tech eras, because AI capability is more rapidly commoditising than distribution.

RELATED FRAMEWORKS IN THIS COURSE

Monetisation Canvas · *Slibrary 20* | **AI Pricing Models** · *Slibrary 20* | **Ecosystem Capture** · *Slibrary 20* | **Value Capture Curve** · *Slibrary 20* | **Porter Five Forces** · *Slibrary 2*

SOURCES (HARVARD REFERENCING STYLE)

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Christensen, C.M. (1997) The Innovator's Dilemma. Boston, MA: Harvard Business School Press.
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Ecosystem Capture Map

SLIBRARY 20 · SHEET 4 / 5

Central platform · satellite stakeholders — winner-takes-most dynamics

PROFESSIONAL DEFINITION

The Ecosystem Capture Map is a hub-and-spoke visualisation of the stakeholders an AI platform must orchestrate: Developers (who build applications), Customers (who consume AI), Partners (who extend distribution), Models (whose weights are hosted), Data (whose contributors supply training corpora), and Regulators (who permit or constrain operations). Platform strategy is the coordinated management of all six relationships simultaneously (Parker, Van Alstyne and Choudary, 2016; Iansiti and Levien, 2004).

WHO DEVELOPED IT

The hub-and-spoke ecosystem view is developed in Iansiti and Levien's The Keystone Advantage (Iansiti and Levien, 2004) and Parker, Van Alstyne and Choudary's Platform Revolution (Parker et al., 2016). Application to AI platforms is in Andreessen Horowitz's 2024 analyses (a16z, 2024) and the MIT Initiative on the Digital Economy (Brynjolfsson and McElheran, 2024).

WHY IT WAS DEVELOPED

AI platforms that focus on only one stakeholder category — e.g. developers — and neglect others — e.g. regulators or data contributors — fail. The map forces consideration of all six relationships and supports investment allocation across ecosystem strands.

FIGURE 1 · SIX-SPOKE ECOSYSTEM MAP

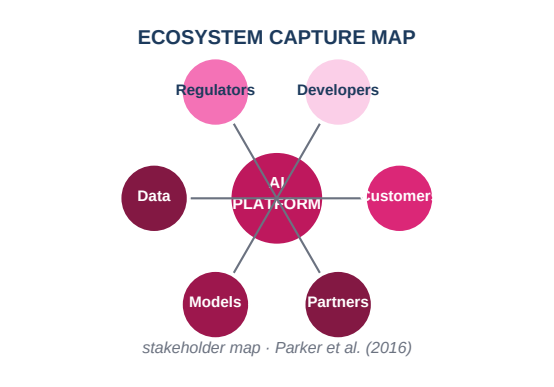

Figure 1. The central AI platform coordinates six stakeholder categories. Each spoke represents a distinct economic and regulatory relationship that must be managed concurrently (Parker et al., 2016).

HOW IT IS USED

Platform strategists audit their investments across the six spokes, identify the weakest spoke, and allocate attention and resources to strengthen it. Periodic mapping reveals structural imbalances that precede platform failure.

WHEN IT IS USED

Platform strategy reviews, investor communications about ecosystem health, regulatory submissions on platform fairness (Digital Markets Act), and competitive analyses of rival platforms.

BY WHOM IT IS USED

Platform leadership, strategy teams, ecosystem partnership leaders, regulatory affairs teams, and investors evaluating platform defensibility.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with developer-first or customer-first platform thinking, the six-spoke view captures the full stakeholder set that AI platforms must serve. Compared with generic ecosystem maps, it highlights AI-specific categories (models and data contributors) that do not appear in software platform ecosystems.

RELATED FRAMEWORKS IN THIS COURSE

Monetisation Canvas · *Slibrary 20* | **AI Pricing Models** · *Slibrary 20* | **Platform vs Product** · *Slibrary 20* | **Value Capture Curve** · *Slibrary 20* | **Porter Five Forces** · *Slibrary 2*

SOURCES (HARVARD REFERENCING STYLE)

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AI Value Capture Curve

SLIBRARY 20 · SHEET 5 / 5

Who captures value across the AI stack — and how it shifts over time

PROFESSIONAL DEFINITION

The AI Value Capture Curve tracks how total value created by AI is distributed across three stack layers — Compute (hyperscalers, GPU makers), Model Labs (frontier-AI developers), and Application Layer (AI-enabled products) — and how that distribution evolves over time. In 2023, compute captured most value; as models commoditise and applications mature, value capture is predicted to shift upward to the application layer (a16z, 2024; Brynjolfsson, 2022).

WHO DEVELOPED IT

Value-capture curves across technology stacks were articulated by Andreessen Horowitz's 2024 State-of-AI reports (a16z, 2024). Theoretical grounding is in value-chain analysis (Porter, 1985) and Brynjolfsson's general-purpose-technology research (Brynjolfsson, 2022). Empirical tracking is by Epoch AI (Epoch AI, 2025) and McKinsey (McKinsey, 2024).

WHY IT WAS DEVELOPED

Investors and firms need a structured view of where in the AI stack value will accrue over time. The curve reveals the pattern — compute dominates initially, then value migrates upward as capabilities commoditise — and informs where to position investment, company formation, and strategic partnerships.

FIGURE 1 · THREE-LAYER EVOLUTION 2023–2027

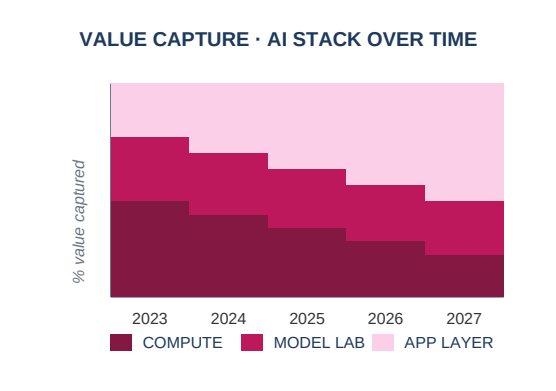

Figure 1. Value-capture share across three AI stack layers from 2023 to 2027. Compute share declines; application-layer share rises as models commoditise (a16z, 2024).

HOW IT IS USED

Strategy teams and investors use the curve to time investments — entering compute positions early, model labs mid-cycle, and application layers as infrastructure matures. Portfolio construction explicitly balances across the three layers.

WHEN IT IS USED

Investment portfolio construction, strategic planning for AI-adjacent businesses, corporate venture investment decisions, and investor communications about AI-era value migration.

BY WHOM IT IS USED

Venture investors, public-markets strategists, corporate development teams, AI platform strategists, and policymakers concerned with AI industrial policy.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with static value-chain analysis, the curve makes explicit the temporal shift — a 2023 value map misleads a 2026 investment decision. Compared with general tech-stack analysis, the AI-specific version identifies the unique position of model labs between infrastructure and application layers.

RELATED FRAMEWORKS IN THIS COURSE

Monetisation Canvas · *Slibrary 20* | **AI Pricing Models** · *Slibrary 20* | **Platform vs Product** · *Slibrary 20* | **Ecosystem Capture** · *Slibrary 20* | **Value Chain** · *Slibrary 2*

SOURCES (HARVARD REFERENCING STYLE)

Andreessen Horowitz (2024) State of Generative AI: Enterprise. Menlo Park, CA: a16z.
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Algorithmic Trading Spectrum

Discretionary · Quantitative · ML · HFT — the intensity continuum

PROFESSIONAL DEFINITION

The Algorithmic Trading Spectrum orders trading approaches by the intensity of algorithmic involvement: Discretionary (human portfolio managers make decisions), Quantitative (rule-based systematic strategies — AQR, Renaissance), ML / Systematic (model-driven — Two Sigma, Bridgewater), and High-Frequency (microsecond latency — Citadel Securities, Virtu). Each position implies distinct data, talent, and infrastructure requirements (Muldowney, 2023; López de Prado, 2018).

WHO DEVELOPED IT

The spectrum is synthesised in Muldowney's Impact of Technology and AI on Investment Management (Muldowney, 2023), formalised in López de Prado's Advances in Financial Machine Learning (López de Prado, 2018), and documented in academic overviews by Harvey, Liechty and Liu (Harvey et al., 2016).

WHY IT WAS DEVELOPED

Investors, regulators, and fund managers conflate 'algorithmic trading' into a single category, obscuring that the four positions have fundamentally different economics, risks, and competitive dynamics. The spectrum forces explicit positioning and enables meaningful benchmarking across firms.

FIGURE 1 · FOUR POSITIONS ON THE SPECTRUM

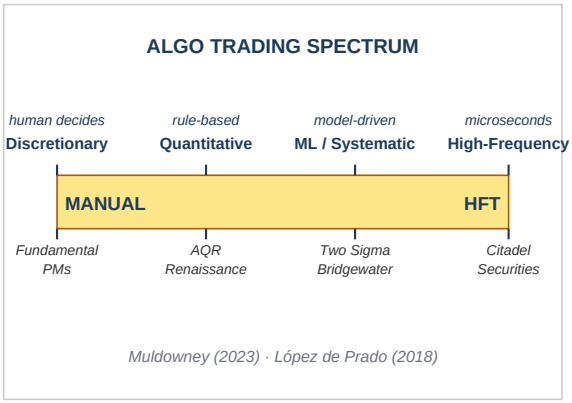

Figure 1. The algorithmic trading spectrum from Discretionary to HFT, with representative firms. Each position requires progressively more data, compute, and infrastructure investment (Muldowney, 2023).

HOW IT IS USED

Asset managers position their strategies explicitly on the spectrum, benchmark against peers at the same position, and make investment in data and technology decisions consistent with their position.

WHEN IT IS USED

Fund strategy reviews, investor communications about approach, regulatory filings (UCITS, AIFMD) that require strategy descriptions, and due-diligence by allocators assessing manager approach.

BY WHOM IT IS USED

Portfolio managers, CIOs, quant strategists, fund-of-funds allocators, institutional investors, and regulators reviewing fund filings.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with binary 'active vs passive' framing, the spectrum reveals the four distinct algorithmic positions. Compared with technology-focused views, it anchors positioning in investment approach rather than infrastructure. The spectrum is central to assessing where AI adds most value within a manager's workflow.

RELATED FRAMEWORKS IN THIS COURSE

Portfolio AI Stack · *Slibrary 21* | **Factor Zoo** · *Slibrary 21* | **Risk Analytics** · *Slibrary 21* | **Investment AI Maturity** · *Slibrary 21* | **Automation vs Augmentation** · *Slibrary 4*

SOURCES (HARVARD REFERENCING STYLE)

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López de Prado, M. (2018) Advances in Financial Machine Learning. Hoboken, NJ: Wiley.
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Portfolio AI Stack

Data · Signals · Optimiser · Output — the four-layer architecture

PROFESSIONAL DEFINITION

The Portfolio AI Stack is a four-layer architecture underlying modern AI-enabled investment management: Data (prices, fundamentals, alternative datasets), Signals (factor models, ML-based alpha generation), Optimiser (risk budgeting, constraint management), and Output (trades, reports, rebalancing instructions). Each layer is a potential source of competitive advantage and a potential outsourcing candidate (BlackRock Aladdin, 2024; Muldowney, 2023).

WHO DEVELOPED IT

The stack is articulated operationally in BlackRock Aladdin's architecture documentation (BlackRock, 2024) and theoretically in Muldowney's analysis (Muldowney, 2023). The signal-optimiser separation derives from Markowitz portfolio theory (Markowitz, 1952) and modern factor investing (Fama and French, 1993).

WHY IT WAS DEVELOPED

Investment firms face build-or-buy decisions at each layer — build proprietary signals or license factor models, build an optimiser or use Aladdin, run in-house data pipelines or use Bloomberg/FactSet. The stack makes these choices explicit and enables coherent architectural strategy rather than ad-hoc vendor accumulation.

FIGURE 1 · FOUR-LAYER STACK

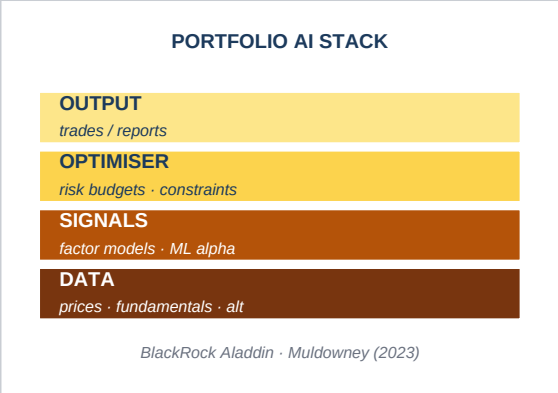

Figure 1. Data at the base, Signals and Optimiser in the middle, Output at the top. Each layer is a distinct build-or-buy decision and a potential source of differentiation (BlackRock Aladdin, 2024).

HOW IT IS USED

CIOs and CTOs map each investment strategy against the four layers, identify where to invest in proprietary capability and where to use platform solutions, and review the architecture annually as the vendor landscape evolves.

WHEN IT IS USED

Technology strategy reviews at asset managers, vendor negotiations with BlackRock, MSCI, FactSet, and Bloomberg, M&A due-diligence on investment-management targets, and platform-versus-build decisions at new fund launches.

BY WHOM IT IS USED

CIOs, CTOs, heads of investment technology, quant research leaders, and consultants advising on investment-management architecture.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with monolithic 'investment system' thinking, the layered stack allows targeted investment. Compared with pure Buy-vs-Build binary, it reveals the hybrid architectures used by modern firms. The stack is the conceptual spine of the BlackRock Aladdin platform that now manages ~$20T in assets.

RELATED FRAMEWORKS IN THIS COURSE

Algo Trading Spectrum · *Slibrary 21* | **Factor Zoo** · *Slibrary 21* | **Risk Analytics** · *Slibrary 21* | **Investment AI Maturity** · *Slibrary 21* | **Data Value Chain** · *Slibrary 7*

SOURCES (HARVARD REFERENCING STYLE)

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Factor Zoo · Classical + ML

Value · Momentum · Quality · Size · Volatility · ML Alpha

PROFESSIONAL DEFINITION

The Factor Zoo is the collection of systematic return drivers identified by empirical finance research — the classical five (Value, Momentum, Quality, Size, Low-Volatility) plus the emerging category of ML Alpha (non-linear signals extracted by machine learning). Over 300 factors have been published academically (Harvey and Liu, 2019), though many are redundant or statistically questionable (Fama and French, 1993; Asness et al., 2013).

WHO DEVELOPED IT

Classical factors were formalised by Fama and French (1993), extended by Carhart (1997) with momentum, and codified in the five-factor model (Fama and French, 2015). The 'Factor Zoo' critique is in Harvey, Liu and Zhu (2016). ML-based alpha extraction is covered in López de Prado (2018) and Gu, Kelly and Xiu (2020).

WHY IT WAS DEVELOPED

Factor investing provides a theoretically grounded, empirically tested framework for systematic strategies. The zoo concept acknowledges that many factors are duplicative or false discoveries, and that careful statistical validation is required before deployment. ML extends the factor approach to non-linear signals but inherits (and amplifies) the overfitting risks.

FIGURE 1 · SIX FACTOR CATEGORIES

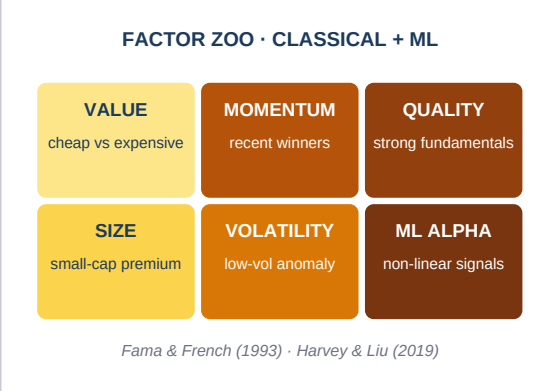

Figure 1. Six factor categories — the classical five plus ML Alpha. Each category has established academic support but widely varying real-world premium persistence (Fama and French, 1993; Harvey and Liu, 2019).

HOW IT IS USED

Quant teams construct factor-diversified portfolios, run factor-attribution analyses on active returns, and select factor exposures consistent with client mandates. ML is used either as a new factor or as a meta-signal for factor timing.

WHEN IT IS USED

Strategy design for systematic funds, factor-based product structuring (smart-beta ETFs), performance attribution of active managers, and academic finance research.

BY WHOM IT IS USED

Quant portfolio managers, factor-investing specialists, ETF product managers, allocators assessing factor exposure, and academic finance researchers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with ad-hoc strategy design, factor investing provides a statistically grounded discipline. Compared with pure price-based strategies, it grounds returns in identifiable risk premia. ML Alpha extends the framework to capture interactions and non-linearities that linear factor models miss — at the cost of reduced interpretability and greater overfitting risk.

RELATED FRAMEWORKS IN THIS COURSE

Algo Trading Spectrum · *Slibrary 21* | **Portfolio AI Stack** · *Slibrary 21* | **Risk Analytics** · *Slibrary 21* | **Investment AI Maturity** · *Slibrary 21* | **Bias-Variance** · *Slibrary 5*

SOURCES (HARVARD REFERENCING STYLE)

Fama, E.F. and French, K.R. (1993) 'Common Risk Factors in the Returns on Stocks and Bonds', Journal of Financial Economics, 33(1), pp. 3–56.
Gu, S., Kelly, B. and Xiu, D. (2020) 'Empirical Asset Pricing via Machine Learning', Review of Financial Studies, 33(5), pp. 2223–2273.
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López de Prado, M. (2018) Advances in Financial Machine Learning. Hoboken, NJ: Wiley.

Risk Analytics · VaR

Value at Risk — the portfolio risk number boards actually read

PROFESSIONAL DEFINITION

Value at Risk (VaR) estimates the maximum expected loss of a portfolio over a specified time horizon at a given confidence level (typically 95% or 99%). VaR is the most widely-used portfolio risk metric, mandated by Basel III for banks and by most institutional investors. ML-based VaR extensions (copulas, quantile regression) address the classical VaR's failure modes in tail events (Jorion, 2007; López de Prado, 2018).

WHO DEVELOPED IT

VaR was developed at J.P. Morgan in the early 1990s with the RiskMetrics methodology (J.P. Morgan, 1996) and popularised by Jorion's authoritative text (Jorion, 2007). Basel II and III enshrined VaR as a regulatory metric (BCBS, 2019). ML extensions are in López de Prado (2018) and Engle (2004).

WHY IT WAS DEVELOPED

Investment committees and boards need a single number that summarises portfolio risk. VaR provides that — a statement like 'our 99% 10-day VaR is $10M' is interpretable across different asset classes and strategies. The single-number convenience has its critics (Taleb) but no alternative has displaced it.

FIGURE 1 · VAR ON A RETURN DISTRIBUTION

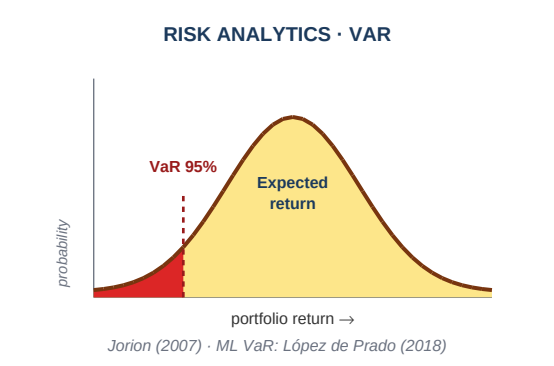

Figure 1. The portfolio return distribution with VaR 95% marked at -1.645 standard deviations. The red tail represents the 5% probability of losses exceeding the VaR threshold (Jorion, 2007).

HOW IT IS USED

Risk teams compute VaR daily for each portfolio and aggregate for the fund; results are reported to the investment committee, board, and regulator. Back-testing compares realised losses to VaR predictions to validate the methodology.

WHEN IT IS USED

Daily portfolio risk reporting, regulatory submissions (Basel III for banks, AIFMD for alternative investment funds), investment committee reviews, and back-testing for model validation.

BY WHOM IT IS USED

Chief Risk Officers, portfolio risk managers, investment committees, regulators (ECB, Fed, FCA), and auditors validating risk models.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with ad-hoc risk descriptions, VaR provides a single interpretable number. Compared with standard-deviation-only measures, it focuses on downside specifically. ML VaR addresses classical VaR's Gaussian-assumption failures — particularly during regime breaks — by using non-parametric quantile methods.

RELATED FRAMEWORKS IN THIS COURSE

Algo Trading Spectrum · *Slibrary 21* | **Portfolio AI Stack** · *Slibrary 21* | **Factor Zoo** · *Slibrary 21* | **Investment AI Maturity** · *Slibrary 21* | **AI Risk Taxonomy** · *Slibrary 15*

SOURCES (HARVARD REFERENCING STYLE)

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Investment AI Maturity Staircase

Manual · Assisted · Augmented · Autonomous · Fully AI

PROFESSIONAL DEFINITION

The Investment AI Maturity Staircase describes the progression of asset managers' AI adoption through five stages: Manual (spreadsheets), Assisted (AI-aided research), Augmented (ML alpha combined with human judgment), Autonomous (model-driven portfolios with human oversight), and Fully AI (end-to-end AI agents). Each stage requires different talent, technology, and governance — and sits at a different competitive position (McKinsey, 2024; Muldowney, 2023).

WHO DEVELOPED IT

The staircase is synthesised from McKinsey's annual State-of-AI reports (McKinsey, 2024), Muldowney's investment-management analysis (Muldowney, 2023), and the Alternative Investment Management Association's AI surveys (AIMA, 2024). The maturity-model paradigm derives from CMMI (Paulk et al., 1993).

WHY IT WAS DEVELOPED

Asset managers often misplace themselves on the AI maturity curve, claiming Fully AI capability when they are Assisted, or claiming to be Manual while in fact heavily reliant on ML signals. The staircase provides an honest self-assessment tool and a roadmap for advancement.

FIGURE 1 · FIVE-STAGE STAIRCASE

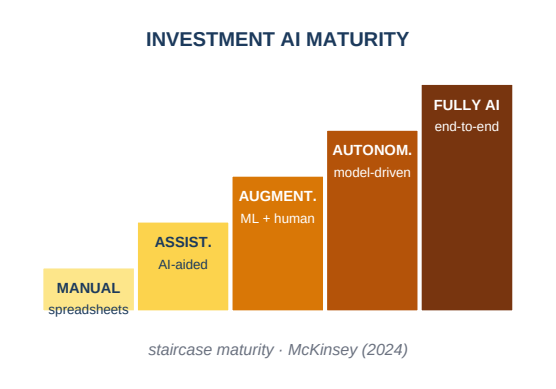

Figure 1. The five-stage maturity staircase. Each step represents materially different talent, infrastructure, and governance requirements. Most firms currently sit at Assisted or Augmented (McKinsey, 2024).

HOW IT IS USED

Executive teams assess current position honestly, define target position, and build a multi-year roadmap that addresses the talent, technology, and governance changes required to advance. The staircase is also used externally in investor and client communications.

WHEN IT IS USED

Strategic planning at asset managers, investor communications about AI capability, M&A due-diligence on investment-management targets, and allocator assessments of manager AI maturity.

BY WHOM IT IS USED

CEOs, CIOs, CTOs, heads of investment technology, investment committees, allocators, and external consultants advising on AI transformation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with marketing claims about AI capability, the staircase enforces operational honesty — each stage has measurable evidence. Compared with generic AI maturity models, it is tuned to investment management's specific capability sequence and talent requirements.

RELATED FRAMEWORKS IN THIS COURSE

Algo Trading Spectrum · *Slibrary 21* | **Portfolio AI Stack** · *Slibrary 21* | **Factor Zoo** · *Slibrary 21* | **Risk Analytics** · *Slibrary 21* | **AI Maturity Staircase** · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

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Investment Process Cycle

Research · Construct · Execute · Monitor · Rebalance · Report

PROFESSIONAL DEFINITION

The Investment Process Cycle is the six-phase loop followed by institutional asset managers: Research (alpha and data discovery), Construct (portfolio design), Execute (trade and manage costs), Monitor (risk and P&L), Rebalance (reallocate to targets), and Report (client and board). AI inserts into each phase differently — research benefits from alt data and ML signals; execution benefits from smart order routing; monitoring benefits from real-time risk analytics (CFA Institute, 2020; Maginn et al., 2007; Muldowney, 2023).

WHO DEVELOPED IT

The six-phase structure is codified in the CFA Institute's curriculum (CFA Institute, 2020), Maginn et al.'s Managing Investment Portfolios (Maginn et al., 2007), and the Global Investment Performance Standards (CFA Institute GIPS, 2020). AI-specific application is analysed in Muldowney (2023).

WHY IT WAS DEVELOPED

Asset managers deploying AI often target a single phase — typically research — and neglect others. The cycle reveals that AI's value is unlocked by integrating across phases, and that bottlenecks in any one phase cap the value of AI in other phases.

FIGURE 1 · SIX-PHASE PROCESS CYCLE

Figure 1. The six-phase investment process loop. AI inserts into each phase with different value-capture mechanisms (CFA Institute, 2020).

HOW IT IS USED

CIOs audit AI deployment across all six phases, identify the phase where additional AI investment has the highest marginal value, and sequence AI initiatives to balance capability across the cycle.

WHEN IT IS USED

Asset-manager AI strategy reviews, investor communications about AI integration, board-level AI transformation reviews, and regulatory submissions on systematic investment processes.

BY WHOM IT IS USED

CIOs, CTOs, Heads of Investment Technology, investment committees, and consultants advising asset managers on AI adoption.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with 'research-first' AI adoption, the cycle forces balanced capability building. Compared with technology-stack views, the process-cycle view keeps the focus on investment outcomes rather than infrastructure.

RELATED FRAMEWORKS IN THIS COURSE

Aladdin Modules · *Slibrary 22* | **Alternative Data** · *Slibrary 22* | **Backtesting** · *Slibrary 22* | **ESG + AI** · *Slibrary 22* | **Algo Trading Spectrum** · *Slibrary 21*

SOURCES (HARVARD REFERENCING STYLE)

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BlackRock Aladdin · Modules

Portfolio · Risk · Compliance · Trading · Operations · Client

PROFESSIONAL DEFINITION

BlackRock Aladdin (Asset, Liability, Debt and Derivative Investment Network) is the dominant investment-management platform, running approximately $21 trillion in assets across BlackRock's own funds and 200+ third-party clients. Its architecture comprises six core modules: Portfolio (analytics and performance), Risk (multi-factor and VaR), Compliance (rule engine), Trading (EMS/OMS), Operations (middle and back office), and Client (reporting). Aladdin is the archetype of an enterprise investment platform (BlackRock, 2024).

WHO DEVELOPED IT

Developed by BlackRock Solutions starting in 1994 and documented in BlackRock's technical and annual report disclosures (BlackRock, 2024). Academic treatment by Haberly et al. (Haberly, MacDonald-Korth, Urban and Wojcik, 2019) analyses Aladdin's competitive and systemic-risk implications. Muldowney (2023) covers its role in investment-management AI integration.

WHY IT WAS DEVELOPED

Investment managers historically cobbled together dozens of vendor systems (risk, trading, reporting, compliance) with poor data interoperability. Aladdin unified them in one platform with a single security master and consistent data model — a decisive competitive advantage that captured the market and, critics argue, now produces systemic-risk concerns through concentration.

FIGURE 1 · SIX CORE MODULES

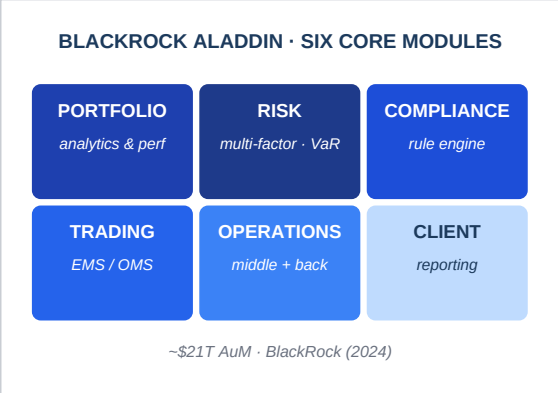

Figure 1. The six core Aladdin modules cover the entire investment-management lifecycle. Approximately $21T in assets are managed on the platform (BlackRock, 2024).

HOW IT IS USED

Asset managers license Aladdin (or competitors — State Street Alpha, MSCI Barra, SimCorp Dimension), configure it for their strategies, and use it as the operational spine for the investment process. AI capabilities are increasingly delivered through the platform rather than built bespoke.

WHEN IT IS USED

Build-vs-buy decisions at new asset managers, platform migrations at existing managers, vendor negotiations, regulatory reviews of operational resilience, and systemic-risk analyses by central banks.

BY WHOM IT IS USED

CTOs, COOs, operations leaders at asset managers, BlackRock Solutions sales teams, and regulators examining investment-management platform concentration.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with fragmented point-solution architectures, Aladdin unifies the investment lifecycle under one platform. Compared with purely-analytical risk tools, it integrates risk into daily operations. The platform model is the template being replicated by competitors and, arguably, extended by AI agents in the next generation of investment platforms.

RELATED FRAMEWORKS IN THIS COURSE

Investment Process Cycle · *Slibrary 22* | **Alternative Data** · *Slibrary 22* | **Backtesting** · *Slibrary 22* | **ESG + AI** · *Slibrary 22* | **Portfolio AI Stack** · *Slibrary 21*

SOURCES (HARVARD REFERENCING STYLE)

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BlackRock (2024) 2024 Annual Report. New York, NY: BlackRock.
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Alternative Data Taxonomy

Satellite · Transaction · Web · Sentiment — beyond price and fundamentals

PROFESSIONAL DEFINITION

Alternative Data is the category of non-traditional information used in investment decisions, classified in four groups: Satellite imagery (parking lots, shipping), Transaction data (credit-card panels, payment flows), Web-derived data (scraped content, social, app usage), and Sentiment data (news NLP, earnings-call analysis). The alt-data market reached ~$4B in 2024 and is growing 30% annually as managers seek edge beyond commoditised price and fundamental data (Denev and Amen, 2020; AlternativeData.org, 2024).

WHO DEVELOPED IT

The category was formalised by Denev and Amen's The Book of Alternative Data (Denev and Amen, 2020), AlternativeData.org's industry reports (AlternativeData.org, 2024), and academic work by Eisfeldt, Kim and Papanikolaou (Eisfeldt et al., 2020). The Alternative Data Council sets industry standards.

WHY IT WAS DEVELOPED

Classical financial data (prices, fundamentals) is commoditised and widely available; alpha is compressed. Alt data offers informational edge for managers willing to build the capability to source, clean, and analyse it. The taxonomy structures the search across four dominant categories and reveals which category fits which strategy.

FIGURE 1 · FOUR ALT-DATA CATEGORIES

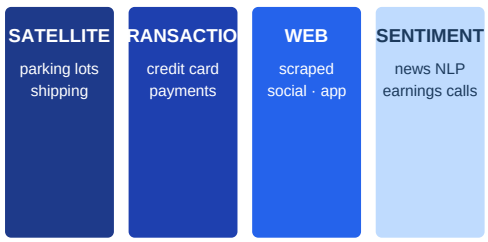

Figure 1. The four major categories of alternative data. Each requires different sourcing, cleaning, and analytical expertise (Denev and Amen, 2020).

HOW IT IS USED

Quant and fundamental managers source alt data from specialist vendors (Orbital Insight, Second Measure, RavenPack), build cleaning and alpha-testing pipelines, and integrate successful signals into production strategies. Compliance review addresses data-provenance and privacy risks.

WHEN IT IS USED

New signal research, strategy-performance attribution reviews, vendor selection for alt-data subscriptions, and due-diligence by allocators on manager data sophistication.

BY WHOM IT IS USED

Quant research teams, alt-data acquisition specialists, CIOs, data-governance officers, compliance teams verifying data provenance, and vendors in the alt-data ecosystem.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with price-and-fundamentals-only investing, alt data extends the information universe dramatically. Compared with ad-hoc data experiments, the taxonomy structures the search and enables systematic comparison. ML techniques are essential for extracting signal from the unstructured nature of most alt-data categories.

RELATED FRAMEWORKS IN THIS COURSE

Investment Process Cycle · *Slibrary 22* | **Aladdin Modules** · *Slibrary 22* | **Backtesting** · *Slibrary 22* | **ESG + AI** · *Slibrary 22* | **Data Value Chain** · *Slibrary 7*

SOURCES (HARVARD REFERENCING STYLE)

AlternativeData.org (2024) Alternative Data Industry Report. New York, NY: Alternative Data Council.
Denev, A. and Amen, S. (2020) The Book of Alternative Data. Hoboken, NJ: Wiley.
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Muldowney, P. (2023) 'The Impact of Technology and AI on Investment Management', Plans, Trusts & Pensions, 4(1), pp. 8–13.

Backtesting Pipeline

Data · Train · Test · Walk-Forward · Live — avoiding look-ahead bias

SLIBRARY 22 · SHEET 4 / 5

PROFESSIONAL DEFINITION

The Backtesting Pipeline is the disciplined sequence used to validate an investment strategy before deployment: Data (historical inputs), Train (in-sample parameter fitting), Test (out-of-sample validation), Walk-Forward (rolling window evaluation), and Live (paper trading before real capital). The pipeline guards against look-ahead bias, overfitting, and data-snooping failures that routinely produce backtested results that do not survive live deployment (López de Prado, 2018; Bailey et al., 2014).

WHO DEVELOPED IT

Codified in López de Prado's Advances in Financial Machine Learning (López de Prado, 2018), Bailey et al.'s 'Pseudo-Mathematics and Financial Charlatanism' (Bailey et al., 2014) on overfitting in finance, and Harvey, Liu and Zhu's 'Evaluating Trading Strategies' (Harvey et al., 2016).

WHY IT WAS DEVELOPED

Backtested investment strategies routinely fail to replicate in live trading because developers explicitly or implicitly use future information (look-ahead), overfit to historical idiosyncrasies, or test many strategies and report only the successful one (data snooping). The disciplined pipeline structurally prevents each failure mode.

FIGURE 1 · FIVE-STAGE PIPELINE

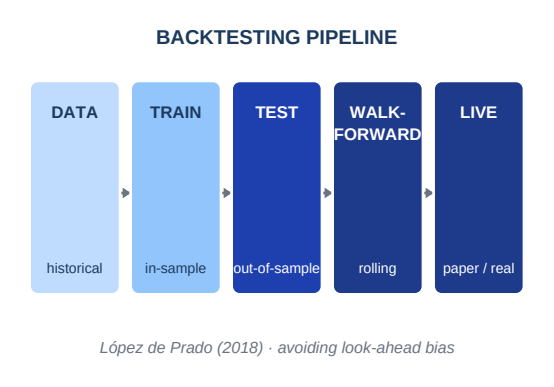

Figure 1. The backtesting pipeline. Each stage produces evidence that must be met before progressing. Walk-Forward is critical to detect regime-dependence (López de Prado, 2018).

HOW IT IS USED

Quant teams structure strategy development around the pipeline, maintain audit logs of tests attempted, and apply multiple-testing corrections when many hypotheses are evaluated. Live paper-trading is used as a final sanity check before capital deployment.

WHEN IT IS USED

New strategy development, periodic re-validation of existing strategies, regulatory submissions on risk-model validation, and due-diligence by allocators assessing manager process rigour.

BY WHOM IT IS USED

Quant researchers, risk managers validating strategy performance, internal auditors reviewing model risk, and allocators assessing manager process quality.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with informal backtesting that treats the entire history as training data, the pipeline preserves out-of-sample evidence. Compared with naive cross-validation, walk-forward captures the time-ordered nature of financial data. López de Prado (2018) provides the ML-era canonical treatment.

RELATED FRAMEWORKS IN THIS COURSE

Investment Process Cycle · *Slibrary 22* | **Aladdin Modules** · *Slibrary 22* | **Alternative Data** · *Slibrary 22* | **ESG + AI** · *Slibrary 22* | **Train-Val-Test** · *Slibrary 5*

SOURCES (HARVARD REFERENCING STYLE)

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López de Prado, M. (2018) Advances in Financial Machine Learning. Hoboken, NJ: Wiley.

Muldowney, P. (2023) 'The Impact of Technology and AI on Investment Management', Plans, Trusts & Pensions, 4(1), pp. 8–13.

ESG + AI Integration

AI at the centre of Environmental · Social · Governance analysis

PROFESSIONAL DEFINITION

ESG + AI integration uses machine learning and alternative data to score corporate performance on Environmental (emissions, resource use), Social (labour, supply chain, community), and Governance (board, audit, ethics) dimensions. AI addresses the core ESG data problem — patchy, inconsistent, unverified disclosure — by extracting signal from text, imagery, and alternative sources. ESG assets under management reached ~$30T in 2024 (Principles for Responsible Investment, 2024; Global Sustainable Investment Alliance, 2024).

WHO DEVELOPED IT

The PRI (Principles for Responsible Investment, 2024) frames the field; the Global Sustainable Investment Alliance (GSIA, 2024) tracks industry data. Academic treatment by Berg, Kölbel and Rigobon (Berg et al., 2022) analyses ESG rating divergence. AI-specific ESG research is emerging through the UN PRI working groups and the MIT Sloan Sustainability Initiative.

WHY IT WAS DEVELOPED

ESG scoring is plagued by low-quality disclosure, rating divergence across providers (Berg et al., 2022), and greenwashing. AI provides scalable tools to verify claims against satellite and text evidence, identify inconsistencies, and produce ratings that are more robust than pure disclosure-based scores.

FIGURE 1 · AI AT THE ESG OVERLAP

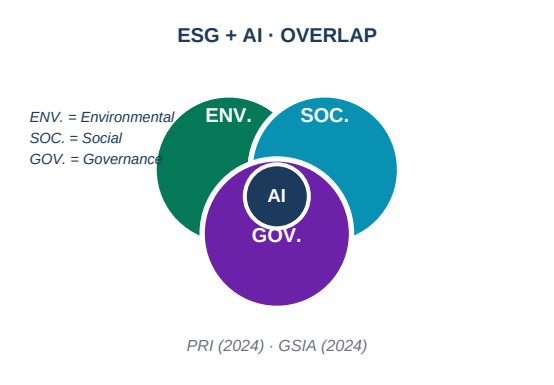

Figure 1. AI sits at the intersection of Environmental, Social, and Governance analysis, extracting signal from diverse data sources where disclosure is weak (PRI, 2024).

HOW IT IS USED

ESG-integrated investment firms build or license AI-enhanced ESG scores, run exclusion and tilt strategies, produce client reporting on ESG exposure, and engage with portfolio companies on material ESG issues identified by AI.

WHEN IT IS USED

ESG mandate fulfilment, sustainability-labelled fund structuring (SFDR Article 8 and 9 in Europe), stewardship and engagement activities, and reporting to pension-fund ESG committees and regulators.

BY WHOM IT IS USED

ESG analysts, ESG data vendors (MSCI ESG, Sustainalytics), portfolio managers of sustainable funds, stewardship teams, and regulators implementing SFDR, TCFD, and CSRD.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with pure disclosure-based ESG ratings, AI-enhanced ESG uses independent verification via satellite imagery, text NLP of sustainability reports, and cross-source consistency checks. Compared with single-provider scoring, AI enables multi-source ensemble ratings that reduce single-point-of-failure bias.

RELATED FRAMEWORKS IN THIS COURSE

Investment Process Cycle · *Slibrary 22* | **Aladdin Modules** · *Slibrary 22* | **Alternative Data** · *Slibrary 22* | **Backtesting** · *Slibrary 22* | **Responsible AI Radar** · *Slibrary 16*

SOURCES (HARVARD REFERENCING STYLE)

Berg, F., Kölbel, J.F. and Rigobon, R. (2022) 'Aggregate Confusion: The Divergence of ESG Ratings', Review of Finance, 26(6), pp. 1315–1344.
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Rogers Diffusion for AI Adoption

SLIBRARY 23 · SHEET 1 / 5

Innovators · Early Adopters · Majority · Laggards

PROFESSIONAL DEFINITION

Rogers' Diffusion of Innovations (Rogers, 2003) classifies adopters of a new technology into five categories by adoption timing: Innovators (2.5%), Early Adopters (13.5%), Early Majority (34%), Late Majority (34%), and Laggards (16%). Applied to AI, the model predicts that most enterprises currently sit at the Early Majority chasm, and that crossing it requires different capabilities than the initial Innovator-stage experiments (Rogers, 2003; Moore, 1991).

WHO DEVELOPED IT

Everett Rogers formulated the framework in Diffusion of Innovations (Rogers, 1962, 5th edn. 2003). Geoffrey Moore extended it in Crossing the Chasm (Moore, 1991) with the analysis of the gap between Early Adopters and Early Majority. AI-specific application is in McKinsey's State-of-AI series (McKinsey, 2024).

WHY IT WAS DEVELOPED

Enterprises often plan AI programmes as if adoption is linear, when empirically adoption follows an S-curve with discontinuities. The framework predicts where resistance and acceleration will occur, and when to deploy Innovator-stage features (unrefined, experimental) versus Majority-stage features (polished, integrated).

FIGURE 1 · ADOPTION CURVE WITH CATEGORIES

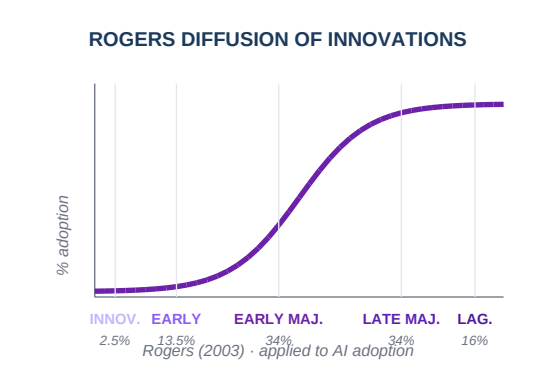

Figure 1. The S-curve of cumulative adoption with Rogers' five categories. Most enterprises currently cluster around Early Majority for AI (Rogers, 2003).

HOW IT IS USED

Transformation leaders identify their current position on the curve, plan chasm-crossing investments (product-market fit, reference customers, case studies), and time capability investments against the expected wave of adopter readiness.

WHEN IT IS USED

AI product launches, enterprise AI strategy, go-to-market planning for AI startups, and board-level discussion of transformation timing.

BY WHOM IT IS USED

Chief Transformation Officers, CMOs, product leaders, founders, and investors assessing AI product-market fit.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with linear rollout plans, the diffusion curve reveals the non-linear adopter dynamics. Compared with purely technology-led deployment, it surfaces the human adoption sequence that actually determines success or failure at each stage.

RELATED FRAMEWORKS IN THIS COURSE

Pilot-to-Scale · *Slibrary 23* | **Change Flywheel** · *Slibrary 23* | **Talent Matrix** · *Slibrary 23* | **Readiness Radar** · *Slibrary 23* | **Perez S-Curve** · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

McKinsey (2024) The State of AI in 2024. New York, NY: McKinsey & Company.
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Pilot-to-Scale Funnel

Ideation · Pilot · Production · Scale — the 10:1 reality

PROFESSIONAL DEFINITION

The Pilot-to-Scale Funnel describes the empirical survival rate of AI initiatives: of 100 ideated, ~70 reach pilot, ~30 reach production, and only ~12 reach enterprise scale. Each stage gate requires capabilities — data, engineering, change management — that many organisations lack. Most programmes fail between Pilot and Production, not at ideation (BCG, 2024; Deloitte, 2024).

WHO DEVELOPED IT

Survival-rate data is compiled by BCG (BCG, 2024), Deloitte's State of AI surveys (Deloitte, 2024), and McKinsey's annual research (McKinsey, 2024). The funnel structure draws on classical R&D pipeline management (Cooper, 2008) applied to AI.

WHY IT WAS DEVELOPED

Firms count pilots as success and celebrate experimentation; the empirical reality is that scaling is the hard part and requires different capability than piloting. Making the funnel explicit forces investment in the stages after pilot — precisely where most programmes fail silently.

FIGURE 1 · FOUR-STAGE SURVIVAL FUNNEL

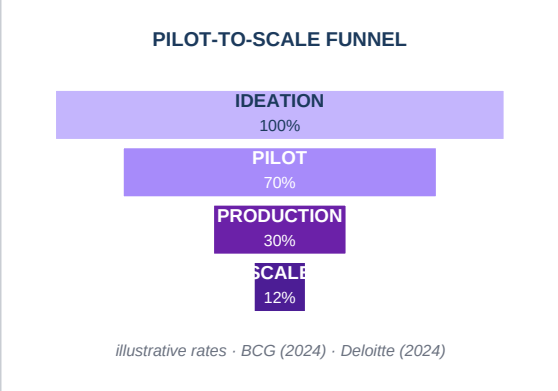

Figure 1. Illustrative survival rates: 100% Ideation → 70% Pilot → 30% Production → 12% Scale. The biggest drop is Pilot → Production (BCG, 2024).

HOW IT IS USED

AI programme managers track conversion rates at each stage gate, identify where the pipeline leaks, and invest capability in the leaking stage rather than the already-strong stages.

WHEN IT IS USED

AI portfolio reviews, transformation programme audits, board-level oversight of AI investment ROI, and consultant benchmarking against industry survival rates.

BY WHOM IT IS USED

Chief AI Officers, transformation leaders, PMO teams, internal auditors, and investors assessing AI programme discipline.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with celebrating pilots, the funnel enforces stage-gate discipline. Compared with generic R&D pipeline views, it calibrates to the specific survival rates observed in AI programmes. The framework is central to moving firms from pilot-fatigue to sustained scale.

RELATED FRAMEWORKS IN THIS COURSE

Diffusion Curve · *Slibrary 23* | **Change Flywheel** · *Slibrary 23* | **Talent Matrix** · *Slibrary 23* | **Readiness Radar** · *Slibrary 23* | **AI Project Failure Funnel** · *Slibrary 3*

SOURCES (HARVARD REFERENCING STYLE)

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Cooper, R.G. (2008) 'Perspective: The Stage-Gate Idea-to-Launch Process — Update, What's New, and NexGen Systems', Journal of Product Innovation Management, 25(3), pp. 213–232.
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AI Transformation Flywheel

Data · People · Process · Culture — reinforcing loop

PROFESSIONAL DEFINITION

The AI Transformation Flywheel is a four-element reinforcing loop driving sustainable AI adoption: Data (quality and access), People (capability and incentives), Process (decision-rights and workflows), and Culture (mindset and trust in AI outputs). Each element strengthens the next — better data attracts better people, who design better processes, which reinforce culture — and momentum compounds (McKinsey, 2024; Kotter, 1996).

WHO DEVELOPED IT

The flywheel framing comes from Jim Collins' Good to Great (Collins, 2001); applied to AI transformation in McKinsey's State-of-AI research (McKinsey, 2024). The underlying change-management logic draws on Kotter's 8-step model (Kotter, 1996) and contemporary digital-transformation literature.

WHY IT WAS DEVELOPED

Transformation programmes that invest in only one element (data, or people, or technology) stall because the other elements constrain progress. The flywheel shows that all four elements must progress together, with each reinforcing the others — there is no shortcut.

FIGURE 1 · FOUR-ELEMENT REINFORCING LOOP

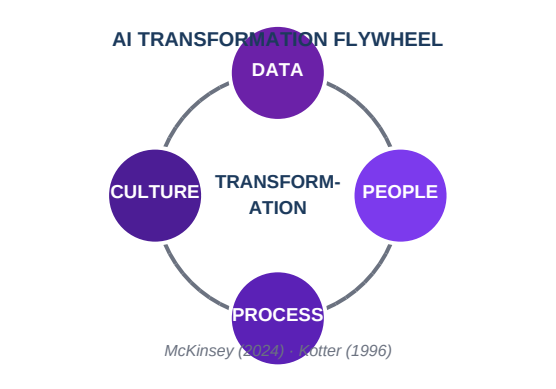

Figure 1. The four flywheel elements — Data, People, Process, Culture — reinforce each other. Momentum compounds once the loop is engaged (McKinsey, 2024).

HOW IT IS USED

Transformation leaders assess current strength of each element, identify the weakest link, and invest disproportionately there to restore balance. Quarterly flywheel reviews monitor the progression of each element.

WHEN IT IS USED

Transformation programme design, annual AI strategy reviews, board-level discussion of transformation progress, and M&A integration planning.

BY WHOM IT IS USED

Chief Transformation Officers, CIOs, Chief People Officers, CEOs, and consultants advising on AI transformation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with element-isolated views (data-led, people-led, process-led), the flywheel enforces systems thinking. Compared with linear transformation models, it captures the compounding nature of momentum — why early progress is slow and later progress accelerates.

RELATED FRAMEWORKS IN THIS COURSE

Diffusion Curve · *Slibrary 23* | **Pilot-to-Scale** · *Slibrary 23* | **Talent Matrix** · *Slibrary 23* | **Readiness Radar** · *Slibrary 23* | **AI Maturity Staircase** · *Slibrary 1*

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AI Talent Matrix

Demand × internal supply — retain, reskill, redeploy, reduce

PROFESSIONAL DEFINITION

The AI Talent Matrix classifies roles by external demand × internal supply, producing four strategic actions: Retain (high demand + high internal), Rehire / Reskill (high demand + low internal — the costliest gap), Redeploy (low demand + high internal — shift to adjacent roles), and Reduce (low demand + low internal — wind down). The matrix is central to AI-era workforce planning (BCG, 2024; McKinsey, 2024).

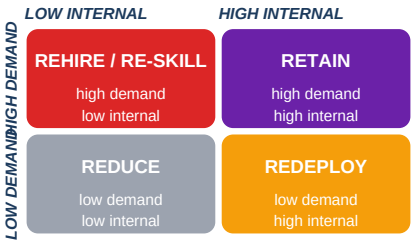
WHO DEVELOPED IT

Talent-matrix frameworks date to strategic HR research (Huselid, 1995); the AI-specific adaptation is developed in BCG's Future of Work research (BCG, 2024) and McKinsey's AI and the Workforce (McKinsey, 2024). Academic grounding in Acemoglu and Restrepo (2018) on AI and employment.

WHY IT WAS DEVELOPED

AI changes role requirements faster than hiring markets can respond; the matrix supports proactive workforce planning that matches investment in reskilling, hiring, and redeployment to business need. Without it, firms default to reactive hiring that is slow and expensive.

FIGURE 1 · DEMAND × SUPPLY QUADRANTS



BCG (2024) · McKinsey (2024)

Figure 1. The four-quadrant talent matrix. High-demand + low-internal (top left) is the critical-gap quadrant requiring aggressive reskilling or hiring (BCG, 2024).

HOW IT IS USED

Chief People Officers map current workforce onto the matrix, design multi-year reskilling and hiring plans targeted at the top-left quadrant, and review quarterly as AI reshapes role requirements.

WHEN IT IS USED

Annual workforce planning, AI transformation programmes, M&A integration (assessing acquired talent), and board-level discussion of human-capital strategy.

BY WHOM IT IS USED

CHROs, heads of talent, L&D leaders, AI transformation leaders, and board-level compensation committees.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with reactive hiring, the matrix enables proactive workforce planning. Compared with one-dimensional 'upskilling' programmes, it targets investment to the gap that matters. AI-specific use is growing as firms realise that AI-era talent requirements are non-linear and require explicit reallocation rather than organic evolution.

RELATED FRAMEWORKS IN THIS COURSE

Diffusion Curve · *Slibrary 23* | **Pilot-to-Scale** · *Slibrary 23* | **Change Flywheel** · *Slibrary 23* | **Readiness Radar** · *Slibrary 23* | **BCG People Pillar** · *Slibrary 4*

SOURCES (HARVARD REFERENCING STYLE)

Acemoglu, D. and Restrepo, P. (2018) 'The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment', American Economic Review, 108(6), pp. 1488–1542.

Boston Consulting Group (2024) Building the AI-Ready Workforce. Boston, MA: BCG.

Huselid, M.A. (1995) 'The Impact of Human Resource Management Practices on Turnover, Productivity, and Corporate Financial Performance', Academy of Management Journal, 38(3), pp. 635–672.

McKinsey (2024) AI and the Workforce. New York, NY: McKinsey & Company.

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AI Readiness Radar

Strategy · Data · Talent · Governance · Tech — five-dimension diagnostic

PROFESSIONAL DEFINITION

The AI Readiness Radar is a five-dimension diagnostic for organisational preparedness: Strategy (clarity of AI priorities), Data (availability and quality), Talent (capability and gap), Governance (risk, ethics, controls), and Tech (infrastructure and tooling). Each dimension is scored on a 0–100 scale against evidence criteria, producing a readiness polygon that reveals balanced or asymmetric preparedness (BCG, 2024; MIT-CISR, 2024).

WHO DEVELOPED IT

Synthesised from BCG's AI Readiness Index (BCG, 2024), MIT Centre for Information Systems Research (MIT-CISR, 2024), and McKinsey's State-of-AI surveys (McKinsey, 2024). The radar visual format is a standard strategy-diagnostic pattern adapted for AI.

WHY IT WAS DEVELOPED

Organisations often have strong capability on some dimensions (e.g. tech infrastructure) and weak capability on others (e.g. governance or data) — and the weakest dimension caps the overall AI value realised. The radar surfaces asymmetry and guides investment to the weakest point.

FIGURE 1 · FIVE-DIMENSION READINESS

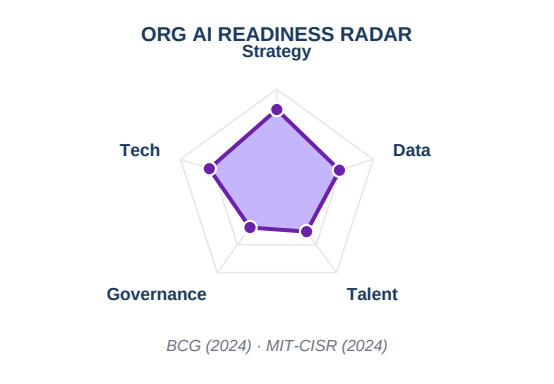

Figure 1. An illustrative readiness polygon. Asymmetry — strong on Strategy and Tech, weaker on Governance and Talent — is the common pattern and the priority for balanced investment (BCG, 2024).

HOW IT IS USED

Transformation leaders score each dimension using evidence, plot the polygon, and build a multi-year roadmap that raises the lowest-scoring dimension rather than further strengthening already-strong dimensions.

WHEN IT IS USED

Annual AI strategy reviews, M&A due-diligence on AI-transformation targets, board-level AI oversight, and external consulting engagements on AI transformation.

BY WHOM IT IS USED

Chief Transformation Officers, CIOs, board AI committees, external consultants, and industry benchmarking bodies.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-score readiness ratings, the radar preserves dimension-level detail. Compared with checklist diagnostics, the polygon visualisation is instantly interpretable at board level. The radar is the most widely-used AI readiness artefact in contemporary transformation consulting.

RELATED FRAMEWORKS IN THIS COURSE

Diffusion Curve · *Slibrary 23* | **Pilot-to-Scale** · *Slibrary 23* | **Change Flywheel** · *Slibrary 23* | **Talent Matrix** · *Slibrary 23* | **AI Maturity Staircase** · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

Boston Consulting Group (2024) The AI Readiness Index. Boston, MA: BCG.
McKinsey (2024) The State of AI in 2024. New York, NY: McKinsey & Company.
MIT Centre for Information Systems Research (2024) Building the AI-Ready Enterprise. Cambridge, MA: MIT-CISR.
Deloitte (2024) State of AI in the Enterprise: 7th Edition. London: Deloitte.

MARTI301 · Six-Unit Integration

Strategy · ML · CX · Finance · IP · Investment — the integrated AI leader

PROFESSIONAL DEFINITION

The Six-Unit Integration Map consolidates the MARTI301 curriculum into a single framework for integrated AI leadership. Each unit — Strategy & Governance (U1), ML & Data (U2), Customer Experience (U3), Finance & Risk (U4), IP & Value Creation (U5), and Investment Management (U6) — contributes a distinct competency that the modern AI leader must combine rather than specialise in. The map is the capstone artefact students produce to demonstrate mastery across the full curriculum.

WHO DEVELOPED IT

Developed for MARTI301 at EU Business School (Haas, 2026) drawing on the integrated AI-leadership literature including McKinsey (McKinsey, 2024), BCG (BCG, 2024), and academic syntheses by Agrawal, Gans and Goldfarb (Agrawal et al., 2022) and Brynjolfsson and McAfee (Brynjolfsson and McAfee, 2014).

WHY IT WAS DEVELOPED

AI leadership is frequently taught as isolated competencies — strategy without ML, ML without governance, finance without IP. The map forces integration: business leaders must navigate all six dimensions simultaneously, and the integrated perspective is the core competency MARTI301 develops.

FIGURE 1 · HEXAGONAL INTEGRATION

Figure 1. Six MARTI301 units radiate from the integrated AI Leader at the centre. Each unit contributes a competency that cannot stand alone (Haas, 2026).

HOW IT IS USED

Students demonstrate mastery by producing an integrated AI strategy for a real firm that explicitly references frameworks from all six units. Practising executives use the map to audit their own portfolio of AI capabilities and identify the unit where investment is weakest.

WHEN IT IS USED

MARTI301 capstone assessment, executive education programmes on integrated AI leadership, and self-audit exercises by AI leaders assessing their competency breadth.

BY WHOM IT IS USED

MARTI301 Masters students, EU Business School faculty, executive programme participants, and practising CAIOs, CIOs, and CDOs assessing their own competency coverage.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with specialist-competency AI education, the integrated map builds cross-functional AI leadership. Compared with purely strategic or purely technical programmes, MARTI301's six-unit design reflects the integrated reality of executive AI decision-making.

RELATED FRAMEWORKS IN THIS COURSE

AI Strategy Canvas · *Slibrary 24* | **Klarna Integrated Case** · *Slibrary 24* | **Practitioner Playbook** · *Slibrary 24* | **Future-Proof Leadership** · *Slibrary 24* | **AI Strategy Pyramid** · *Slibrary 1*

SOURCES (HARVARD REFERENCING STYLE)

Agrawal, A., Gans, J. and Goldfarb, A. (2022) Power and Prediction: The Disruptive Economics of Artificial Intelligence. Boston, MA: Harvard Business Review Press.

Boston Consulting Group (2024) Where's the Value in AI? Build for the Future 2024. Boston, MA: BCG.

Brynjolfsson, E. and McAfee, A. (2014) The Second Machine Age: Work, Progress, and Prosperity in a Time of Brilliant Technologies. New York, NY: W.W. Norton.

McKinsey (2024) The State of AI in 2024. New York, NY: McKinsey & Company.

AI Strategy Canvas

Nine lenses — the single page for AI strategy

PROFESSIONAL DEFINITION

The AI Strategy Canvas is a nine-cell single-page structure capturing the essential strategic decisions for AI: AI (priorities), Data (quality and access), Talent (build/buy/grow), Risk (governance), Value (ROI and P&L), IP (protection), Tech (infrastructure), Change (organisation), and Metrics (board KPIs). Adapted from the Business Model Canvas (Osterwalder and Pigneur, 2010), it produces the single artefact most useful for AI strategy conversations at board level.

WHO DEVELOPED IT

Adapted for MARTI301 from the Business Model Canvas (Osterwalder and Pigneur, 2010) and the AI Strategy tools in BCG's Build-for-the-Future research (BCG, 2024) and McKinsey's State-of-AI (McKinsey, 2024). Academic framing of single-page strategy in Martin (2010).

WHY IT WAS DEVELOPED

AI strategies often sprawl into 80-slide decks that obscure the essential choices. The canvas forces compression into one page, revealing where choices are clear and where they are fudged. Boards and investors can evaluate the strategy in minutes rather than hours.

FIGURE 1 · NINE-CELL CANVAS

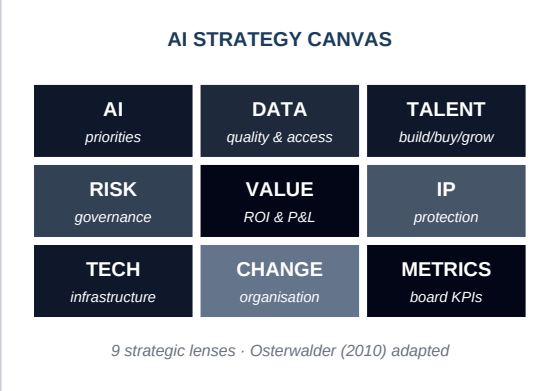

Figure 1. The nine-cell AI strategy canvas. Each cell represents a distinct strategic decision that must be made explicit before deployment (adapted from Osterwalder and Pigneur, 2010).

HOW IT IS USED

AI leaders populate each of the nine cells, present the canvas to board or investment committee, and revise iteratively. The canvas becomes the reference artefact for all subsequent strategy discussions.

WHEN IT IS USED

Annual AI strategy planning, board AI reviews, investor presentations on AI strategy, and consultant-led AI strategy sprints.

BY WHOM IT IS USED

Chief AI Officers, CIOs, CEOs, strategy consultants, and investors assessing AI strategy quality.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with long-form strategy documents, the canvas enforces concision. Compared with generic business strategy tools, it is specifically calibrated to the AI-era strategic trade-offs — data, talent, IP, governance — that classical strategy tools do not foreground.

RELATED FRAMEWORKS IN THIS COURSE

Six-Unit Integration · *Slibrary 24* | **Klarna Integrated Case** · *Slibrary 24* | **Practitioner Playbook** · *Slibrary 24* | **Future-Proof Leadership** · *Slibrary 24* | **Monetisation Canvas** · *Slibrary 20*

SOURCES (HARVARD REFERENCING STYLE)

Boston Consulting Group (2024) Where's the Value in AI? Build for the Future 2024. Boston, MA: BCG.
Martin, R.L. (2010) 'The Execution Trap', Harvard Business Review, 88(7-8), pp. 64–71.
McKinsey (2024) The State of AI in 2024. New York, NY: McKinsey & Company.
Osterwalder, A. and Pigneur, Y. (2010) Business Model Generation: A Handbook for Visionaries, Game Changers, and Challengers. Hoboken, NJ: Wiley.

Klarna · Integrated Case Study

2022–2025 AI transformation — cross-unit application

PROFESSIONAL DEFINITION

The Klarna Integrated Case traces the fintech's 2022–2025 AI transformation as a cross-unit case study: AI-agent launch (2022), workforce reduction of 38% (2023), ~$40M cost savings (2024), human-rebalance (2025), and planned agentic expansion. The case exercises frameworks from all six MARTI301 units — strategy, ML, customer experience, finance, IP, and investment — in a single real-world context (Klarna, 2022–2025; Financial Times, 2024; Reuters, 2025).

WHO DEVELOPED IT

Primary sources include Klarna's annual reports (Klarna, 2022; 2023; 2024), CEO public statements (Siemiatkowski, 2024; 2025), and industry coverage in the Financial Times (FT, 2024) and Reuters (Reuters, 2025). Academic analysis is emerging in INSEAD and Harvard Business School case-study treatments.

WHY IT WAS DEVELOPED

Klarna is the most publicly documented case of an AI-driven workforce transformation in a regulated fintech, and the subsequent partial reversal (human-rebalance 2025) makes it unusually rich as a teaching case. It covers strategy, ML, CX, finance, IP, and investment simultaneously.

FIGURE 1 · FOUR-YEAR TIMELINE

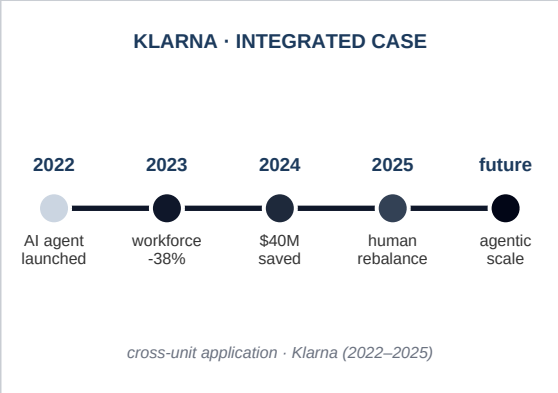

Figure 1. Key Klarna AI milestones 2022–2025. The 2025 human-rebalance following earlier aggressive automation provides critical nuance for case analysis (Klarna, 2022–2025).

HOW IT IS USED

Students apply frameworks from each MARTI301 unit to the Klarna timeline, producing multi-unit analyses of the same situation. Practitioners use the case as a reference point for AI-driven transformation decisions in their own organisations.

WHEN IT IS USED

MARTI301 assessments (midterm, final, resit), executive education AI transformation discussions, and consulting engagements where Klarna is a frequent reference case.

BY WHOM IT IS USED

MARTI301 students, EUBS faculty, executive programme participants, AI transformation consultants, and fintech industry analysts.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-unit cases (strategy-only or ML-only), the Klarna case exercises integration across units. Compared with static cases, the 2025 human-rebalance provides a built-in critique that forces students to engage with the limits of aggressive AI automation rather than assume linear progress.

RELATED FRAMEWORKS IN THIS COURSE

Six-Unit Integration · *Slibrary 24* | **AI Strategy Canvas** · *Slibrary 24* | **Practitioner Playbook** · *Slibrary 24* | **Future-Proof Leadership** · *Slibrary 24* | **AI Value Drivers** · *Slibrary 13*

SOURCES (HARVARD REFERENCING STYLE)

Financial Times (2024) 'Klarna's AI-Led Workforce Reduction', Financial Times, 4 March.
Klarna Bank AB (2024) Annual Report 2024. Stockholm: Klarna.
Reuters (2025) 'Klarna Reverses Course, Hires Back Customer Service Staff', Reuters, 14 May.
Siemiatkowski, S. (2024) 'Interview with the CEO on AI Transformation', Bloomberg, 28 January.

Practitioner Playbook

Diagnose · Strategise · Prototype · Scale · Sustain

PROFESSIONAL DEFINITION

The Practitioner Playbook is a five-step sequence synthesising MARTI301's frameworks into a reusable workflow for AI leaders: (1) Diagnose — use the Readiness Radar to assess current state; (2) Strategise — produce an AI Strategy Canvas; (3) Prototype — pilot with clear KPIs; (4) Scale — move to production with governance and monitoring; (5) Sustain — measure and iterate. Each step has supporting frameworks drawn from the course's 120-sheet library.

WHO DEVELOPED IT

Developed for MARTI301 at EU Business School (Haas, 2026) drawing on McKinsey's Transform-Scale-Sustain frameworks (McKinsey, 2024), BCG's Build-for-the-Future research (BCG, 2024), and academic change-management literature (Kotter, 1996).

WHY IT WAS DEVELOPED

Practitioners need an action-oriented synthesis that is shorter than the full 120-sheet framework library. The five-step playbook provides the minimum viable structure for executing AI transformation without requiring the practitioner to re-derive the workflow from first principles.

FIGURE 1 · FIVE-STEP PLAYBOOK

Figure 1. Five sequential steps, each with its dominant framework from the MARTI301 library. The playbook condenses the course into an executable sequence (Haas, 2026).

HOW IT IS USED

AI leaders follow the five steps as the default sequence for any new AI initiative, adapting framework choices within each step to the specific context. The playbook is shared across transformation teams as a common language.

WHEN IT IS USED

Every new AI initiative, transformation programme kick-offs, and onboarding of new AI leadership hires who need to align quickly with organisational method.

BY WHOM IT IS USED

Chief AI Officers, transformation consultants, AI programme managers, and newly-appointed AI leaders in transformation roles.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with project-management-generic workflows, the playbook is tuned to AI's specific capability dependencies. Compared with full framework catalogues, it supplies the minimum actionable structure — enough to act, little enough to remember.

RELATED FRAMEWORKS IN THIS COURSE

Six-Unit Integration · *Slibrary 24* | **AI Strategy Canvas** · *Slibrary 24* | **Klarna Integrated Case** · *Slibrary 24*
Future-Proof Leadership · *Slibrary 24* | **Pilot-to-Scale** · *Slibrary 23*

SOURCES (HARVARD REFERENCING STYLE)

Boston Consulting Group (2024) Where's the Value in AI? Build for the Future 2024. Boston, MA: BCG.
Kotter, J.P. (1996) Leading Change. Boston, MA: Harvard Business Review Press.
McKinsey (2024) The State of AI in 2024. New York, NY: McKinsey & Company.
Cooper, R.G. (2008) 'Perspective: The Stage-Gate Idea-to-Launch Process', Journal of Product Innovation Management, 25(3), pp. 213–232.

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Future-Proof AI Leadership

Adaptive strategy · Responsible ethics · Excellent execution

PROFESSIONAL DEFINITION

Future-Proof AI Leadership is a three-tier synthesis of the competencies required to lead through accelerating AI change: Adaptive Strategy (sense and respond to rapid change), Responsible Ethics (earn trust and manage safety), and Excellent Execution (ship and iterate). The three tiers must all be present simultaneously — strength in one does not compensate for weakness in another. This is the capstone synthesis of MARTI301.

WHO DEVELOPED IT

Synthesised for MARTI301 (Haas, 2026) drawing on Teece's dynamic capabilities literature (Teece, 2007), responsible-AI academic work (Floridi et al., 2018), and execution-excellence research (Martin, 2010). The three-tier structure aligns with classical leadership models (Northouse, 2018).

WHY IT WAS DEVELOPED

AI leaders who are strong strategists but weak on execution fail to deliver; strong executors who neglect ethics face regulatory and reputational failure; ethical and executional leaders who are strategically inflexible are overtaken by competitors. All three tiers are non-negotiable for future-proof AI leadership.

FIGURE 1 · THREE-TIER PYRAMID

Figure 1. Excellent Execution is the broad base; Responsible Ethics sits in the middle; Adaptive Strategy is the apex. All three are required (Haas, 2026).

HOW IT IS USED

MARTI301 graduates commit to developing all three tiers, use the pyramid to audit their own leadership periodically, and evaluate peers and direct reports against the integrated model rather than single-tier strengths.

WHEN IT IS USED

MARTI301 capstone reflection, annual self-assessment by AI leaders, leadership-development programmes, and performance reviews of AI leadership roles.

BY WHOM IT IS USED

MARTI301 students upon completion, practising AI leaders, board AI committees assessing AI leadership, and executive recruiters evaluating AI leadership candidates.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with single-dimension leadership models (strategic, ethical, or execution-focused), the pyramid insists on integration. Compared with generic leadership frameworks, it is tuned to the specific demands of AI leadership — accelerating change, high stakes, and novel ethical challenges. This is the synthesis MARTI301 leaves its graduates with as they begin their AI leadership journey.

RELATED FRAMEWORKS IN THIS COURSE

Six-Unit Integration · *Slibrary 24* | **AI Strategy Canvas** · *Slibrary 24* | **Klarna Integrated Case** · *Slibrary 24* | **Practitioner Playbook** · *Slibrary 24* | **Responsible AI Radar** · *Slibrary 16*

SOURCES (HARVARD REFERENCING STYLE)

Floridi, L., Cowls, J., Beltrametti, M., et al. (2018) 'AI4People — An Ethical Framework for a Good AI Society', Minds and Machines, 28(4), pp. 689–707.
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Teece, D.J. (2007) 'Explicating Dynamic Capabilities: The Nature and Microfoundations of (Sustainable) Enterprise Performance', Strategic Management Journal, 28(13), pp. 1319–1350.